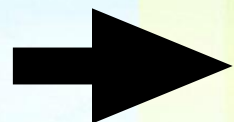


Business Process Automation with VBA and Python

Mr. Eddie Chow / 10 October 2026



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Introduction to business process automation

Business Process automation with VBA

Business process automation with Python

Introduction to project management for business process automation

Development and implementation of business process automation

Final Group Presentation





HKUSPACE
香港大學專業進修學院
HKU School of Professional and Continuing Education

Lecturer Biography

<https://www.linkedin.com/in/eddie-chow-a3860464/>



- Founder of InnoVi Limited which using the latest technologies to conduct digital transformation in construction, real estate, agriculture.
- Product Owner of Visionlabs App, an cloud-based NoCode AI Platform Builder that helps companies streamlining decision-making processes.
- International IIDF Expert at Internet Initiatives Development Fund (IIDF), helping Russian entrepreneurs transform the AI startup into global companies
- AI Tech Speaker, Mentor at various corporation, association including Asia Safety Management Summit (ASMS) 2024, CityU, HKBU, Microsoft, Whub, FasterCapital, PPMA
- Area : Data science, Business Automation, Agentic AI, Software Development
- 10+ years' development experience on computer vision, machine learning, system, software

Assessment

- **Continuous Assessment**

In-class discussions and computer assignments which include the usage of different applications & methodologies

- **Final Assessment**

20-mins (2-3 persons) group project presentation which integrate knowledge in this module, apply different AI methodologies and software to business applications of process automation



What is Process Automation, RPA, IPA?
Overview of the technological building blocks related to process automation
Contemporary tools for process automation
Introduction to VBA
Introduction to Python
Challenges and opportunities of business process automation
Business implications of process automation



Intended Learning Outcomes

1. describe the contemporary trends of process automation and explain the opportunities and challenges of business process automation;
2. outline key steps in process automation project management and illustrate the significance of each step;
3. apply computational tools to implement process automation;
4. discuss the development of process automation and practical cases for business.

Memory Test?

- NO Midterm / Final exams

Essays?

- NO research paper
- NO project report

Practical Work!

- 2 in-class coding exercises
- 1 Group Project



Taking
Exams

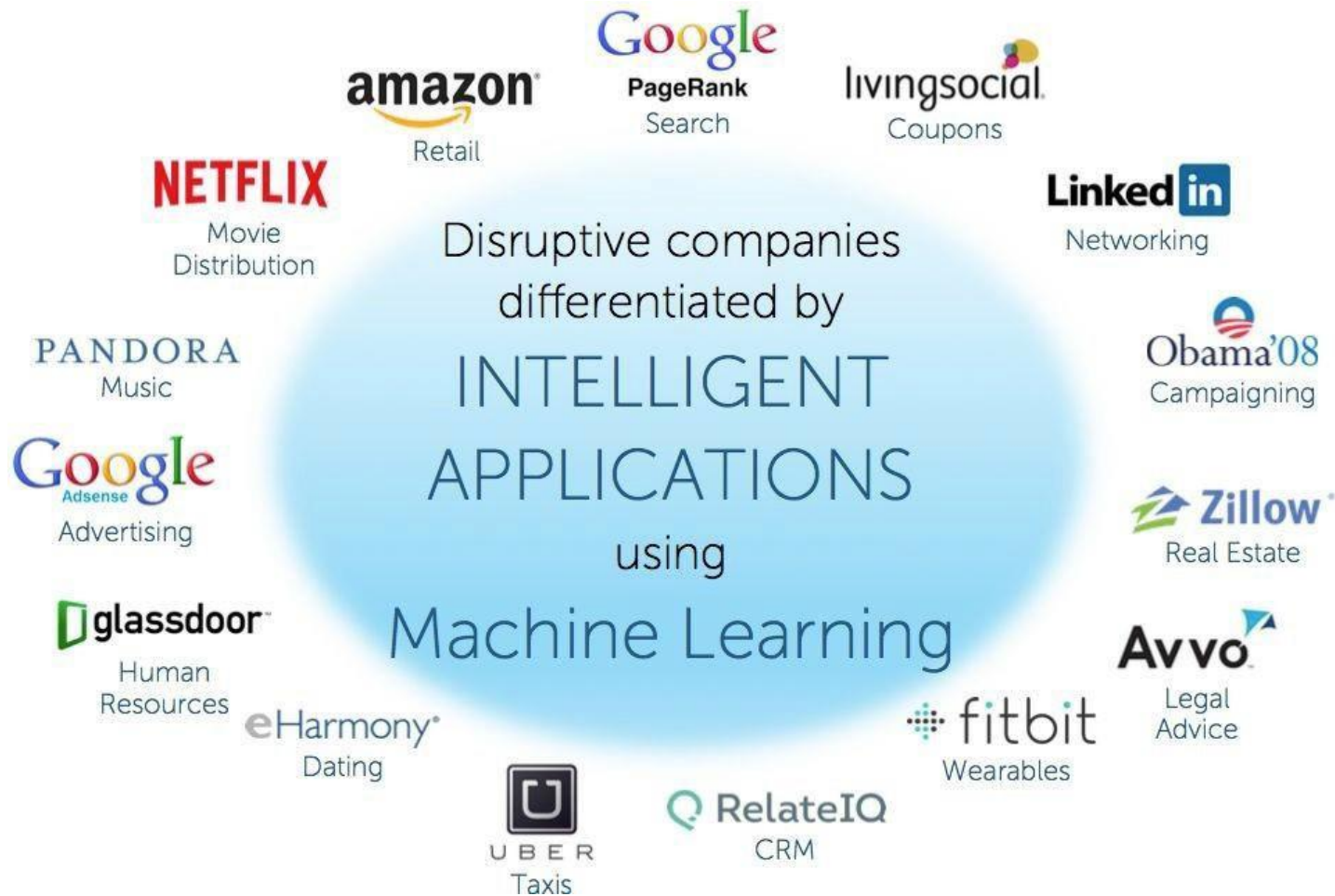
Building
Solutions

Lecture 1 - Business Process Automation with VBA and Python



Introductions: Getting To Know You
Forming Groups of 2 - 3

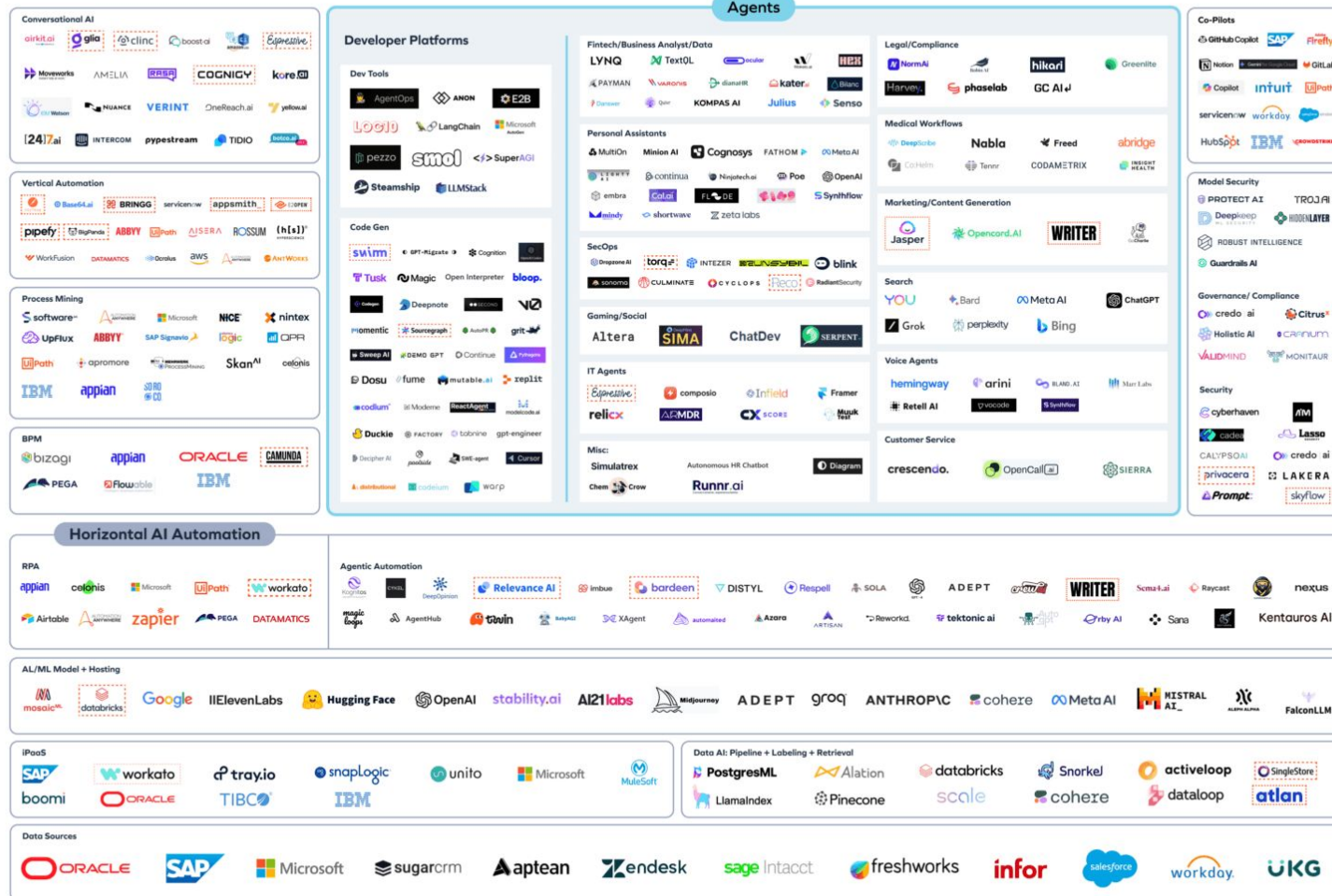
The new disruptors



The new disruptors

AI Automation Market Map

INSIGHT
PARTNERS



<https://www.insightpartners.com/ideas/ai-agents-disrupting-automation/>

What is Business Automation

From ChatGPT...

JA

What is business automation? asking for a friend :)



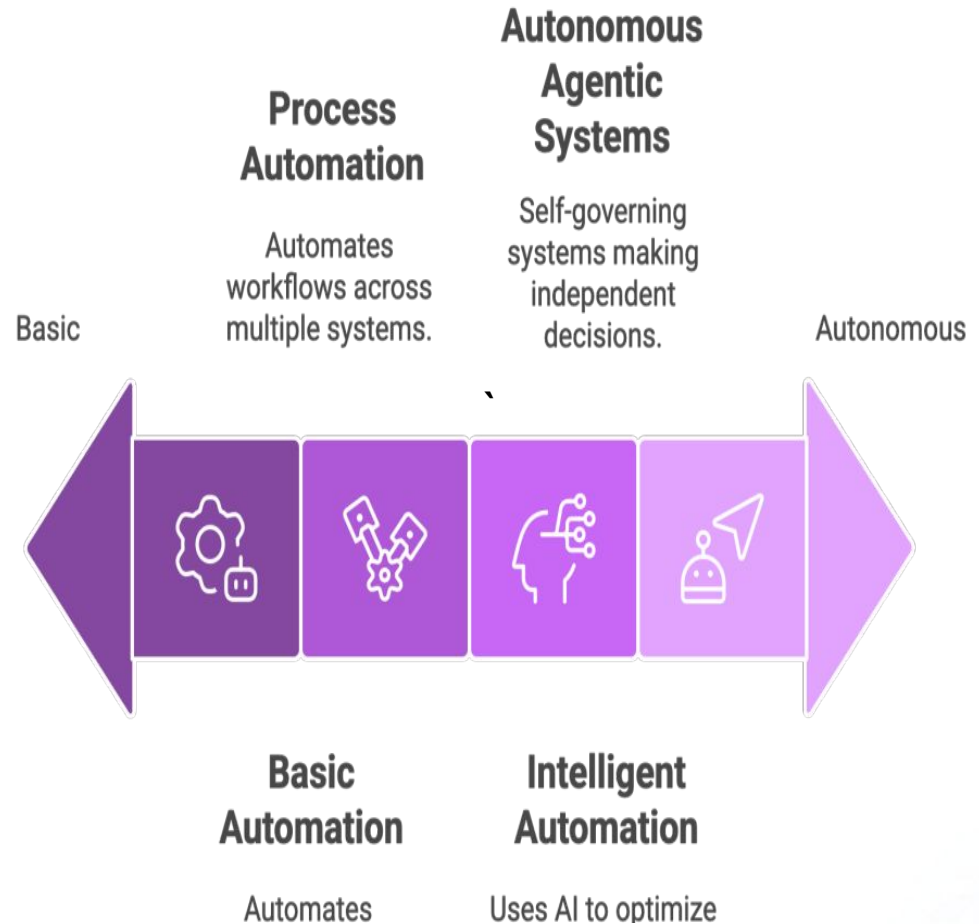
Business automation refers to the process of using technology to automate repetitive or routine tasks and processes within a business, such as data entry, invoicing, and inventory management. It can also involve the integration of various software applications and systems to streamline workflows and improve overall efficiency.

Automation should
connect people,
processes, and data
into one repeatable
workflow



Four Level of Business Automation

1. Basic Automation
2. Process Automation
3. Intelligent Process Automation
4. Autonomous Agentic Systems



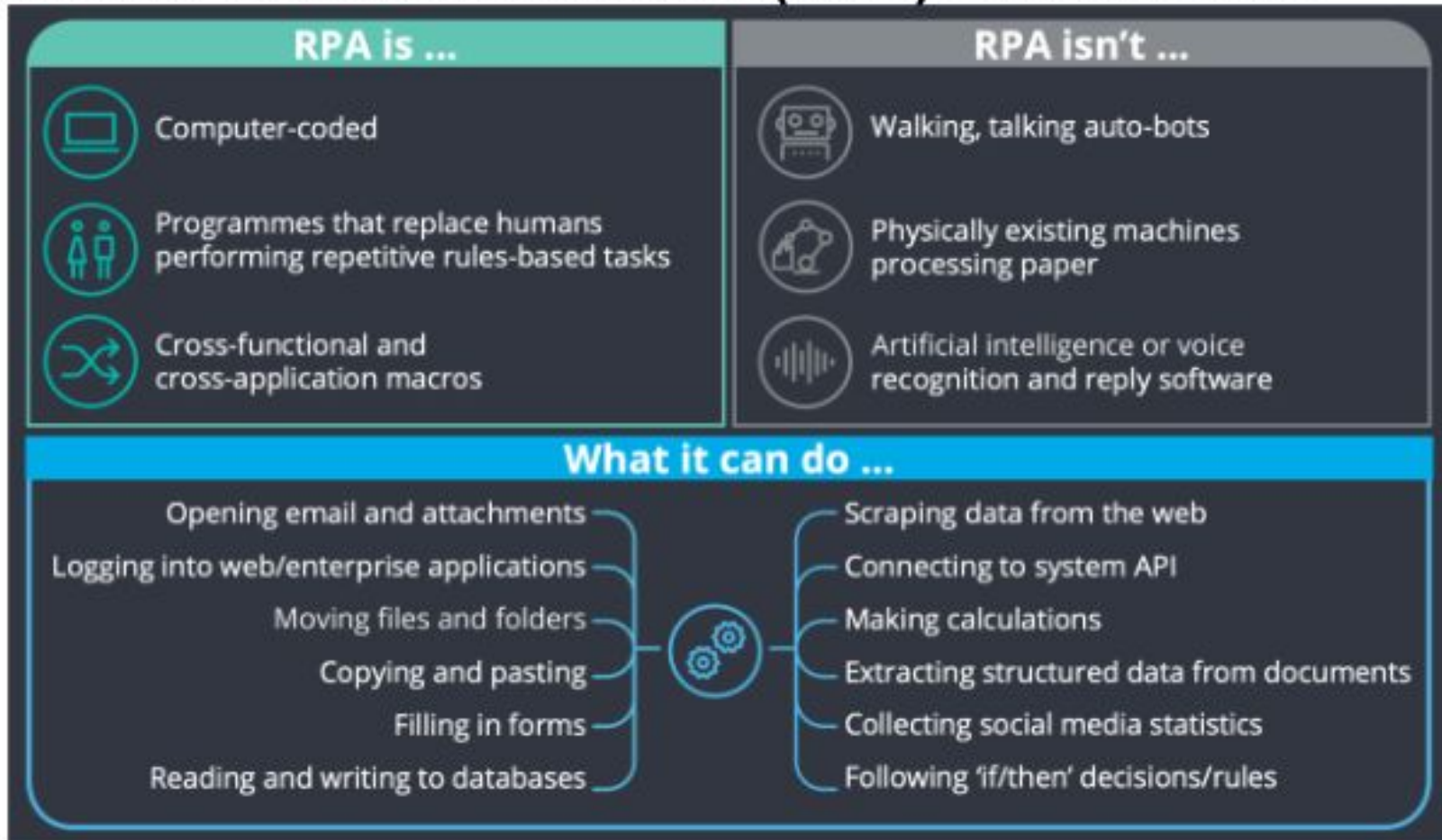
Level 1: Basic Automation

- ✓ **Task-Level Execution:** Automates simple, highly structured, individual tasks (e.g., data entry, batch file renaming, report generation).
- ✓ **Rule-Based Logic:** Operates strictly on deterministic "if-this-then-that" rules without flexibility or cognitive judgment.
- ✓ **Core Technologies:** Excel macros, basic Robotic Process Automation (RPA) bots, scheduled CRON jobs, and simple webhooks.
- ✓ **Low Complexity & Risk:** Highly predictable outputs with zero capability to handle exceptions or unformatted inputs.



Level 1: Basic Automation

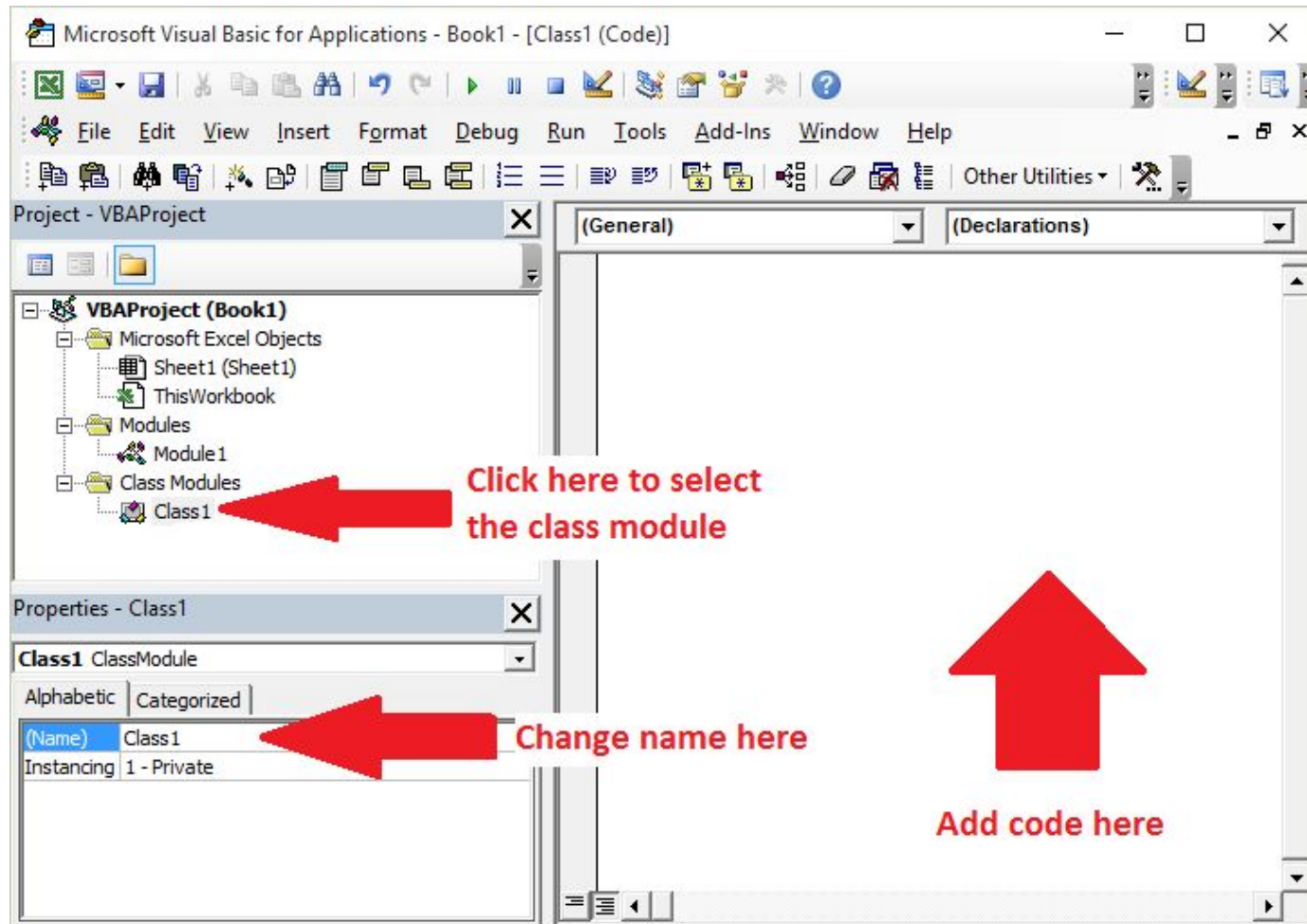
Robotic Process Automation (RPA) - Introduction



1. Deloitte. (2018). *The robots are ready. Are you?*

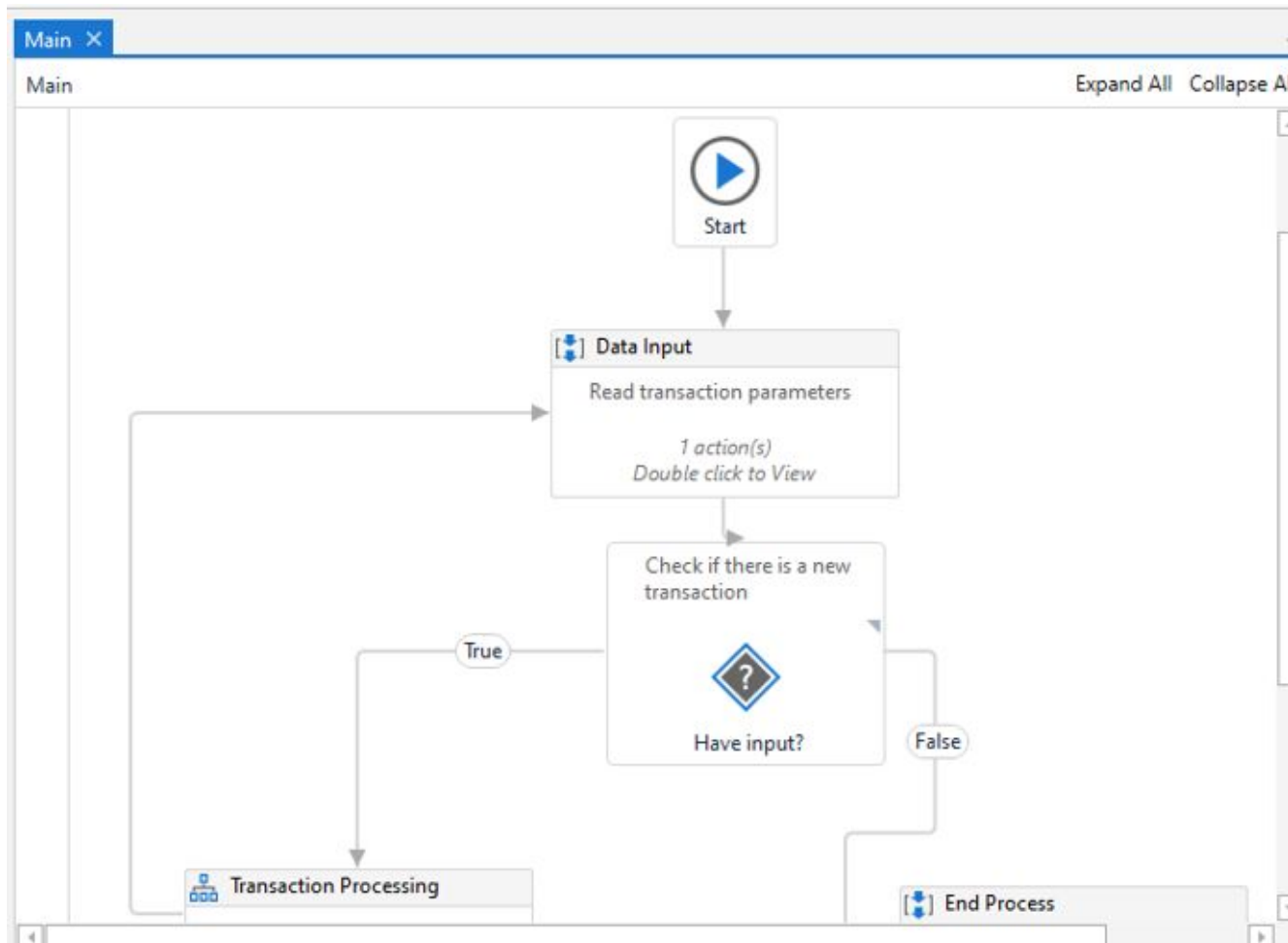
<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

Basic Automation Software Example 1 : Excel VBA



<https://learn.microsoft.com/zh-tw/office/vba/library-reference/concepts/getting-started-with-vba-in-office>

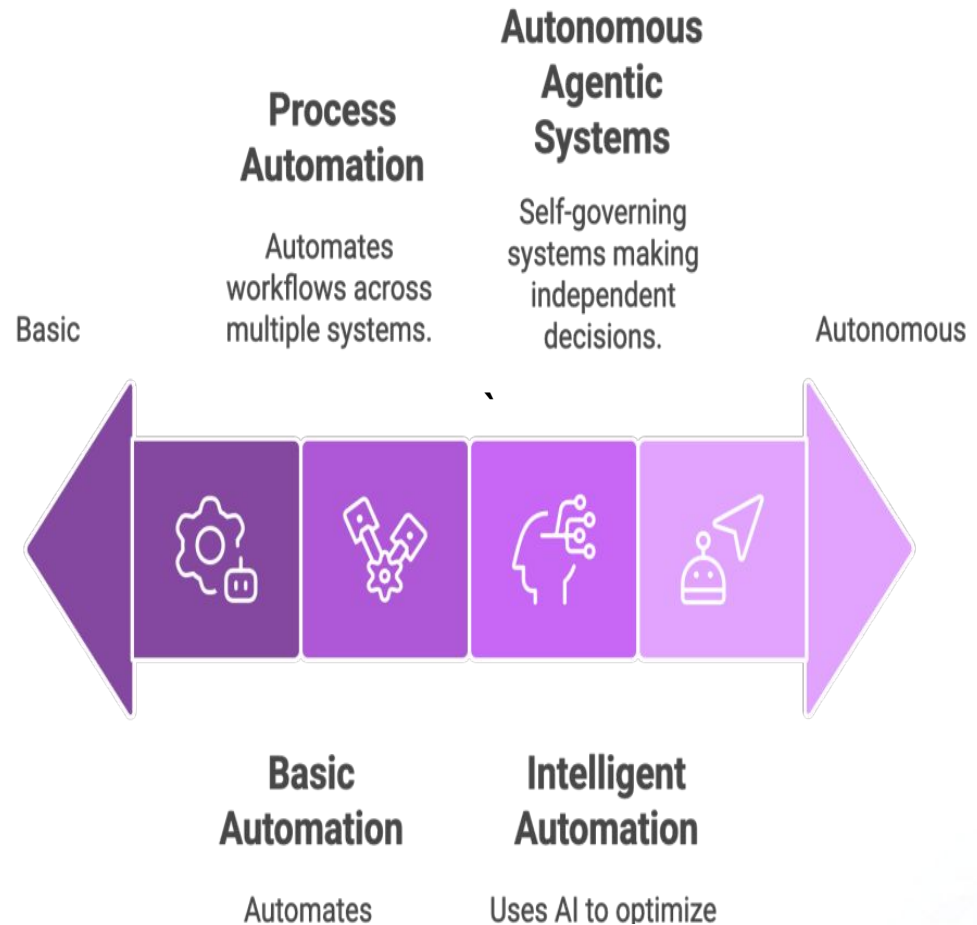
Basic Automation Software Example 2 : UiPath



<https://www.uipath.com/rpa/robotic-process-automation>

Four Level of Business Automation

1. Basic Automation
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Level 2: Process Automation

Cross-System Workflow Automation

Connects multiple discrete tasks across distinct software applications to automate end-to-end business workflows seamlessly.

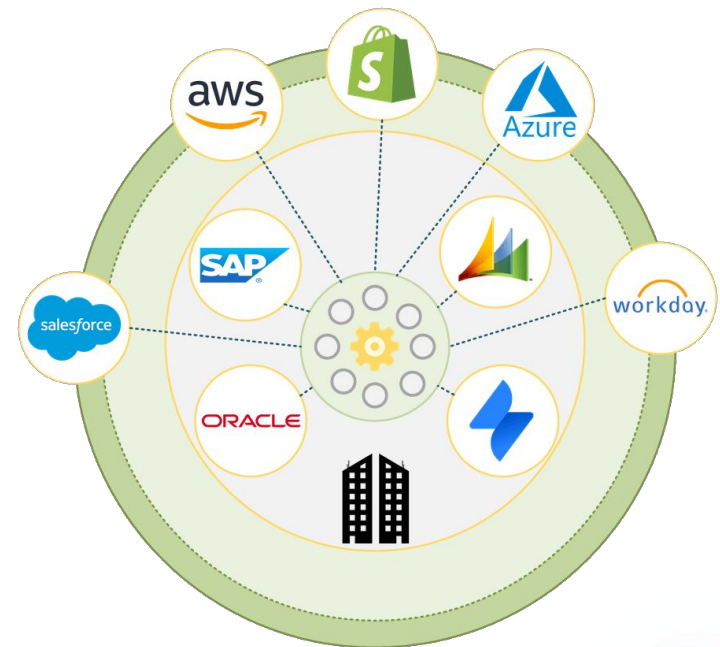
-  **BPM Systems:** Integrates ERP, CRM, and financial databases via robust API connections.
-  **Event Triggers:** Automatically passes data across functional teams upon specific operational milestones.
-  **Exception Routing:** Flags anomalies directly to human operators while keeping routine pathways automated.



Process Automation : System Integration



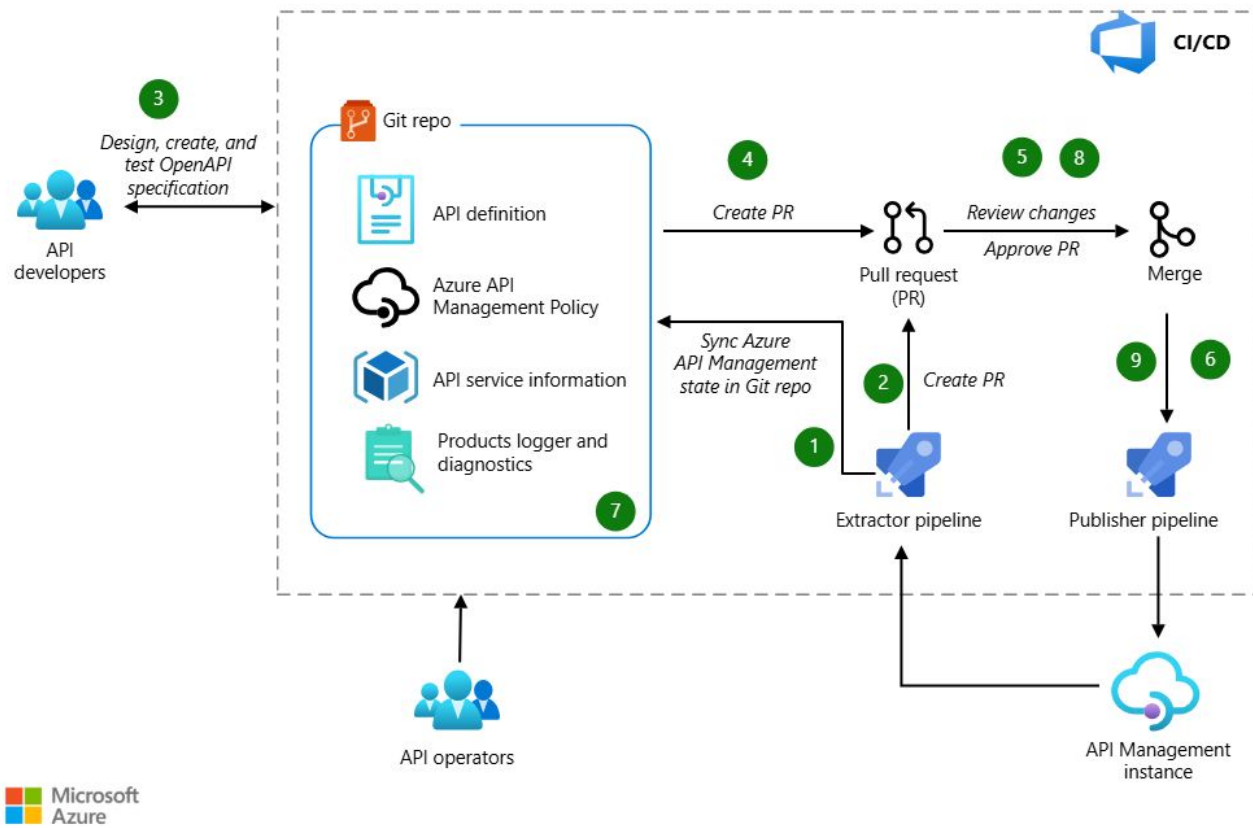
<https://fitsmallbusiness.com/crm-integration/>



<https://iwconnect.com/making-multiple-systems-work-as-one-through-integration/>

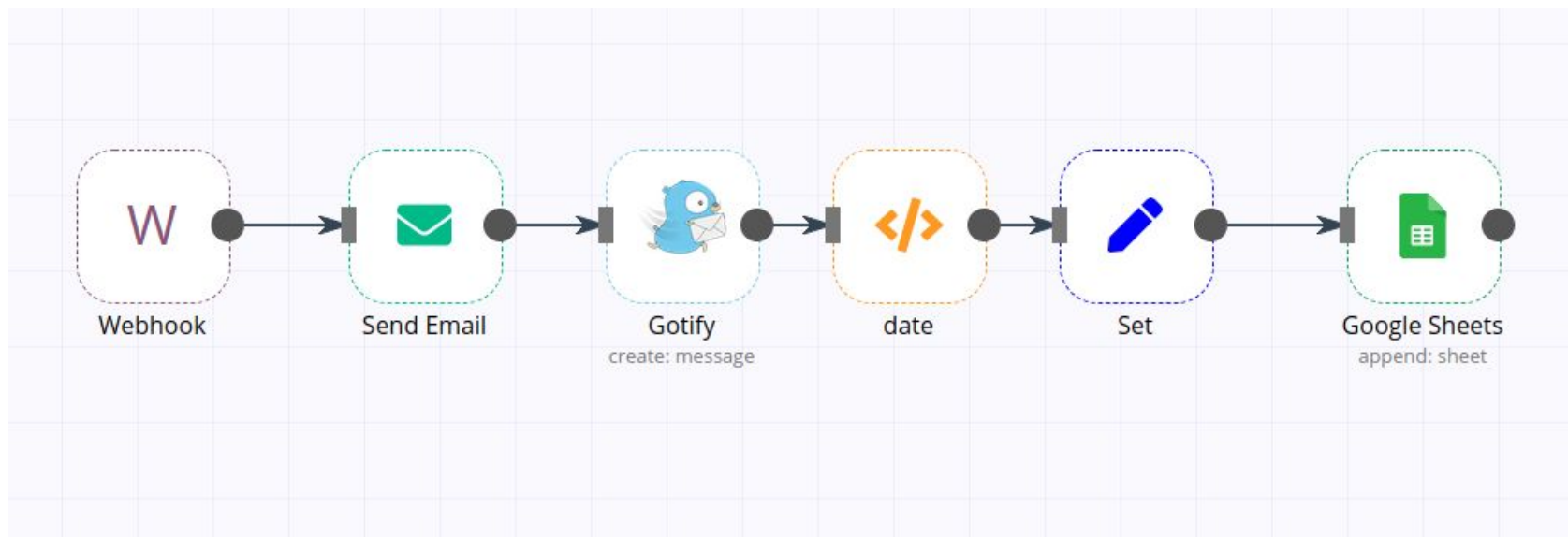
Process Automation : System Integration

<https://learn.microsoft.com/en-us/azure/architecture/example-scenario/devops/automated-api-deployments-apiops>



Process Automation Software Example 1 : n8n

Simple enough to ship in hours, sophisticated enough to scale. n8n lets you automate business processes without limits on your logic.



<https://n8n.io/features/>

Process Automation Software Example 2 : make.com

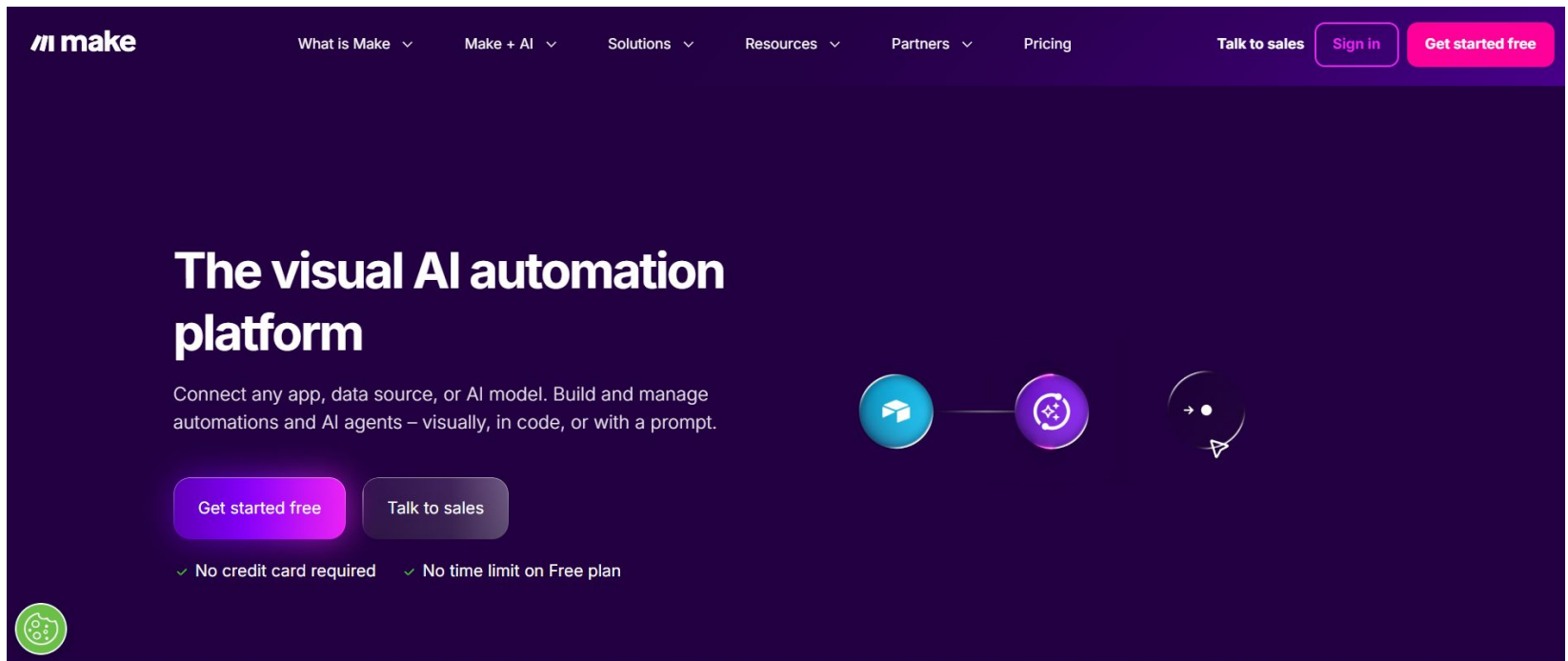
Connect any app, data source, or AI model. Build and manage automations and AI agents – visually, in code, or with a prompt.



<https://www.crmautomationtools.com/automation-tools/make-com/>

Practical Exercise - Weather Automation (Make.com)

Go to <https://www.make.com/en>



Click “Get started free” or enter <https://www.make.com/en/login>

Practical Exercise - Weather Automation (Make.com)

Sign in with Google

My Organization

Organization settings + Create a new scenario

ORGANIZATION TEAMS USERS SUBSCRIPTION PAYMENTS INSTALLED APPS VARIABLES SCENARIO PROPERTIES

Dashboard

Current plan - Free	
Price	\$0.00
Operations/mo	1,000
Unused operations	998
Billing interval	Monthly

Upgrade

Usage reset
2025年8月5日 11:36:41

Active scenarios 0/2

Operations ~2/1,000 (0% used)

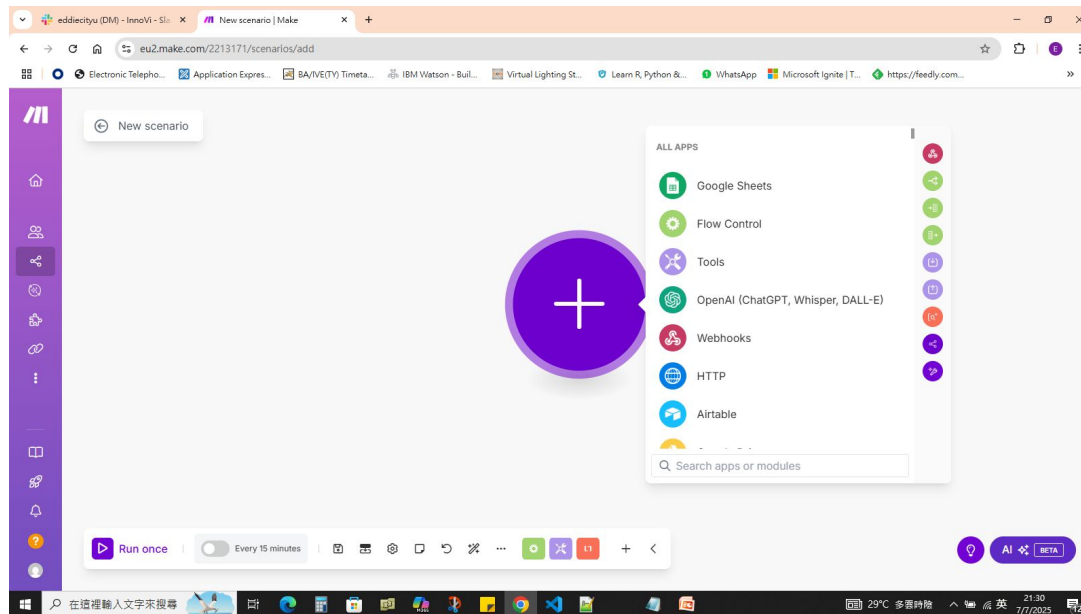
Data transfer ~592.0 B/512.0 MB (0% used)

Search for help

Click “Create a new scenario”

Practical Exercise - Weather Automation (Make.com)

An interface with module is displayed

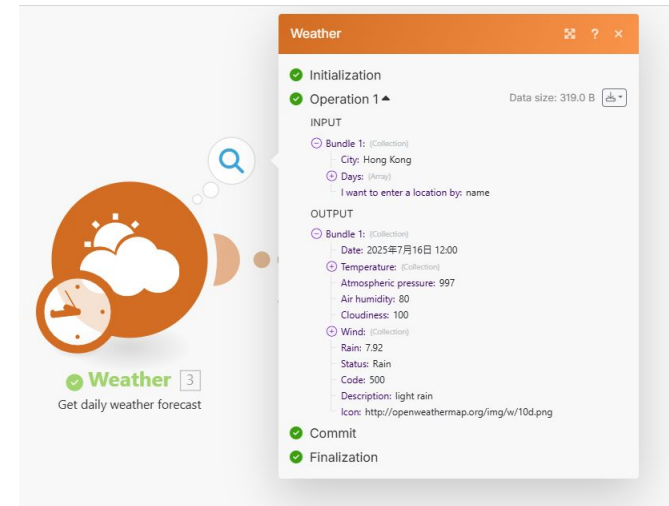
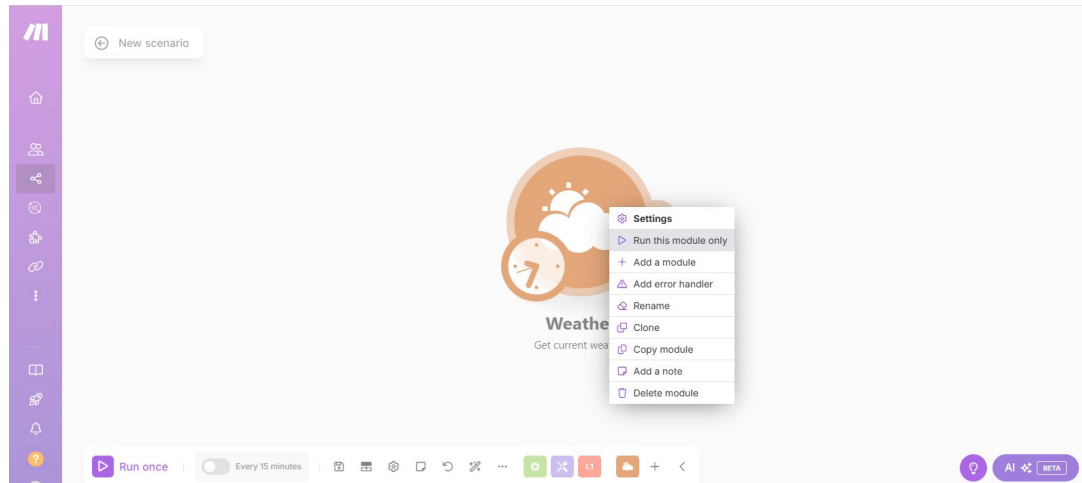


Create Module “Weather” -> Click “Get daily weather forecast”

Practical Exercise - Weather Automation (Make.com)

Enter Days “9 days from today” and City “Hong Kong”
-> Click “Save”

Right-Click and select “Run module”



You can view an output file in JSON format

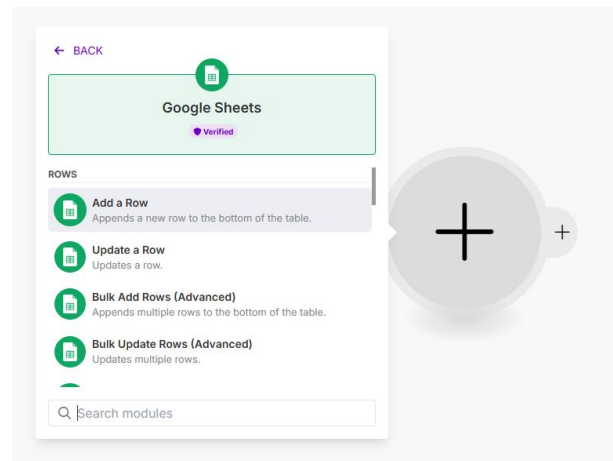
Practical Exercise - Weather Automation (Make.com)

Open drive.google.com -> create an google sheet named "Weather Data"

Create an header as follow:

Date	Temperature	Air humidity	Wind	Rain
------	-------------	--------------	------	------

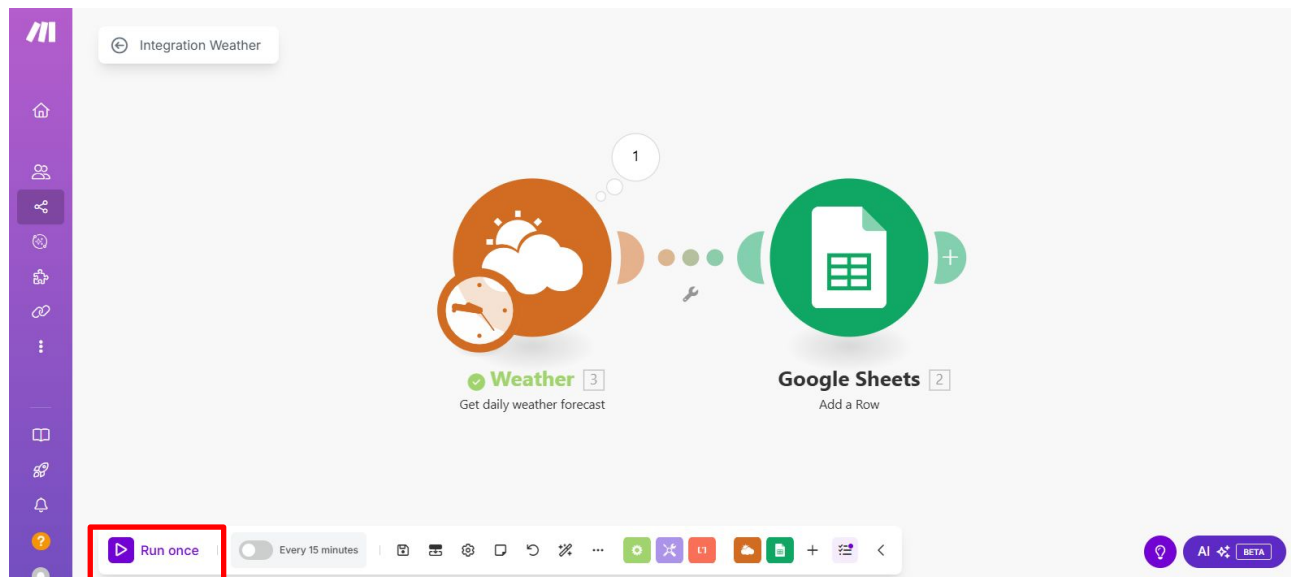
Return to make interface, in the right side of Weather icon, Click "Add another module" -> create "Google Sheets" -> Select "Add a Row"



Automation for updating weather data in google sheet

In Spreadsheet ID, click button “Click here to choose file”

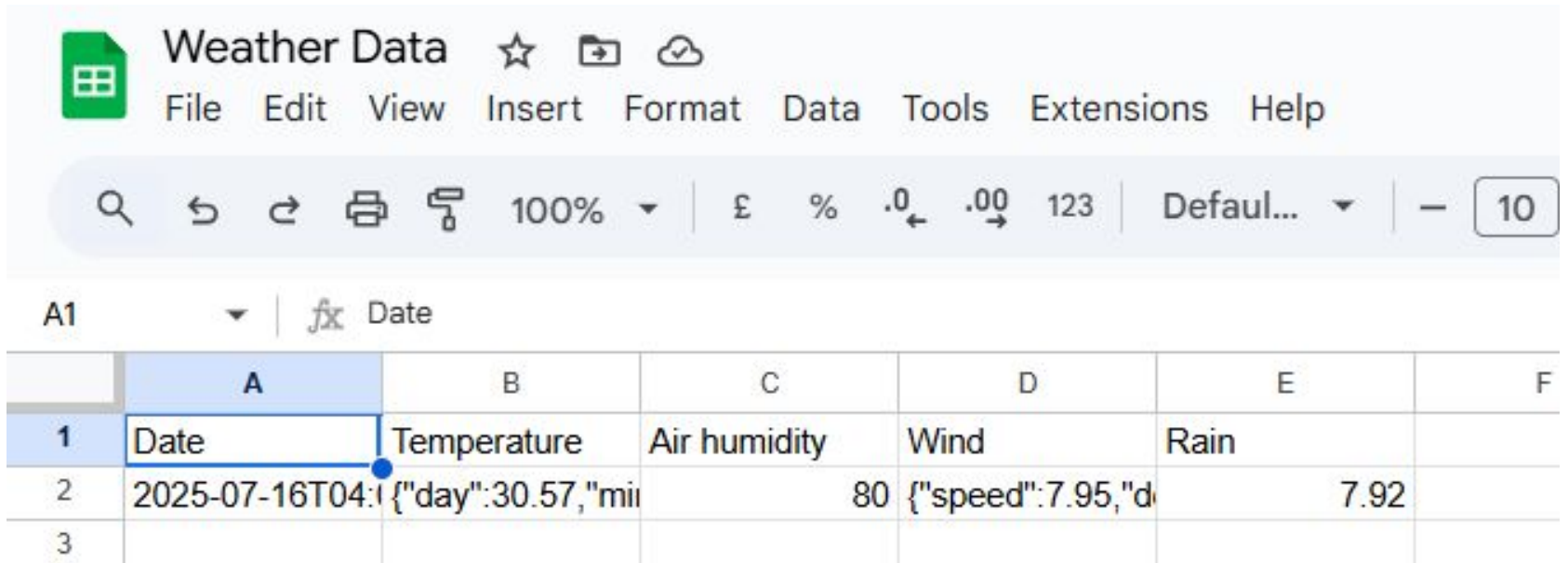
1. Select “Weather Data” -> Enter Sheet Name “Sheet1”
2. Click Date (A), select “3. Date” and other 4 columns
3. Click “Save”



In the button, click “Run once”

Automation for updating weather data in google sheet

Open the google sheet named “Weather Data”, the data is inserted in the second row as following:

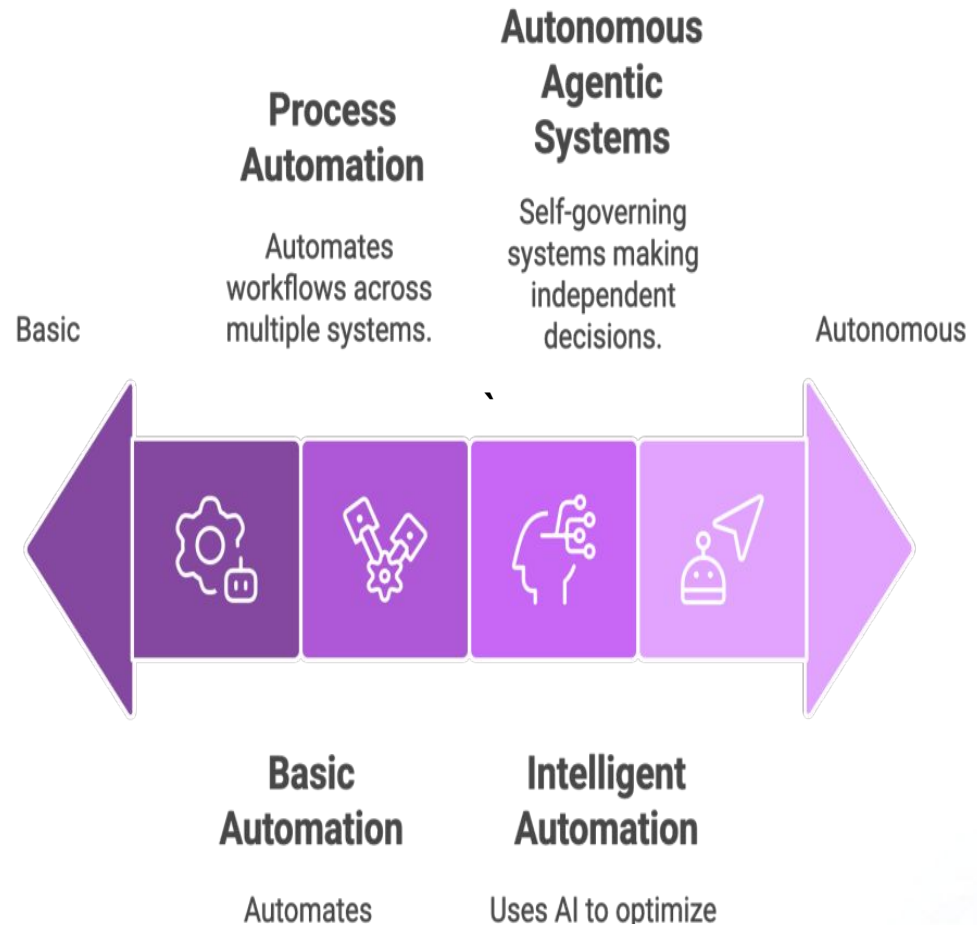


The screenshot shows a Google Sheet titled "Weather Data" with a menu bar (File, Edit, View, Insert, Format, Data, Tools, Extensions, Help) and a toolbar with various icons. The sheet contains a table with the following data:

	A	B	C	D	E	F
1	Date	Temperature	Air humidity	Wind	Rain	
2	2025-07-16T04:10	{ "day": 30.57, "mi	80	{ "speed": 7.95, "d	7.92	
3						

Four Level of Business Automation

1. Basic Automation
2. Process Automation
3. Intelligent Process Automation
4. Autonomous Agentic Systems



Level 3: Intelligent Process Automation

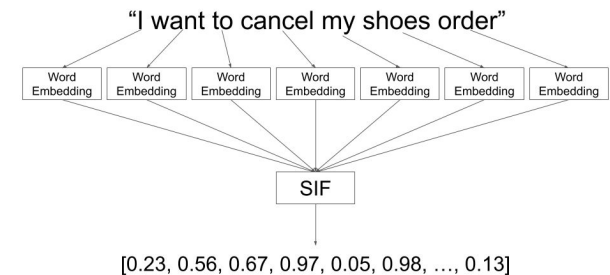
Human Workflow Orchestration with SOP(Standard Operating Procedure)

 **Cognitive Data Processing:** Extracts and categorizes unstructured information from PDFs, emails, handwriting, and voice audio.

 **Machine Learning Integration:** Uses trained ML models for predictive scoring, fraud detection, and dynamic pricing decisions.

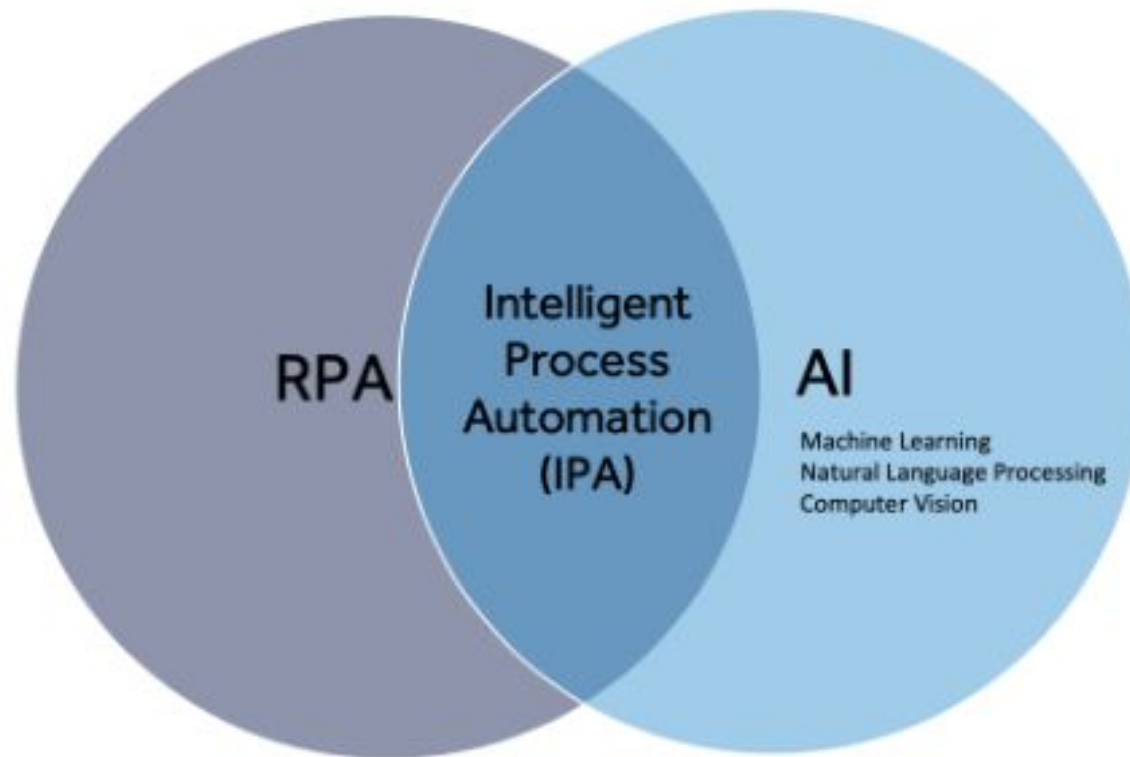
 **Natural Language Understanding:** Powers intelligent virtual assistants capable of sentiment analysis and context-aware responses.

 **Adaptive Decisioning:** Adjusts execution paths dynamically based on probability confidence thresholds.



Level 3: Intelligent Process Automation

Intelligent Process Automation (IPA)



1. Kosmopoulos, C. (2021, May 11). *What is Intelligent Process Automation (IPA)?* | Blueprint. What Is Intelligent Process Automation (IPA)? | Blueprint. <https://www.blueprintsys.com/blog/rpa/what-is-intelligent-process-automation-ipa>

Level 3: Intelligent Process Automation

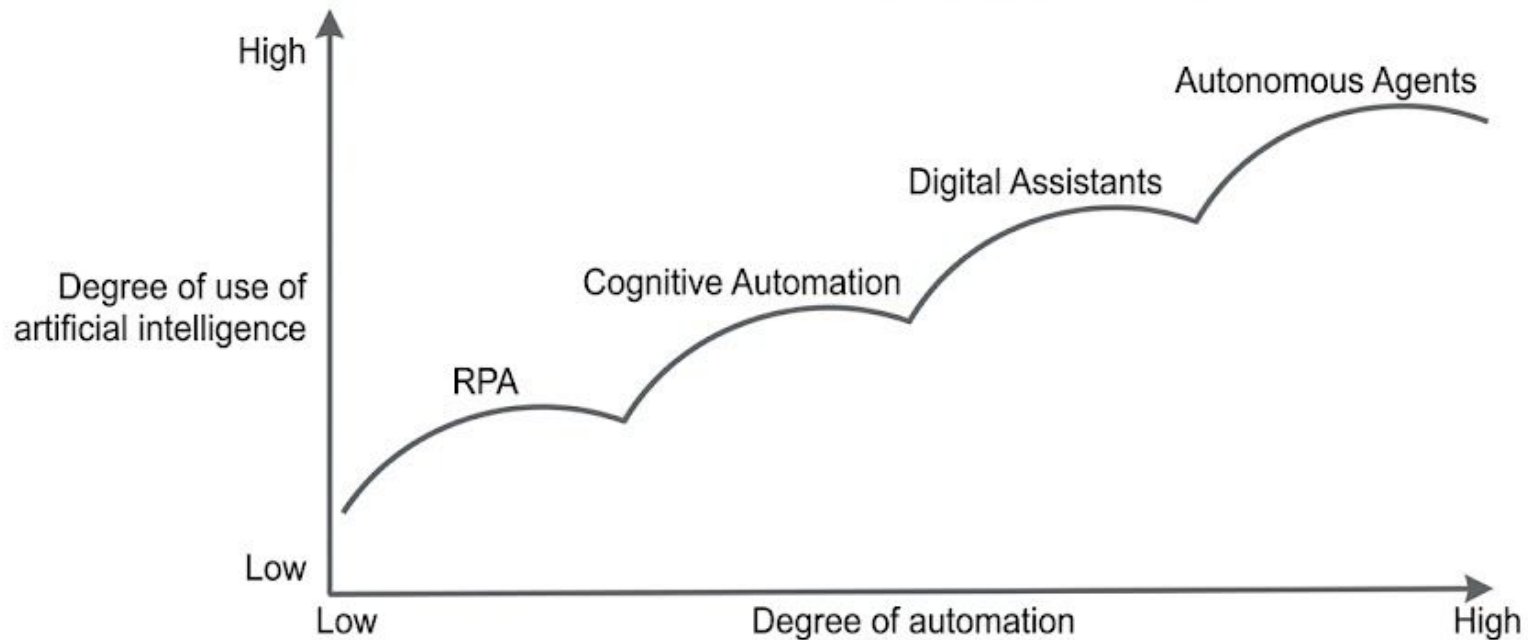


Fig. 2.2 Classification of RPA according to the degree of automation and the use of artificial intelligence (based on Ostrowicz 2018, p. 4)

1. Smeets, M., Erhard, R., & Kaußler, T. (2021, July 30). *Robotic Process Automation (RPA) in the Financial Sector: Technology - Implementation - Success for Decision Makers and Users*. Springer. <https://doi.org/10.1007/978-3-658-32974-7>

Intelligence Automation Software Example 1: Microsoft Power Automate

Link - microsoft.com/en/power-platform/products/power-automate

← Back Every week, post a message in a chat or channel. Save Test Copilot New designer

»

Recurrence

Post message in a chat or channel

Welcome to Copilot in Power Automate

Be more efficient than ever with AI assistance. Simply tell Copilot what you want to do and it will help you get started.

1 of 3 Next

Every week, post a message in a chat or channel.

✓ Connected to **Microsoft Teams**

Ask a question or describe how you want to change this flow

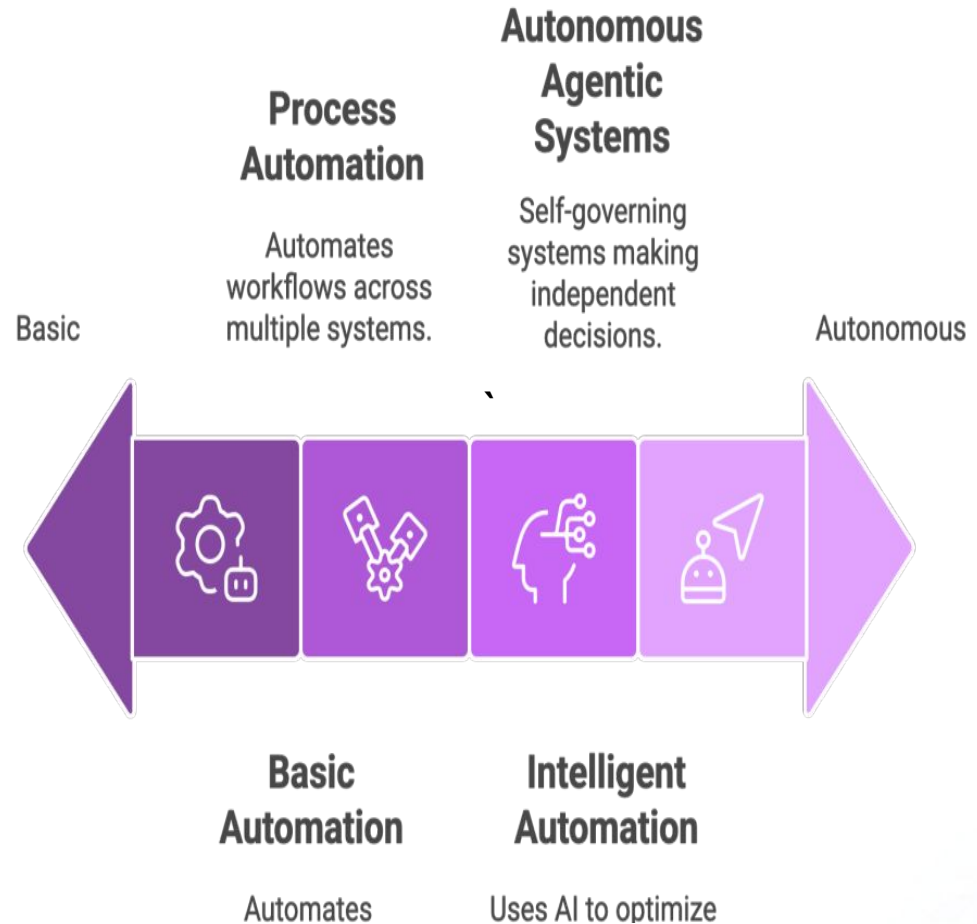
0/2000

Make sure AI-generated content is accurate and appropriate before using.

<https://learn.microsoft.com/en-us/power-automate/create-cloud-flow-using-copilot#create-a-cloud-flow-using-copilot>

Four Level of Business Automation

1. Basic Automation
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Level 4: Autonomous Agentic Systems



Goal-Oriented Reasoning

Agents receive high-level objectives (e.g., "Resolve supply bottleneck"), autonomously breaking them down into multi-step strategies.



Multi-Agent Orchestration

Specialized agents collaborate—one performs market research, another analyzes risk, while a third executes transaction tools.



Self-Correction & Adaptation

If an API fails or new data arrives, the agent reflects on error outputs, adjusts its execution loop, and retries alternative paths.



Human-in-the-Loop Governance

Operates continuously within pre-established safety guardrails, escalating high-stakes approvals while autonomously executing routine objectives.

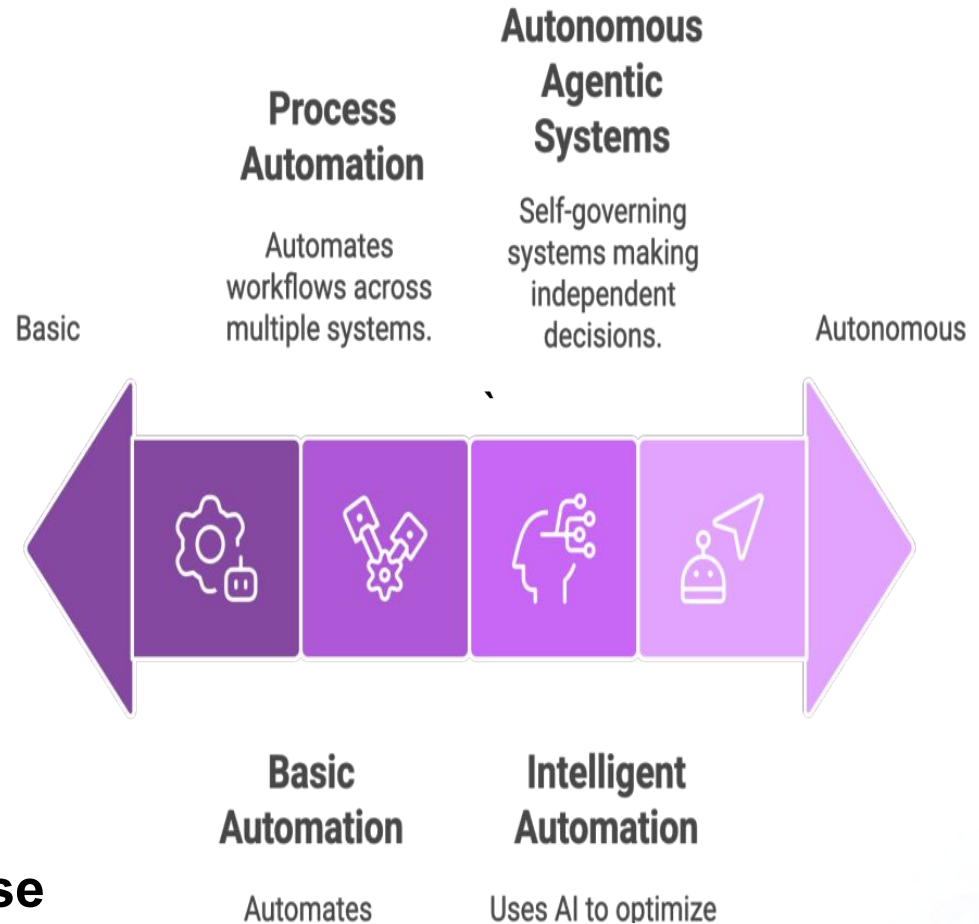
Types of Business Automation - Checkpoint

1. Daily email template to enter Market commentary
2. HR time off workflow
3. Online supermarket order processing
4. Voice-automated credit and hotline
5. Financial Process from Purchase requisition
Purchase order processing, Invoice processing, Expense claim to Audit and compliance reporting

Four Level of Business Automation

1. Basic Automation
2. Process Automation
3. Intelligent Process Automation
4. Autonomous Agentic Systems

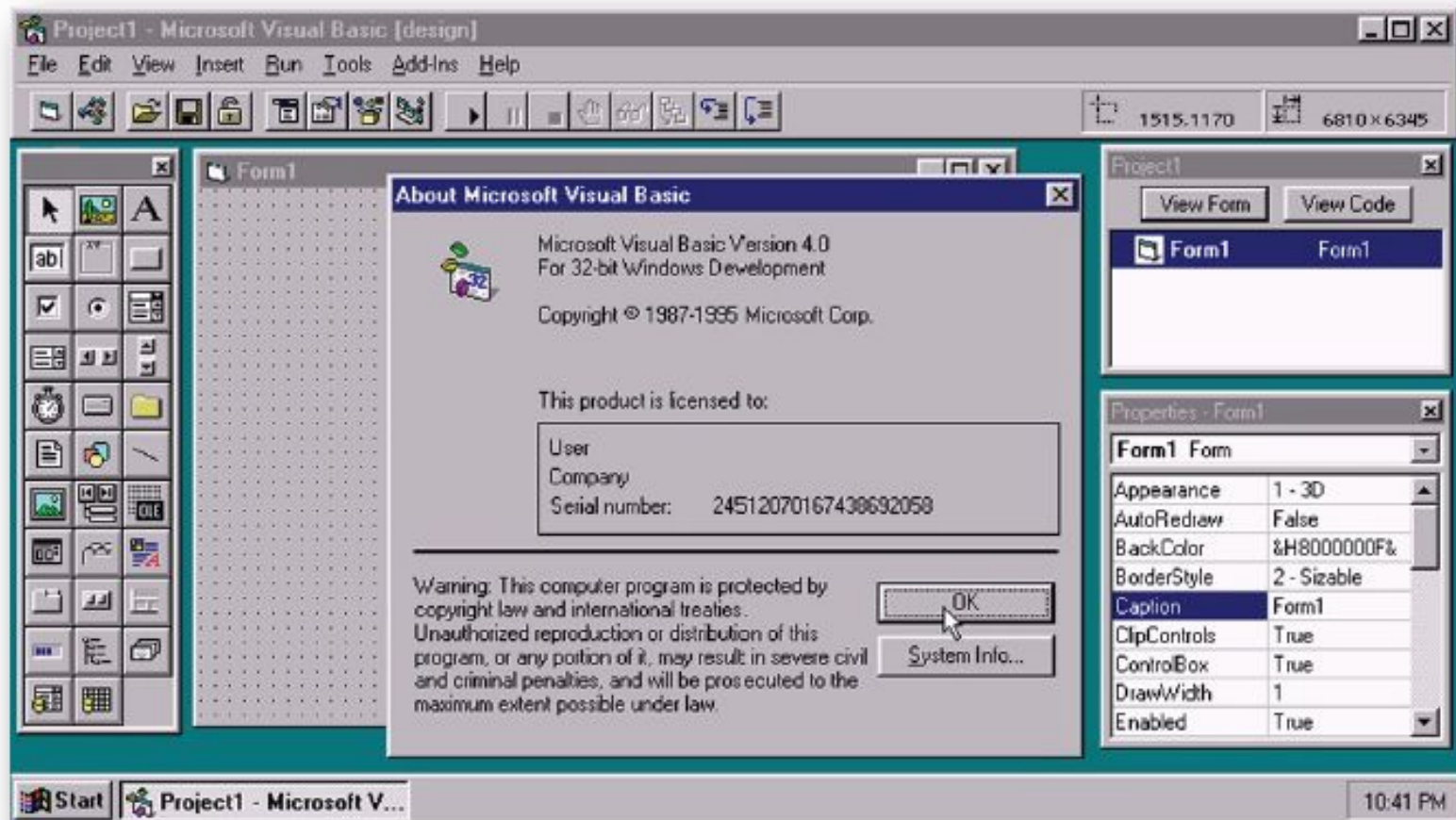
We dive into real practise



Business Process Automation with VBA

VBA - Visual Basic

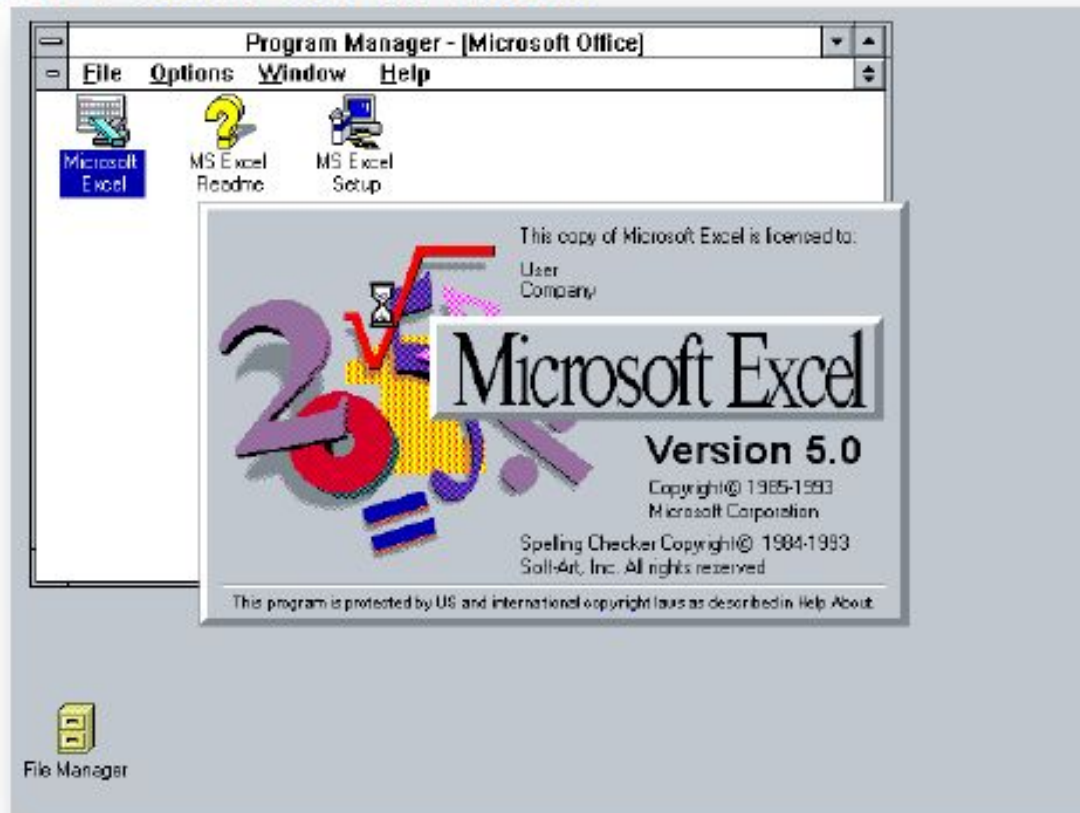
- VB was “The” Microsoft way to write programs



Business Process Automation with VBA

VBA - Visual Basic for Applications

- Introduced to Excel 5.0 in 1993

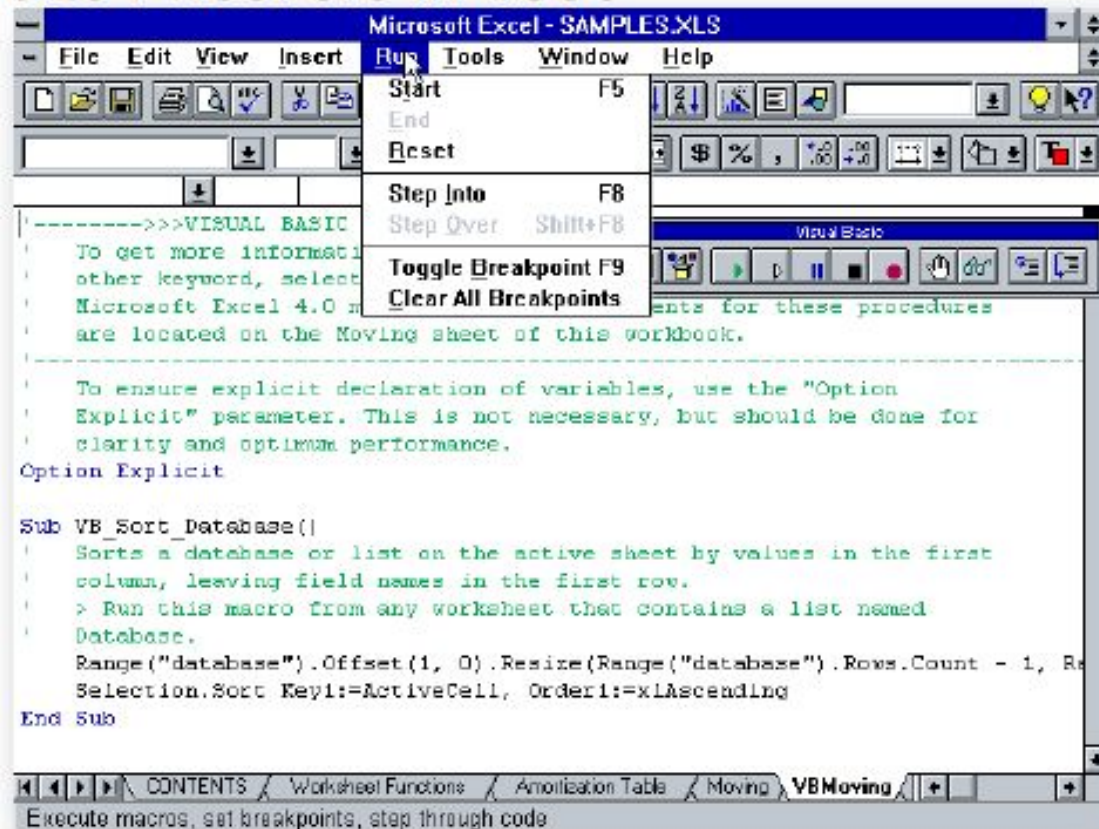


1. Microsoft Excel 5.x. WinWorld. <https://winworldpc.com/product/microsoft-excel/5x>

Business Process Automation with VBA

VBA - Visual Basic for Applications

- Introduced to Excel 5.0 in 1993

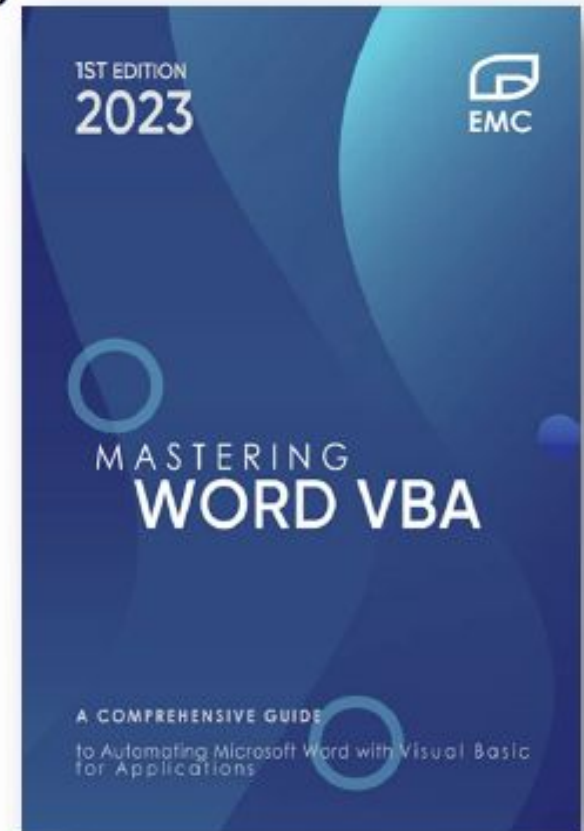


1. Microsoft Excel 5.x. WinWorld. <https://winworldpc.com/product/microsoft-excel/5x>

Business Process Automation with VBA

VBA - Visual Basic for Applications

- Gradually extended to Microsoft Office
- VB classic → ended in 2008
- VB.NET → evolution ended in 2020¹
- VBA → still supported (e.g., M365)



1. Team, N. (2020, March 11). *Visual Basic support planned for .NET 5.0*. Visual Basic Blog. <https://devblogs.microsoft.com/vbteam/visual-basic-support-planned-for-net-5-0/>

Business Process Automation with VBA

VBA – Why?

- Automation
- User interaction
- Widely used in the industry



1. O. (2022, June 8). *Getting started with VBA in Office*. Getting Started With VBA in Office | Microsoft Learn. <https://learn.microsoft.com/en-us/office/vba/library-reference/concepts/getting-started-with-vba-in-office>

Business Process Automation with VBA

VBA - Automation

- Macros → Microsoft's response to automation
- Naming:
 - Micro-instruction vs Macro-instruction
 - 1 key to perform series of commands / actions

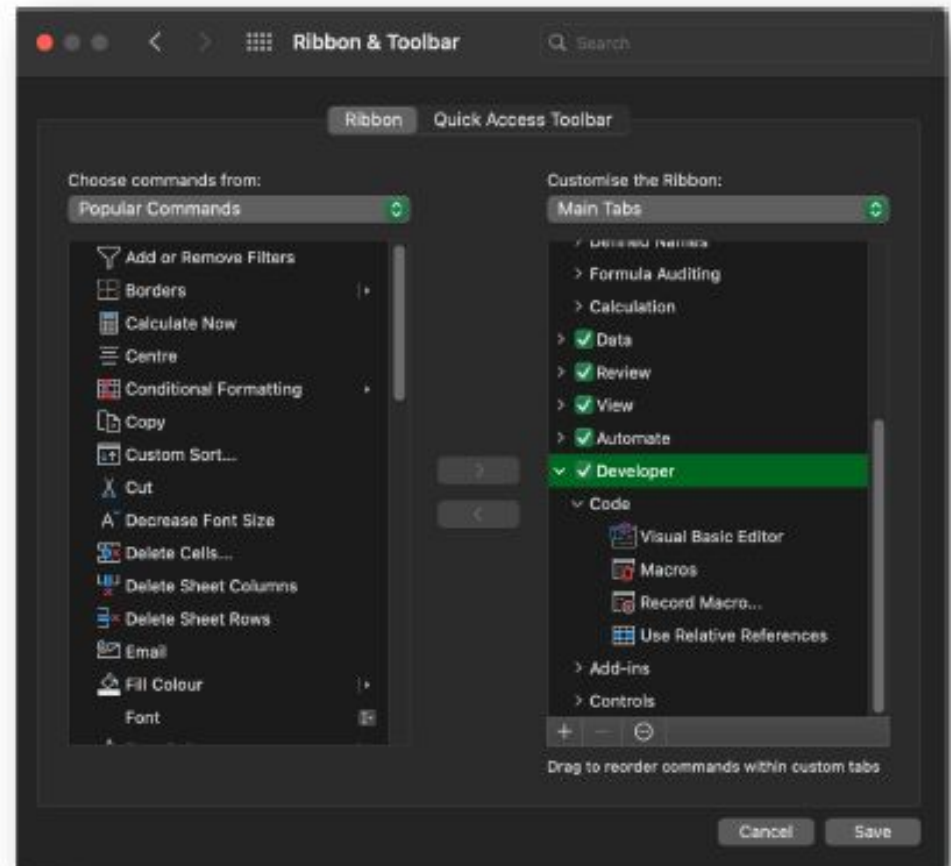


1. Why is Excel VBA called "Macros"? Why Is Excel VBA Called "Macros"? <https://www.excelforum.com/the-water-cooler/791161-why-is-excel-vba-called-macros.html>

Business Process Automation with VBA

Macros – Where?

- “Developer” tab
 - Hidden by default
 - File
 - ➔ Options
 - ➔ Customize Ribbon
- Turn on “Developer”

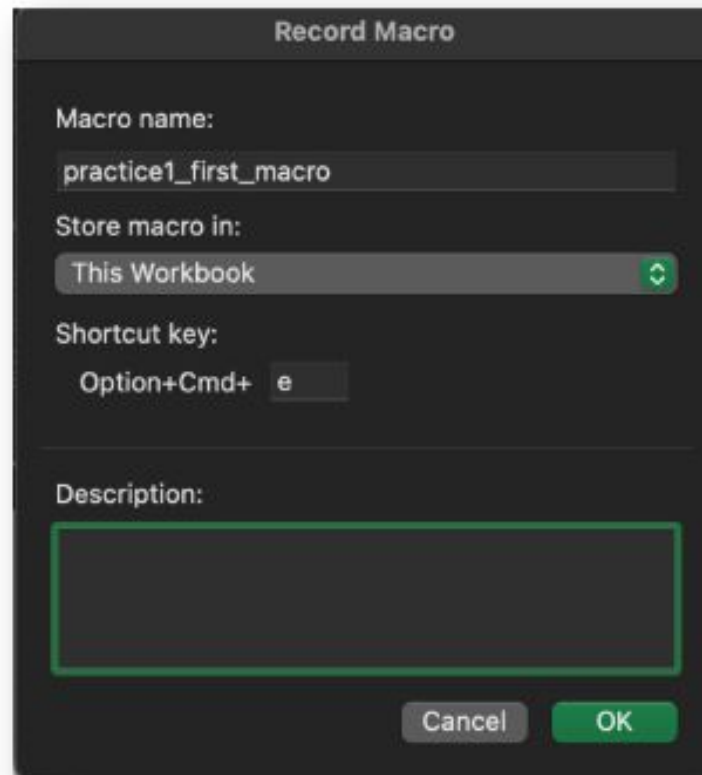


1. *Show the Developer tab - Microsoft Support.* Show the Developer Tab - Microsoft Support.
<https://support.microsoft.com/en-us/office/show-the-developer-tab-e1192344-5e56-4d45-931b-e5fd9bea2d45>

Business Process Automation with VBA

Practice 1 - Your first macro

- Type in a name (use “_” instead of spaces)
- Set a shortcut key
- Click OK



Business Process Automation with VBA

Practice 1 - Your first macro

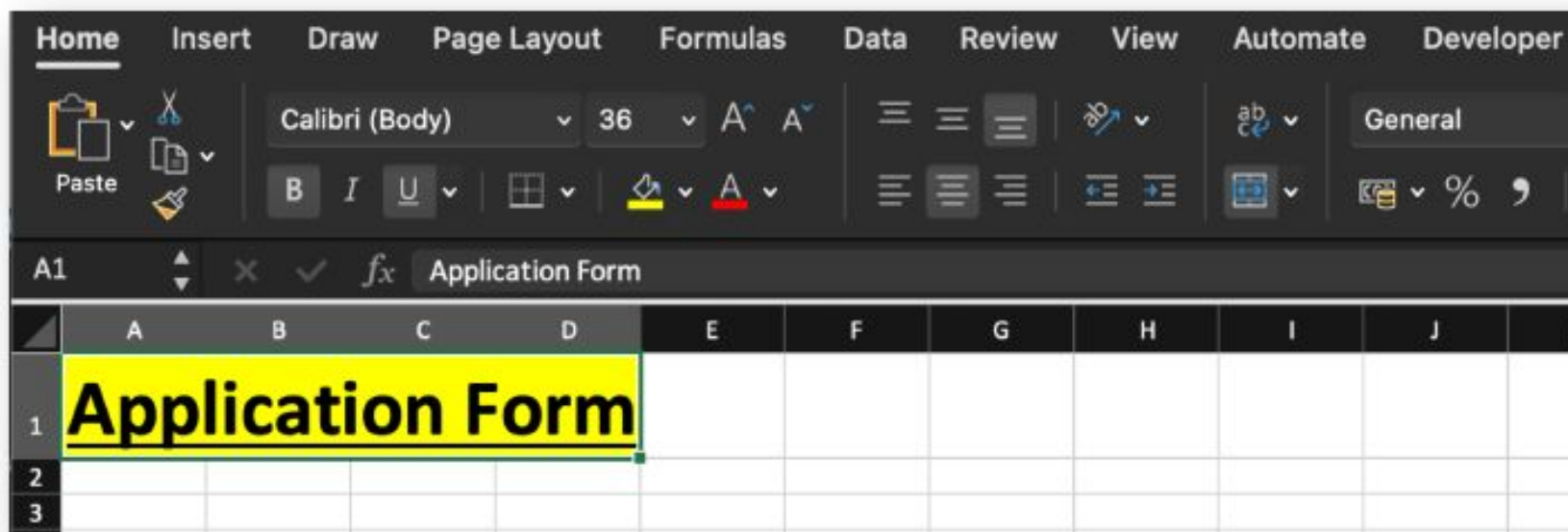
- The recording has started
- Add a new sheet with the “+” at the bottom-left



Business Process Automation with VBA

Practice 1 - Your first macro

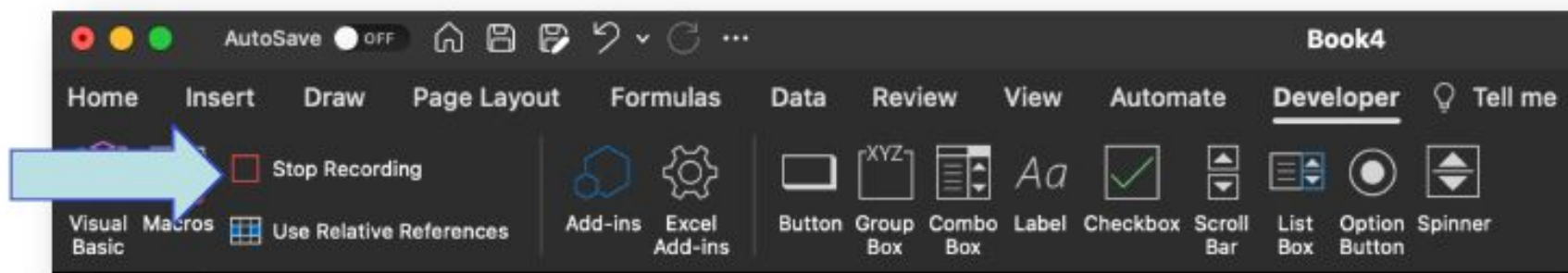
- Go back to “Home” tab
- Select Cells A1 to D1 → Merge
- Enter “Application Form” at the top-left
- Style it: e.g., Font size + Bold + Underline



Business Process Automation with VBA

Practice 1 - Your first macro

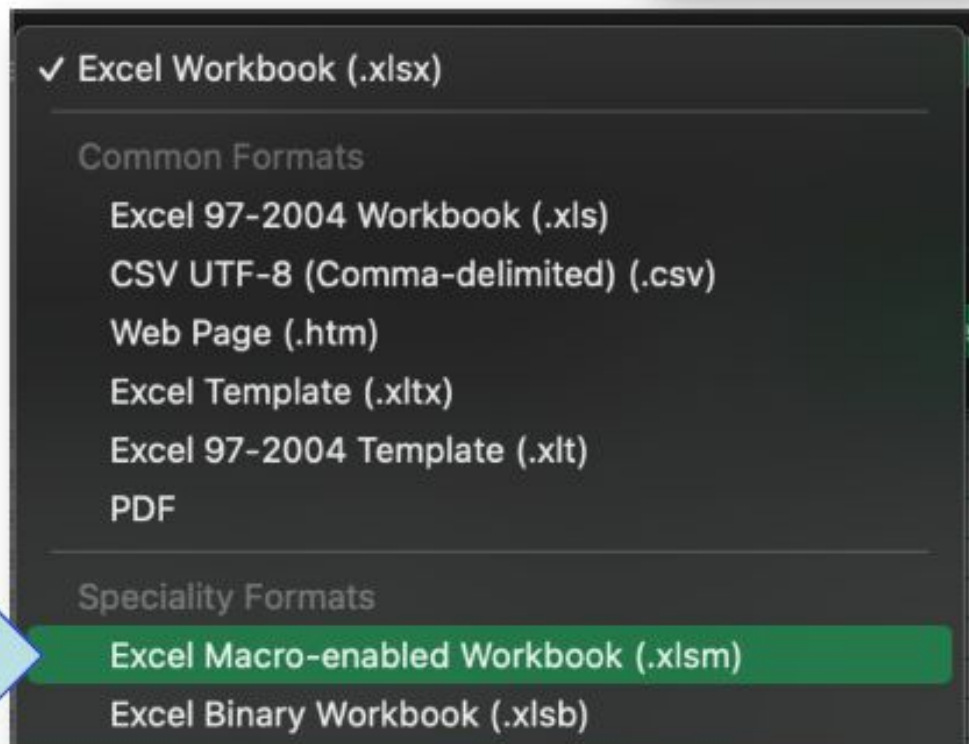
- Go back to “Developer” tab
- Hit “Stop Recording”



Business Process Automation with VBA

Practice 1 - Your first macro

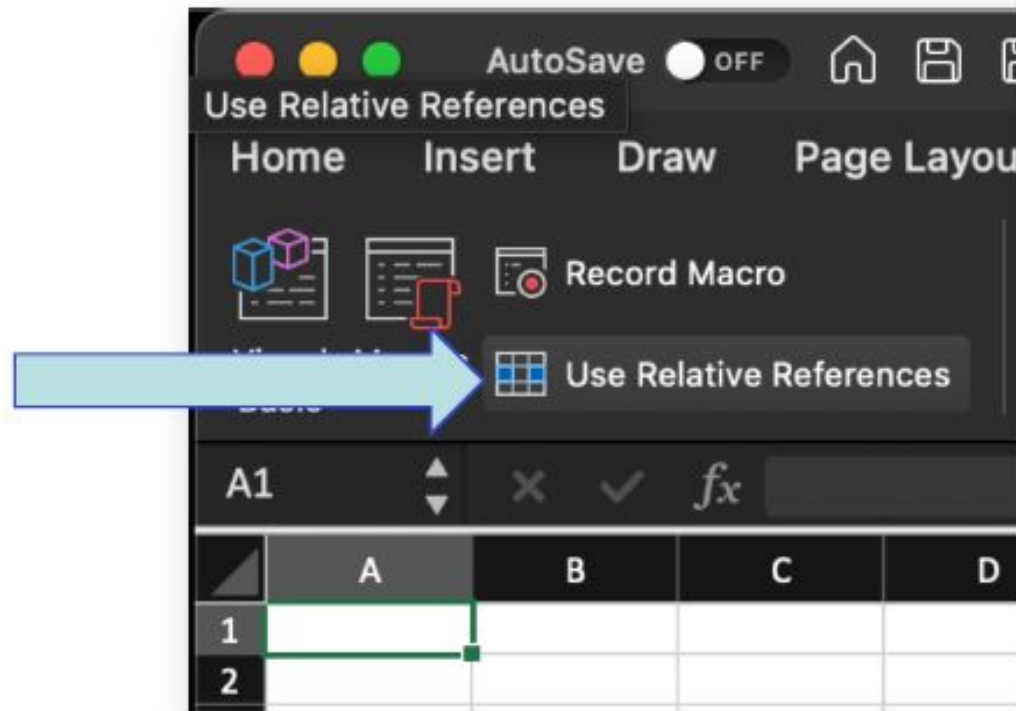
- Save as .xlsm to preserve your macros!



Business Process Automation with VBA

Relative Reference

- Record and execute based on “current selection”
- Click BEFORE “Record Macro”



Practise 2 - Sales Transaction Raw Data Preparation

1. Open a new excel workbook named “Sales_Transaction_012026”
2. Enter heading content “ID, Brand, Product Name, Unit Prices, Quantity, Total Prices”
3. Type following content into rows 1 to 5.

ID	Brand	Product Name	Unit Prices	Quantity	Total Prices
SM_001	Samsung	samsung s25	4000	2	
I_001	Iphone	iPhone 17 Pro	8000	1	
SM_002	Samsung	samsung s26 Pro	3000	1	
I_002	Iphone	iphone 17	4000	2	

4. Start Recording Macro with name “SalesRecord_Macro” to duplicate record with next month record.
5. Rename sheet name and add additional record.

Practise 3 - Sales Transaction Data Cleaning

1. Add 1 more columns “Purchase Date” with value

ID	Purchase Date	Brand	Product Name	Unit Prices	Quantity	Total Prices
SM_001	1012026	Samsung	samsung s25	4000	2	
I_001	3012026	Iphone	iPhone 17 Pro	8000	1	
SM_002	3012026	Samsung	samsung s26 Pro	3000	1	
I_002	5012026	Iphone	iphone 17	4000	2	

2. Conduct Macro with name “SalesRecord_Macro” to create new worksheet with new name “Cleaned_Sales_Transaction_012026”.
3. Conduct data cleaning with standard format.
E.g.
 - a. Heading (id,purchase_date,product_name, ...)
 - b. Date format(2026-05-01)
 - c. Currency format (HKD\$10,000.00)

Data Type



Common Data Types in Excel - Overview

Seven key data types commonly used in Excel cells



Text

Alphanumeric strings,
labels, names



Number

General numeric
values, decimals



Date & Time

Dates, times, datetime
values



Currency

Monetary values with
currency symbol



Percentage

% values shown as 0-100%,
formatted with %



Boolean

TRUE / FALSE
logical values



Error types

#N/A, #VALUE!, #DIV/0!, etc.

Tip: Excel automatically detects data types, but you can set them via Home → **Number Format**

Data Type Details

Understanding Number Formats — Key Data Types, Examples, Formats, and Use Cases for Excel

Data Type	Example	Format	Use Case
 General	12345	No specific format	Default; plain text/numbers, no formatting applied
 Number	1,234.50	1,234.50 (2 decimals)	Standard numeric values; measurements, quantities, counts
 Currency	\$1,234.50	\$#,##0.00	Monetary values; pricing, sales revenue, costs
 Accounting	1 \$ 1,234.50	\$ 1,234.50 (aligned)	Financial statements; aligned currency for clean reports
 Percentage	25.50%	25.50% (percentage format)	Rates, growth, discounts, proportions as percent
 Fraction	1 1/4	# ?/? (fraction format)	Measurements; cooking, engineering, construction dims
 Scientific	1.23E+04	1.23E+04 (scientific notation)	Very large/small values; scientific, engineering data

 **Tip:** Press CTRL + 1 to open the Format Cells dialog and customize number formats under the Number category.

Business Process Automation with VBA

VBA - Why not just use macros?

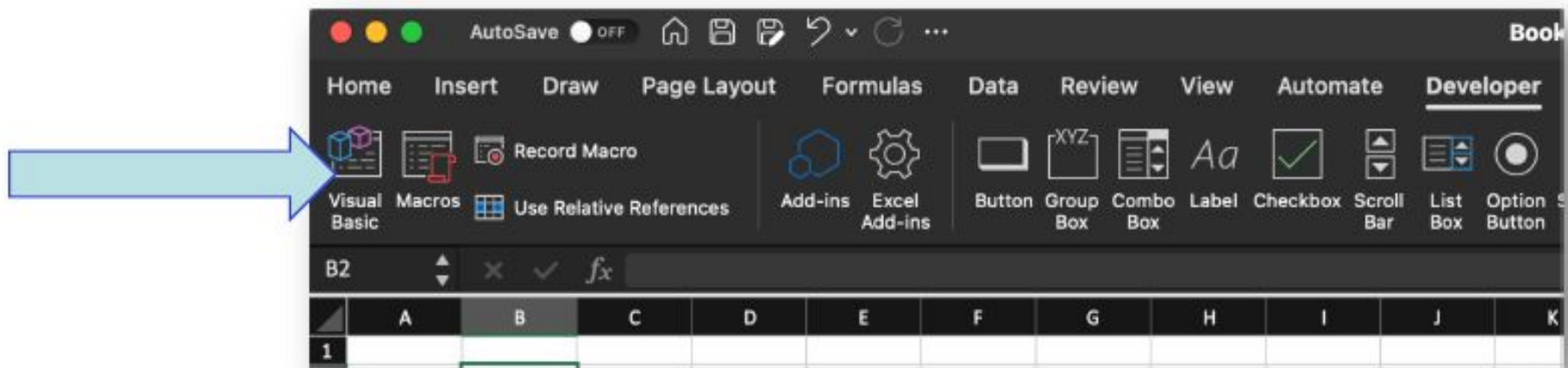
- Macros → great for repetition
- Need → More flexibility and interactivity
- Solution → Dig into the codes



Business Process Automation with VBA

Entering Visual Basic Mode

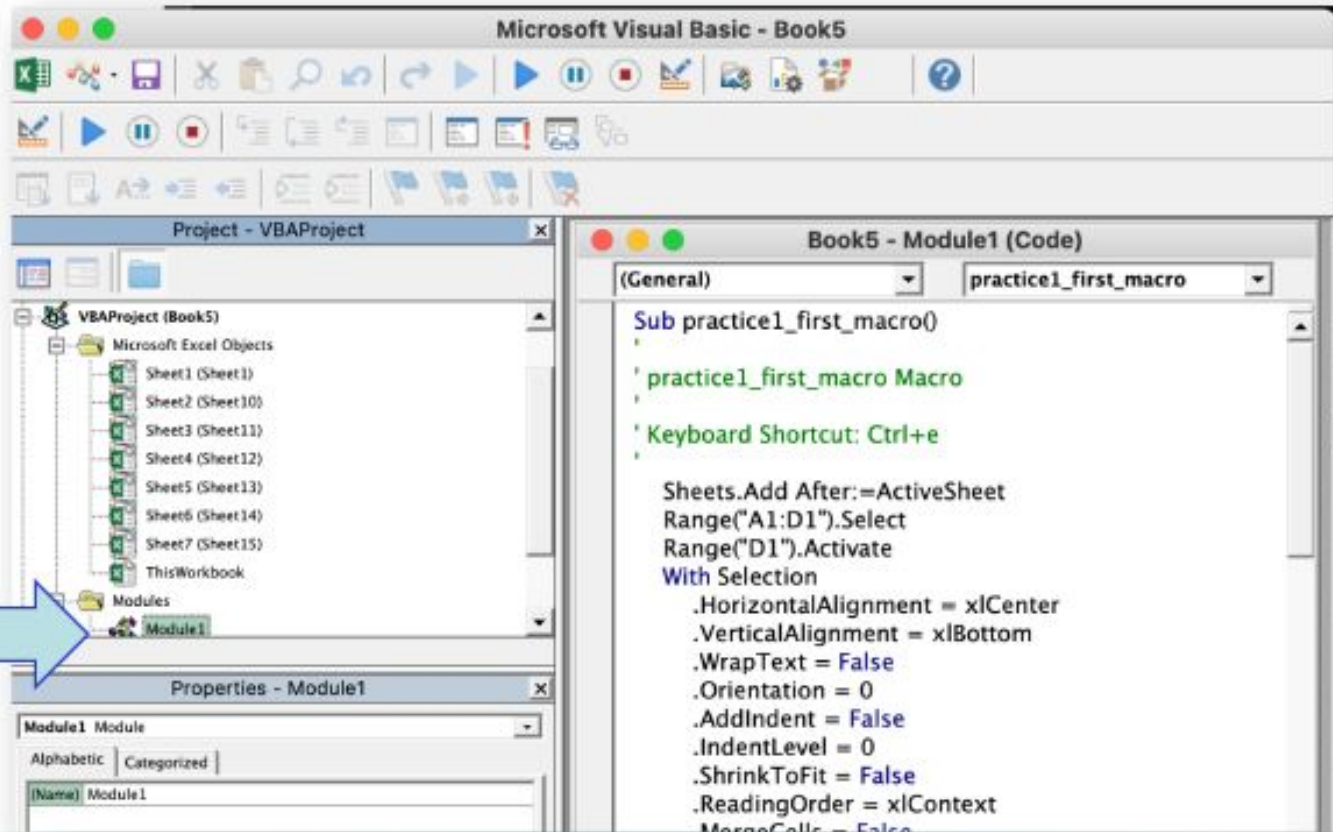
- Developer tab → Visual Basic



Business Process Automation with VBA

Entering Visual Basic Mode

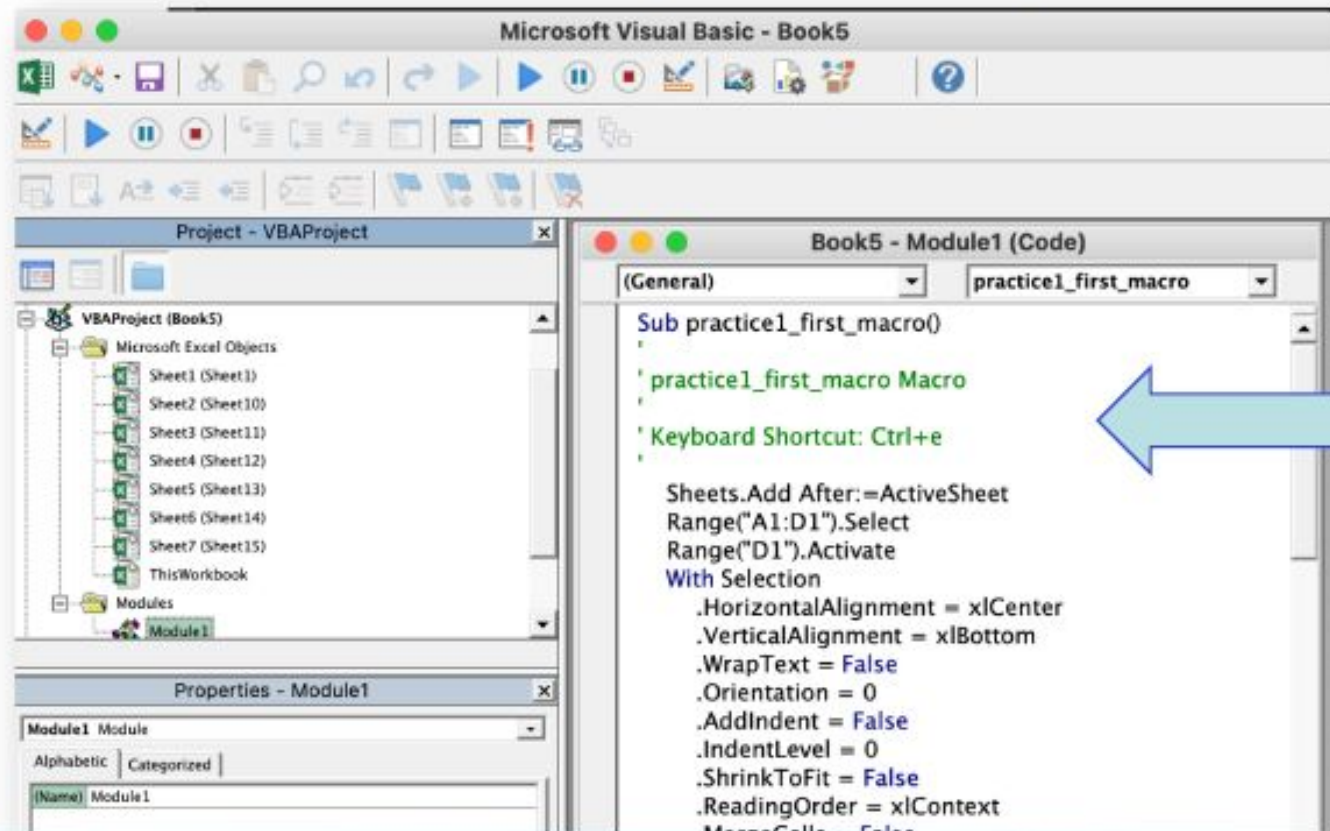
- Integrated Development Environment (IDE)
- Modules = where codes are usually stored



Business Process Automation with VBA

Entering Visual Basic Mode

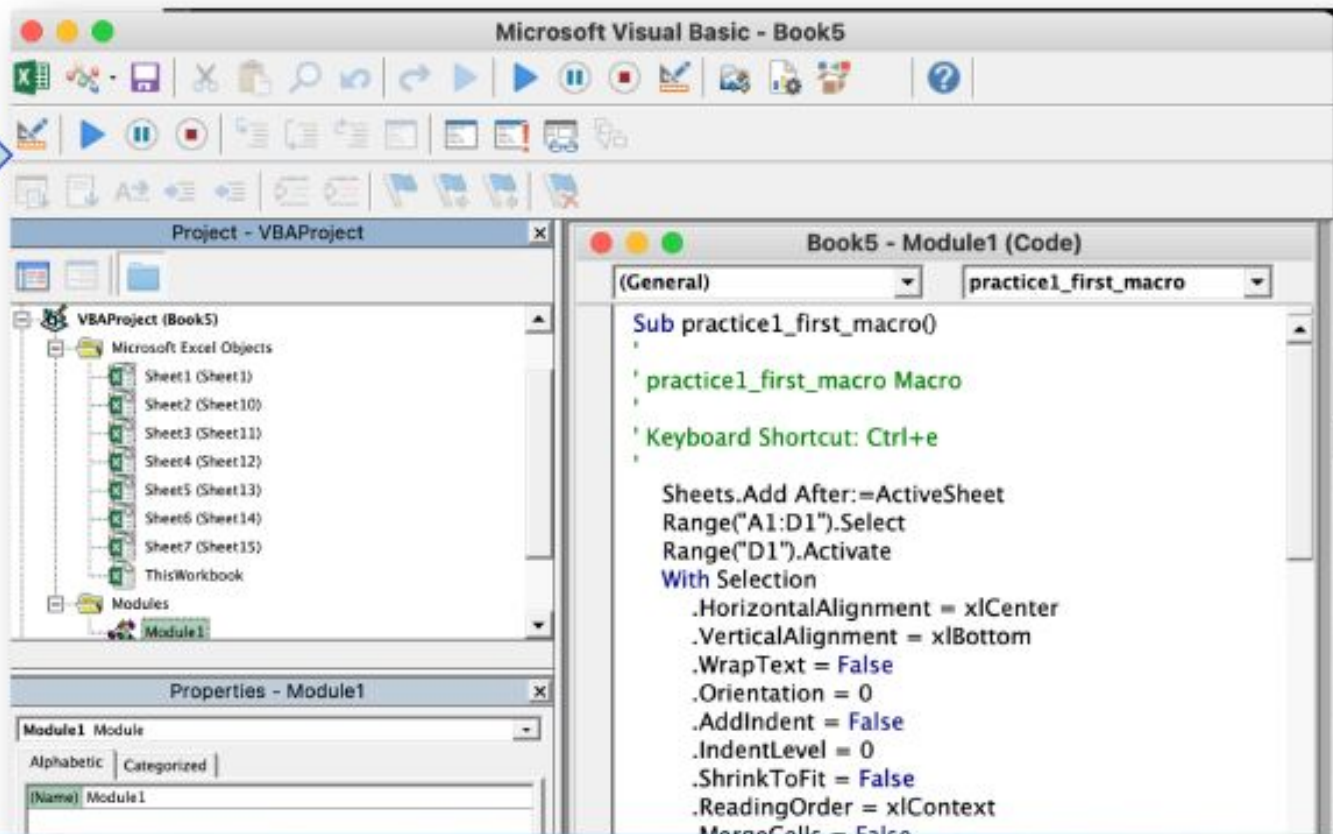
- Main editor on the right
- Modules → 1 or more “Sub” (sub-routines) or “Function”



Business Process Automation with VBA

Entering Visual Basic Mode

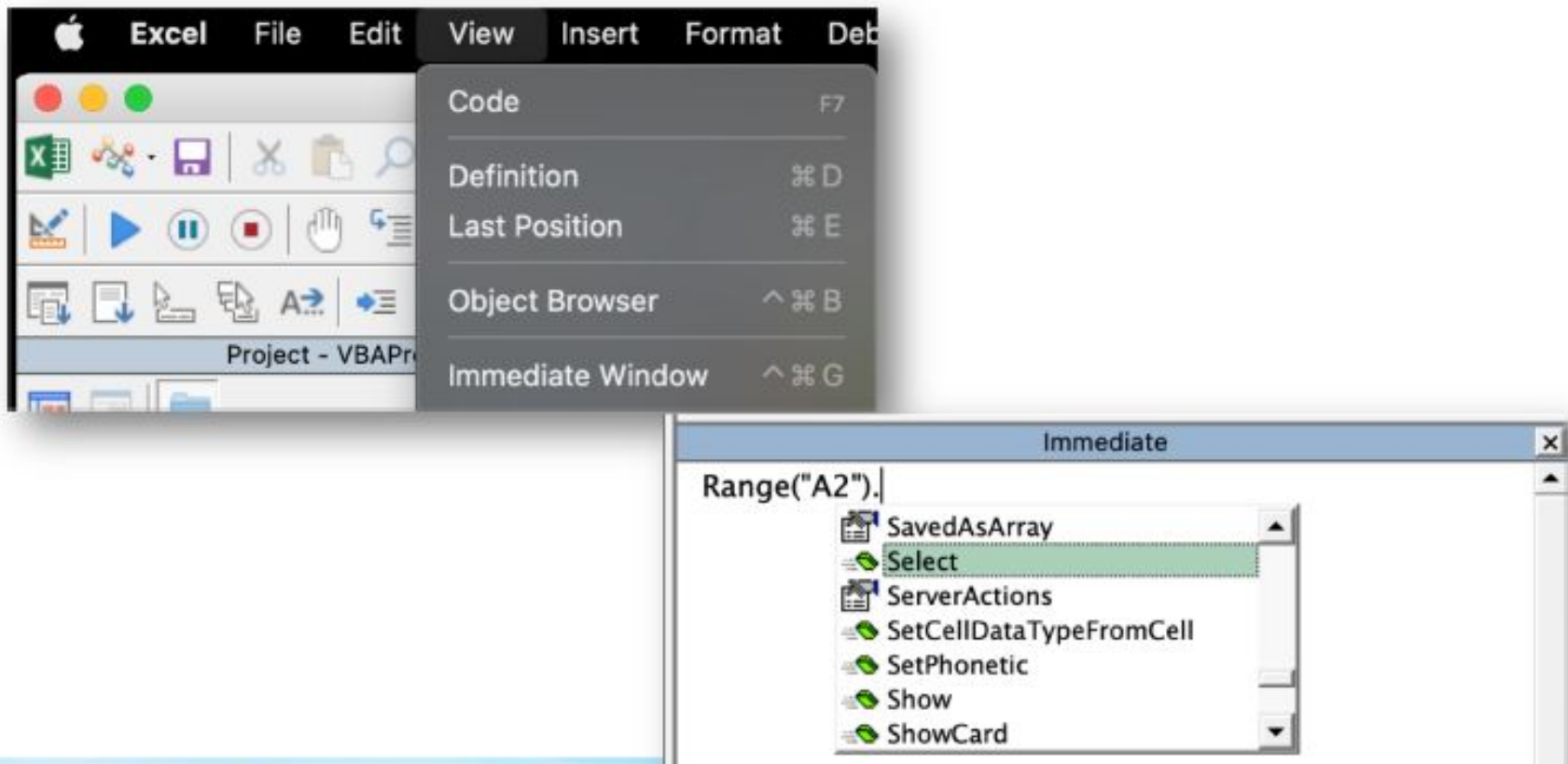
- Debugging toolbar at the top



Business Process Automation with VBA

Immediate Window

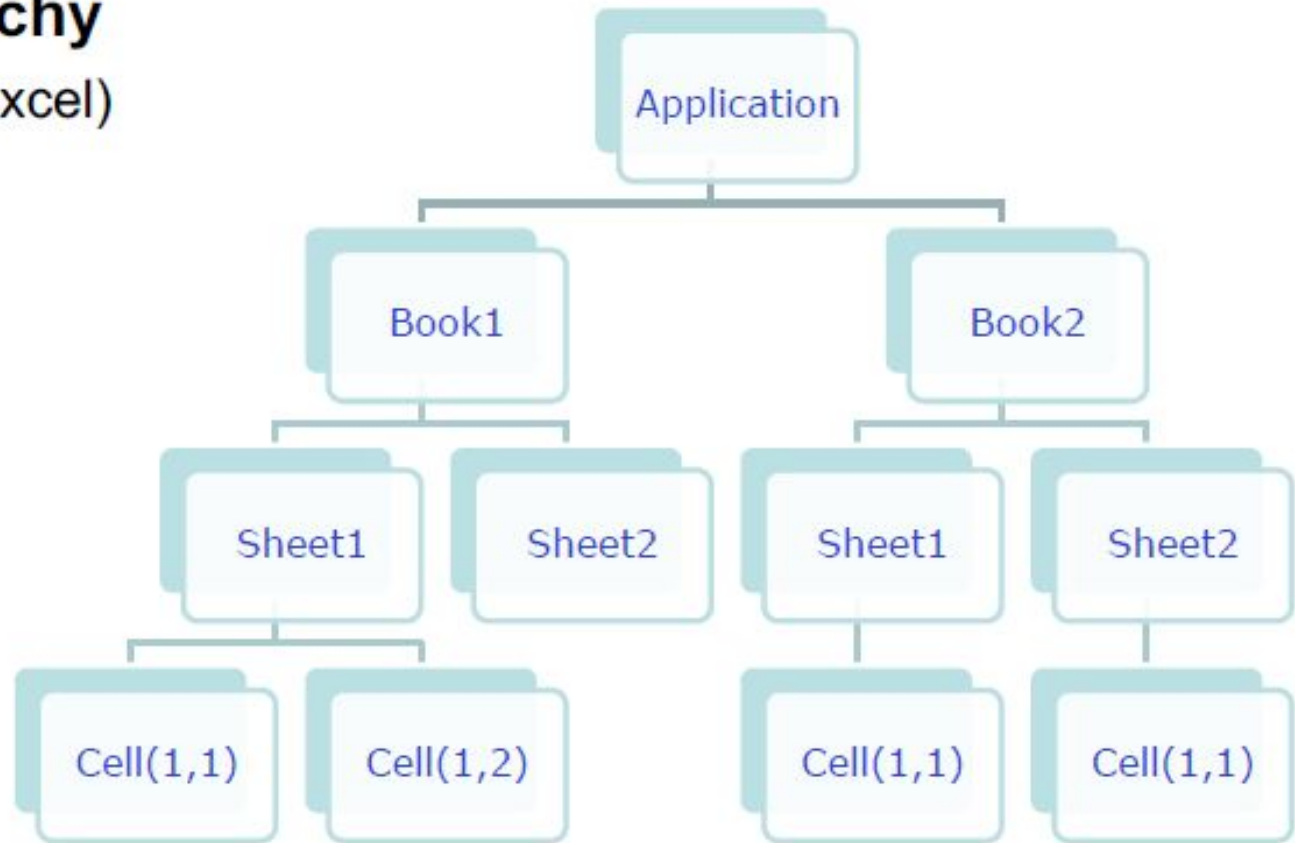
- View → Immediate Window
- Useful for trying out codes (Thanks to Auto-complete)



Business Process Automation with VBA

Object hierarchy

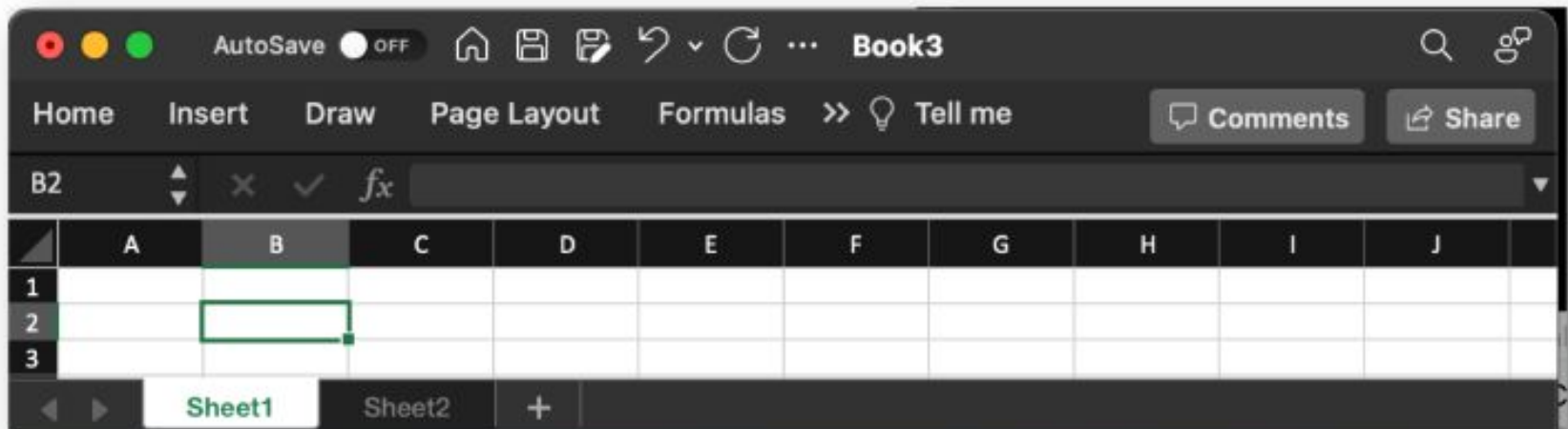
- Application (Excel)
- Workbooks
- Worksheets
- Cells



Business Process Automation with VBA

Object reference – Call something by name

- English: This man / that girl / “Mr. Chan”
- VBA
 - Application → Excel.Application / Outlook.Application / etc
 - Workbooks("Book1")
 - Worksheets("Sheet1") / Worksheets(1)
 - Cells(2,3)
 - Range("B2")



Business Process Automation with VBA

Object reference - Implied naming

- English: Here / Today
- VBA
 - `ActiveWorkbook`
 - `ActiveSheet`
 - `Selection`
 - If you write:
 - `Range("A1:D1").Select`
 - From that moment onwards:
 - `Selection` → Same as `Range("A1:D1")`
- * `ThisWorkbook`
 - This = Where the macro is stored
 - Active = Changeable by user

Business Process Automation with VBA

Object reference - Offset

- English: Is it that building? No, the one on the right
- VBA
 - `Range("B2").Offset(1, 2)`
 - `Worksheets(ActiveSheet.Index + 1).Select`

B2				
	A	B	C	D
1				
2		B2		
3				
4				

Diagram illustrating the VBA formula `Range("B2").Offset(1, 2)`. A blue arrow labeled '1' points down from cell B2 to cell C2. A blue arrow labeled '2' points right from cell C2 to cell D2. The formula bar shows the result of the offset operation.



Me

Watashino migini senseiga imasu
わたしの右に先生がいます
T here is a teacher on my right

coto
Japanese
Academy

Business Process Automation with VBA

Object reference – Offset

- Region
 - `Selection.CurrentRegion.Select`
- Boundary
 - `Range("A1").End(xlDown).Select`

B2				
			x	✓
			f _x	Item
	A	B	C	
1				
2		Item	Price	
3		A	1	
4		B	2	
5		C	3	
6		D	4	
7				

	A	
1		1
2		2
3		3
4		4
5		5
6		6

?Range("A1").End(xlDown).Row
6

Business Process Automation with VBA

Object properties – Actions

- English
 - Jackie runs / That person sings
- VBA (Actions are also known as Methods)
 - ActiveSheet.PrintOut
 - Workbooks("Book5").PrintPreview
 - Worksheets("Sheet15").Select



Business Process Automation with VBA

Object properties – More actions

- VBA – Some methods require parameters



- Copy & Paste
 - `Range("B2").Copy Range("B3")`
- Export to PDF
 - `ActiveSheet.ExportAsFixedFormat Type:=xlTypePDF, Filename:="demo.pdf"`

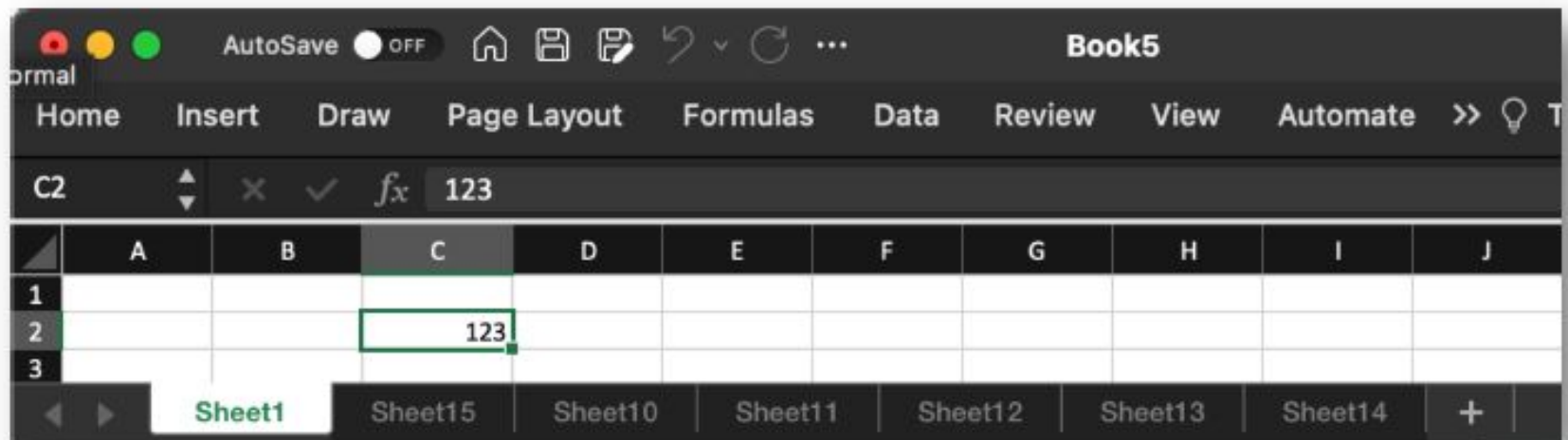
	A	B	
1			
2		testing	
3		testing	
4			

Name	Size	Kind
demo.pdf	13 KB	PDF Document
Book5.xlsm	28 KB	Micros... (.xlsm)

Business Process Automation with VBA

Object properties – Value

- English
 - HSBC → Account 123-654321-012 → Value → \$1000
- VBA
 - `Workbooks("Book5").Worksheets("Sheet1").Cells(2,3).Value = 123`



Business Process Automation with VBA

Object properties – Other properties

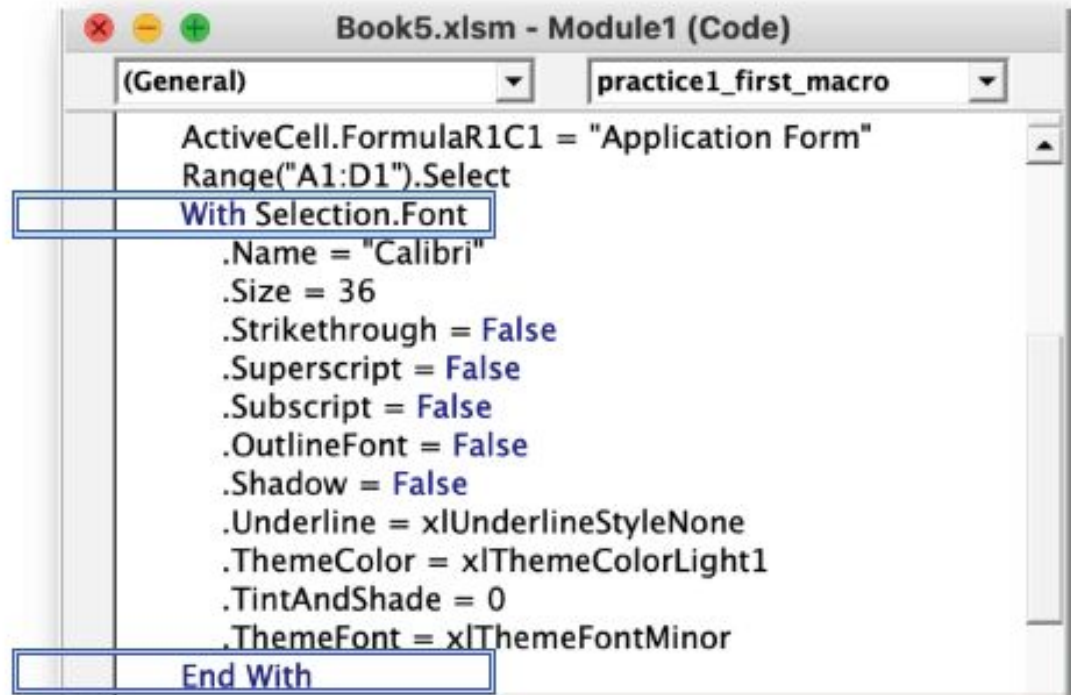
- VBA
 - Worksheets("Sheet1").Range("B2").Font.Size = 36
 - Worksheets("Sheet1").Range("B2").Font.Color = vbRed

	A	B	C
1			
2		testing	
3			

Business Process Automation with VBA

Object properties – Editing many properties

- “With” statement
 - Reduce redundancy
 - Example from Practice 1



The screenshot shows the VBA Editor window titled "Book5.xlsm - Module1 (Code)". The "General" tab is selected, and the macro "practice1_first_macro" is chosen. The code in the editor is as follows:

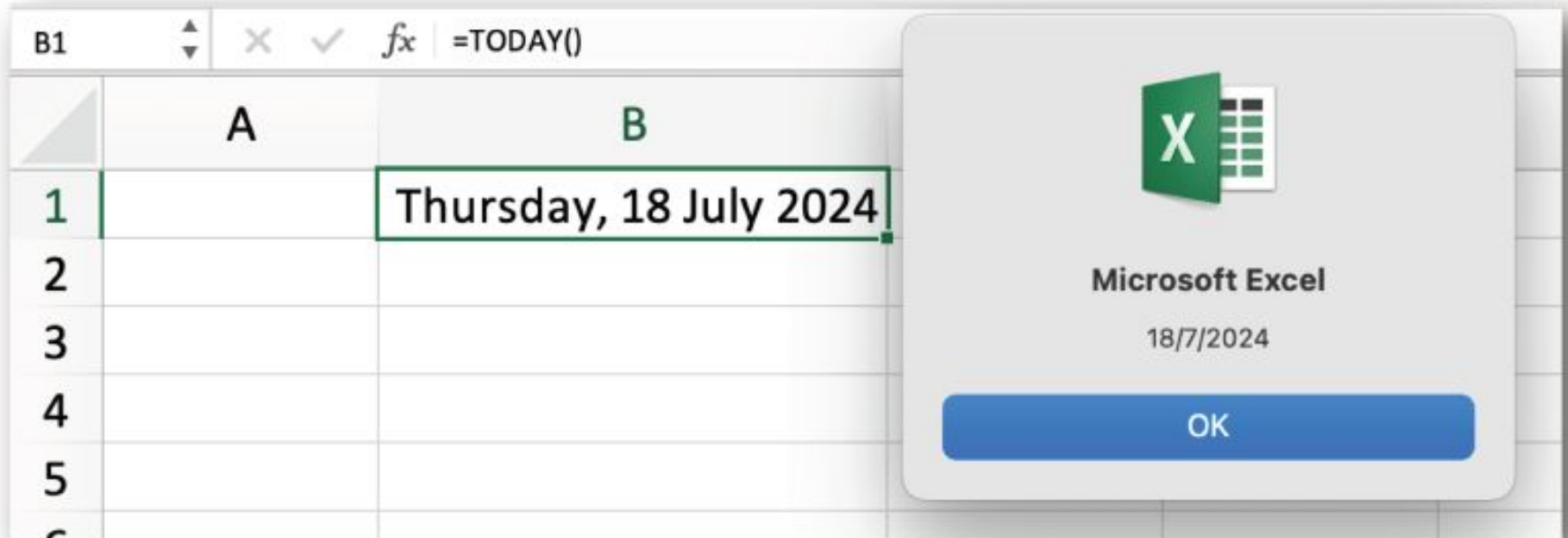
```
ActiveCell.FormulaR1C1 = "Application Form"  
Range("A1:D1").Select  
With Selection.Font  
    .Name = "Calibri"  
    .Size = 36  
    .Strikethrough = False  
    .Superscript = False  
    .Subscript = False  
    .OutlineFont = False  
    .Shadow = False  
    .Underline = xlUnderlineStyleNone  
    .ThemeColor = xlThemeColorLight1  
    .TintAndShade = 0  
    .ThemeFont = xlThemeFontMinor  
End With
```

The "With Selection.Font" line and the "End With" line are highlighted with blue boxes.

Business Process Automation with VBA

Object properties – Value check

- Use “MsgBox”
 - `MsgBox(Worksheets("Sheet1").Range("B1").Value)`

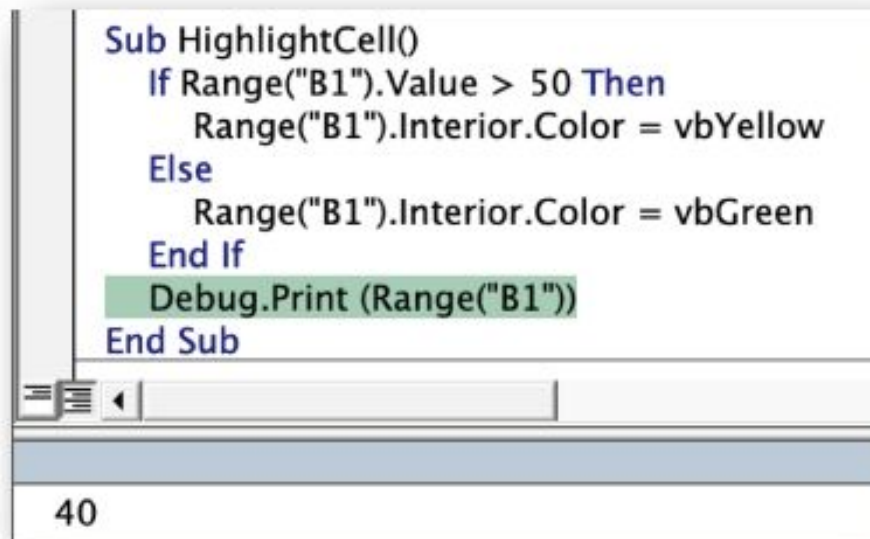


Business Process Automation with VBA

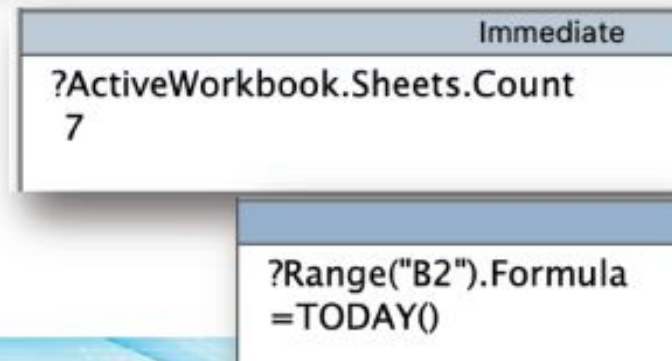
Object properties – Value check

- Use “Debug.Print()”

```
Sub HighlightCell()  
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If  
    Debug.Print (Range("B1"))  
End Sub
```



- Use “?” in the immediate window
 - ?ActiveWorkbook.Sheets.Count
 - ?Range("B2").Formula

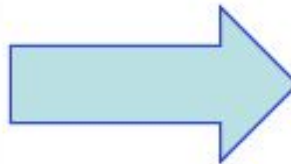


Business Process Automation with VBA

Checkpoint – Value swapping

- How to swap (switch) values between 2 cells?

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2		DEF
3		ABC

Business Process Automation with VBA

Checkpoint – Value swapping

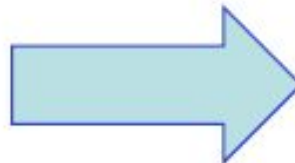
- How to swap (switch) values between 2 cells?
 - 1 solution...

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2	ABC	ABC
3	DEF	DEF

1) Copy



	A	B
1		
2		DEF
3		ABC



3) Clean up

	A	B
1		
2	ABC	DEF
3	DEF	ABC

2) Overwrite

Business Process Automation with VBA

Variables – Motivation

- Storage of values without changing the sheets
- Meaningful references during calculation

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2		DEF
3		ABC

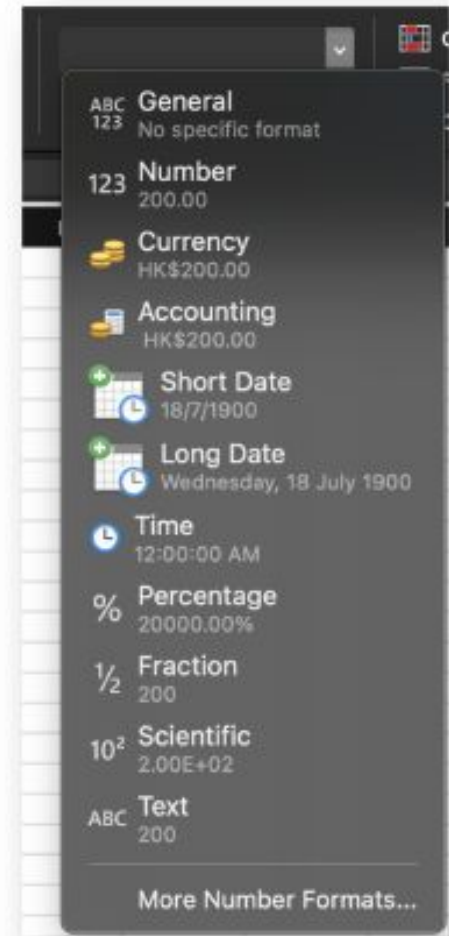
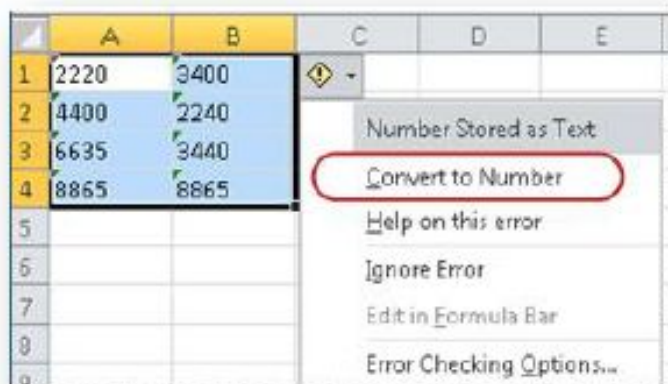


```
temp_integer = Range("B2").Value  
Range("B2").Value = Range("B3").Value  
Range("B3").Value = temp_integer
```

Business Process Automation with VBA

Variables – Data Types

- Excel is good at detecting types
 - Numbers / Date / Text
- Though sometimes can still be wrong



1. Fix text-formatted numbers by applying a number format - Microsoft Support. Fix Text-formatted Numbers by Applying a Number Format - Microsoft Support. <https://support.microsoft.com/en-us/office/fix-text-formatted-numbers-by-applying-a-number-format-6599c03a-954d-4d83-b78a-23af2c8845d0>

Business Process Automation with VBA

Variables – Data Types

- Best to define variables with types

Commonly Used VBA Data Types		
Data Type	Bytes Used	Range of Values
Integer	2	-32,768 to 32,767
Long	4	-2.147 Billion to 2.147 Billion
Single	4	45 digits (left or right of decimal point)
Double	8	308 digits (left or right of decimal point)
Currency	8	-922 T to 922 T (4 digits after decimal point)
Date	8	1/1/100 to 12/31/9999
Boolean	2	True or False
String	1 per char	Varies according to number of characters
Variant	Varies	Any data type

Business Process Automation with VBA

Variables – Data Types

- Integer
 - Dim year As Integer
 - year = 2024
- Double
 - Dim pi As Double
 - pi = 3.14125



- Smart conversion during calculation

```
?TypeName( "A" & 3 )  
String
```

```
? "A"&3  
A 3
```

```
?TypeName(2)  
Integer
```

```
?TypeName(3.3)  
Double
```

```
?TypeName(2 * 3.4)  
Double
```

```
?TypeName(5-3)  
Integer
```

Variables – Data Types

- Date
 - Dim today As Date
 - today = DateSerial(2024, 7, 1)
 - today = Date()
- String
 - Dim message As String
 - message = "Welcome!"
- Boolean
 - Range("A1").Font.Bold = True
 - Range("A1").Font.Underline = False



Business Process Automation with VBA

Common functions - Numbers

- IsNumeric() → Check if a text is number
- CInt() / CDbl() → Text to number

```
?IsNumeric("123")
```

```
True
```

```
?IsNumeric("ABCD")
```

```
False
```

```
?CInt("123")
```

```
123
```

```
?CInt("123.45")
```

```
123
```

```
?CDbl("123.45")
```

```
123.45
```

```
?CDbl("123")
```

```
123
```

Business Process Automation with VBA

Common functions - Numbers

- Round() → Rounding to a certain decimal place

```
?Round(5.1234)
```

```
5
```

```
?Round(5.1234, 2)
```

```
5.12
```

- Log() / Exp() → For financial / engineering calculation

```
?exp(2)
```

```
7.38905609893065
```

```
?log(10)
```

```
2.30258509299405
```

```
?log(exp(1))
```

```
1
```

Business Process Automation with VBA

Common functions - Date

- DateSerial → Create a date literal

```
?DateSerial(2024, 12, 25)  
25/12/2024
```

- Date() → Today's date

```
?Date()  
18/7/2024
```

- Year() / Month() / Day() → Extract

```
?Year(DateSerial(2024, 12, 25))  
2024
```

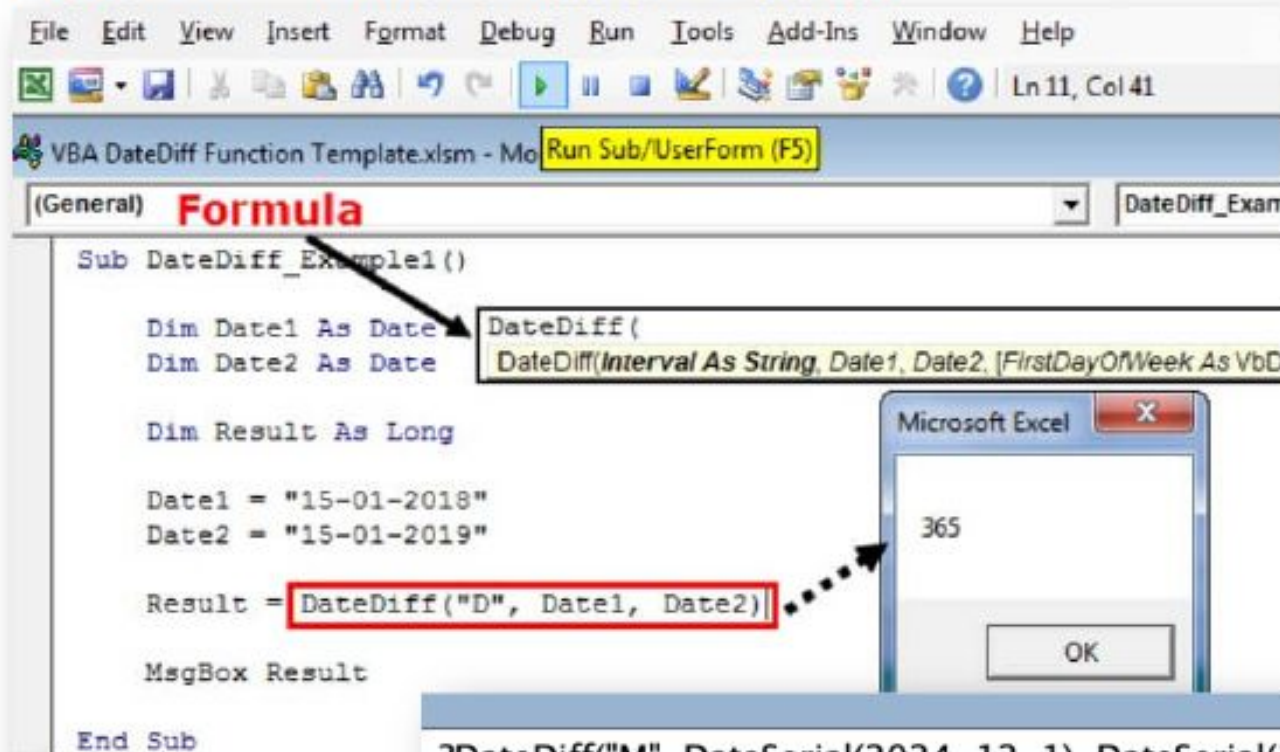
```
?Month(DateSerial(2024, 12, 25))  
12
```

```
?Day(DateSerial(2024, 12, 25))  
25
```

Business Process Automation with VBA

Common functions - Date

- DateDiff – Difference of 2 dates



```
?DateDiff("M", DateSerial(2024, 12, 1), DateSerial(2024, 12, 10))  
0
```

```
?DateDiff("M", DateSerial(2024, 11, 30), DateSerial(2024, 12, 1))  
1
```

Business Process Automation with VBA

Common functions - String

- CDate() → Text to Date

```
?CDate("2023-03-25")  
25/3/2023
```

```
?CDate("April 1, 2023")  
1/4/2023
```

```
?CDate("25/2")  
25/2/2023
```

- Format() → Date to Text

```
?Format(DateSerial(2023, 3, 25), "M")  
3
```

```
?Format(DateSerial(2023, 3, 25), "MM")  
03
```

```
?Format(DateSerial(2023, 3, 25), "MMM")  
Mar
```

```
?Format(DateSerial(2023, 3, 25), "MMMM")  
March
```

```
?Format(DateSerial(2023, 3, 25), "DD-MMM-YYYY")  
25-Mar-2023
```

Business Process Automation with VBA

Common functions - String

- Left() / Right() → Substring

```
?Left("0005.HK", 4)  
0005
```

```
?Right("0005.HK", 2)  
HK
```

- Split() → Tokenizing into a string array

```
?Split("A,B,C", ",")(0)  
A
```

```
?Split("A,B,C", ",")(2)  
C
```

```
?UBound(Split("A,B,C", ","))  
2
```

Business Process Automation with VBA

Common functions - String

- Trim() → Remove empty spaces
 - Note: "" gives 1 quote
- InStr(x, y) → Find y in x
 - Case sensitive

```
?"" & Trim(" Hello! ") & ""  
"Hello!"
```

```
?InStr("Search here", "Search")  
1
```

```
?InStr("Search here", "here")  
8
```

```
?InStr("Search here", "nothing")  
0
```

```
?InStr("Search here", "search")  
0
```

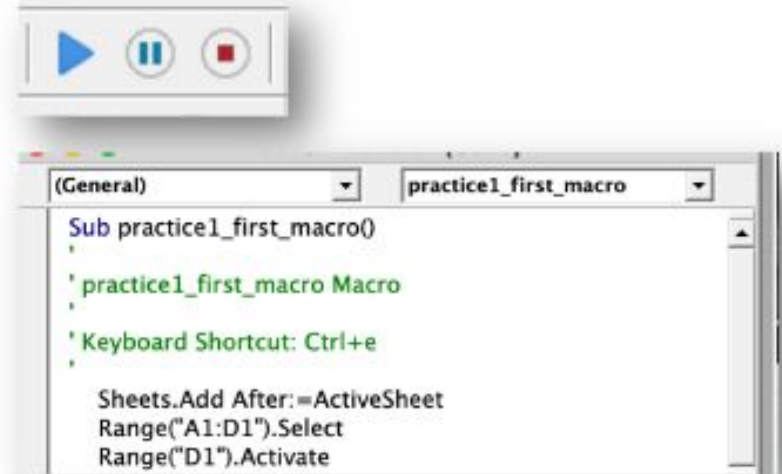
```
?Lcase("HeLlO")  
hello
```

```
?InStr(Lcase("Search here"), "search")  
1
```

Business Process Automation with VBA

Sub-routines

- Macros recorded are wrapped with “Sub ... End Sub”
- Sub-routines are just reusable codes
- Within the scope of a sub-routine
 - Click the “Play” button to execute



Business Process Automation with VBA

Control flow – If, Then, Else

- Example: speed check

	A	B
1	Speed	40
2		

	A	B
1	Speed	50
2		

	A	B
1	Speed	70
2		

```
Sub HighlightCell()
```

```
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If
```

```
End Sub
```

Excel Function Definition - Syntax & Structure

Definition

A VBA Function is a procedure that performs operations and returns a single value to the calling code. Functions are used to create custom calculations that can be called from Excel worksheets or other VBA code.

Syntax

```
1 Function FunctionName(parameters) As ReturnType
    ' Comments: perform the task here
    FunctionName = returnValue
    'Assign the function result to the function name'
End Function
```



Key Points:

- ✓ The function returns a value by assigning it to the FunctionName
- ✓ Use the As keyword to specify the return data type (e.g. Integer, Double, String)
- ✓ Must end with End Function to close the function



Example — Calculate Discount

```
Function CalculateDiscount(price As Double,
discountPercent As Double) As Double

    If discountPercent < 0 Or discountPercent > 1 Then
        CalculateDiscount = -1 ' Return -1 as error for invalid
                                input
    Else
        CalculateDiscount = price * (1 - discountPercent)
        Apply the discount
    End If
End Function
```



Usage in Excel:

```
=CalculateDiscount(100, 0.15)
→ Returns 85 (100 × 0.85)
```



Reusable
call from VBA or Excel formulas



Can be used like built-in
Excel functions



Placed in a Standard Module via
Insert > Module

Note: Functions return a value; Subs perform actions without returning a value

Business Process Automation with VBA

Control flow – If, Then, Else

- Mark → Grade

<i>Mark</i>	<i>Letter Grade</i>
90+	A+
85-89	A
80-84	A-
77-79	B+
73-76	B
70-72	B-
67-69	C+
63-66	C
60-62	C-
57-59	D+
50-56	D
0-49	F

Business Process Automation with VBA

Control flow – Select, Case

- Range check

?GetGrade(90)
A

?GetGrade(80)
B

```
Function GetGrade(score As Integer) As String
    Select Case score
        Case Is >= 90
            GetGrade = "A"
        Case Is >= 80
            GetGrade = "B"
        Case Is >= 70
            GetGrade = "C"
        Case Is >= 60
            GetGrade = "D"
        Case Else
            GetGrade = "F"
    End Select
End Function
```

Business Process Automation with VBA

Control flow – Select, Case

- Grade → Marks

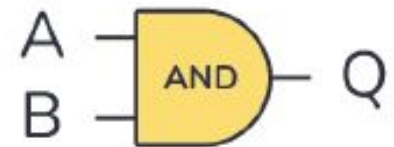
```
Dim letter_grade As String
Dim grade_point As Double

Select Case letter_grade
Case "A", "A+"
    grade_point = 4.0
Case "A-"
    grade_point = 3.7
Case "B+"
    grade_point = 3.3
...
Case Else
    grade_point = 0.0
End Select
```

Business Process Automation with VBA

Control flow – Conditions

- AND
 - (True AND True) → True
 - e.g., Month validity: $1 \leq \text{month}$ AND $\text{month} \leq 12$
- OR
 - (True OR False) → True
 - e.g., Game Area: $\text{gold} > 1000$ OR $\text{level} > 10$
- NOT
 - NOT $x=0$ is same as $x \neq 0$



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Business Process Automation with VBA

Loops – Using “For”

- Iterating through the rows
- What is changing?
 - A1 ➔ A2 ➔ A3 ➔ A4...



	A	B	C
1	1		
2	2		
3	3		
4	4		
5	5		
6	6		
7	7		
8	8		
9	9		
10	10		

```
Sub LoopThroughRows()  
    For i = 1 To 10  
        Range("A" & i).Value = i  
    Next i  
End Sub
```

Business Process Automation with VBA

Loops – Using “For”

- Iterating through the columns
- Again, track the change: A1 → B1 → C1 → ...
- Option 1: Make use of offset()



	A	B	C	D	E	F	G	H	I	J
1	1	2	3	4	5	6	7	8	9	10
2										

```
Sub LoopThroughColumns()  
    For i = 1 To 10  
        Range("A1").Offset(0, i - 1).Value = i  
    Next i  
End Sub
```

Business Process Automation with VBA

Loops – What is happening?

- Temporary variable + “Array”

```
Sub LoopThroughRows()  
    For i = 1 To 10  
        Range("A" & i).Value = i  
    Next i  
End Sub
```

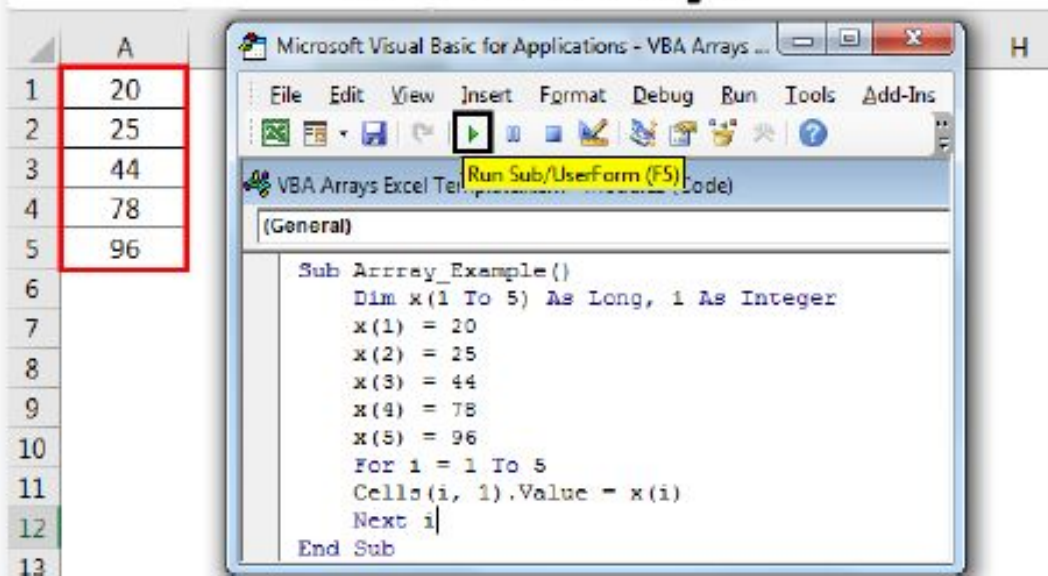
```
Sub LoopThroughColumnsWithCells()  
    Dim i As Integer: i = 1  
    For Each cell In Range("A1:J1").Cells  
        cell.Value = i  
        i = i + 1  
    Next cell  
End Sub
```

Business Process Automation with VBA

Loops – What is happening?

- Array = many variables

VBA Arrays



The screenshot shows an Excel spreadsheet on the left with column A containing the values 20, 25, 44, 78, and 96 in rows 1 through 5, respectively. The cells are highlighted with a red border. On the right, the Microsoft Visual Basic for Applications editor is open, displaying a VBA module named 'VBA Arrays Excel Te'. The code in the editor is as follows:

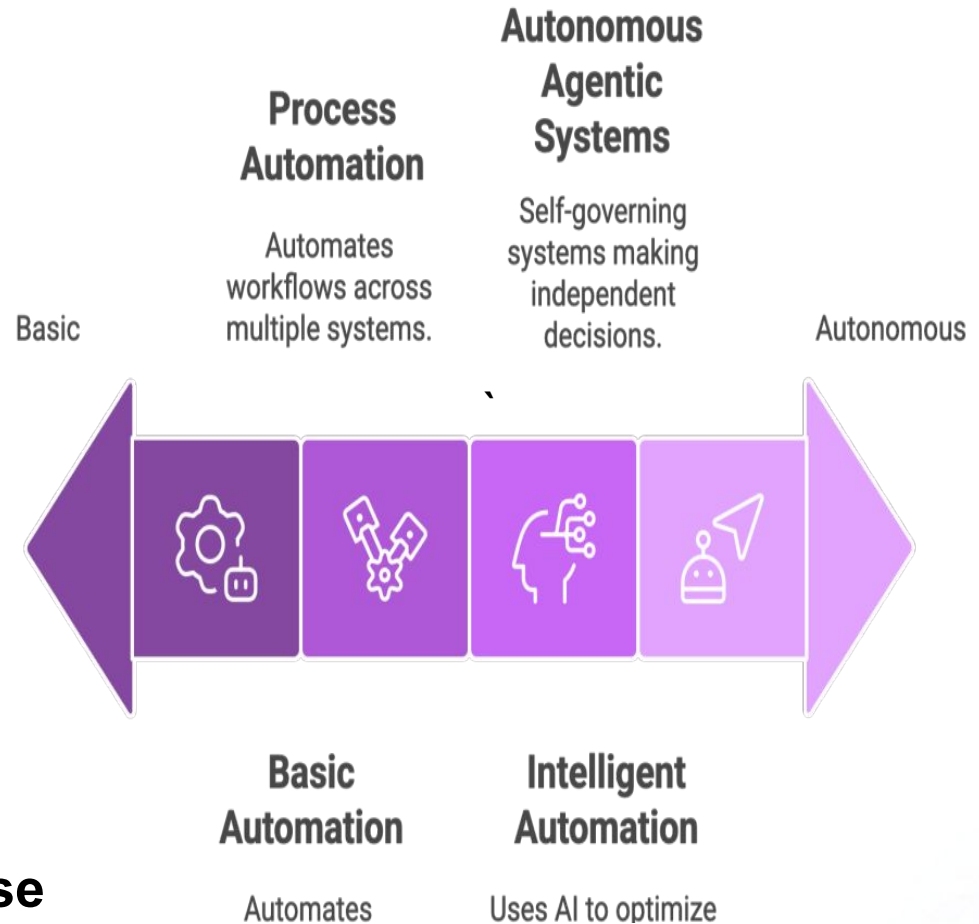
```
Sub Array_Example()  
    Dim x(1 To 5) As Long, i As Integer  
    x(1) = 20  
    x(2) = 25  
    x(3) = 44  
    x(4) = 78  
    x(5) = 96  
    For i = 1 To 5  
        Cells(i, 1).Value = x(i)  
    Next i  
End Sub
```



Four Level of Business Automation

1. Basic Automation
2. Process Automation
3. Intelligent Process Automation
4. Autonomous Agentic Systems

We dive into real practise



Business Process Automation with Python

Python – Origin

- Released in 1991 (same as Visual Basic)
- Free and open-source



Business Process Automation with Python

Motivation – Soft Limits

- Protective measures not matching modern days

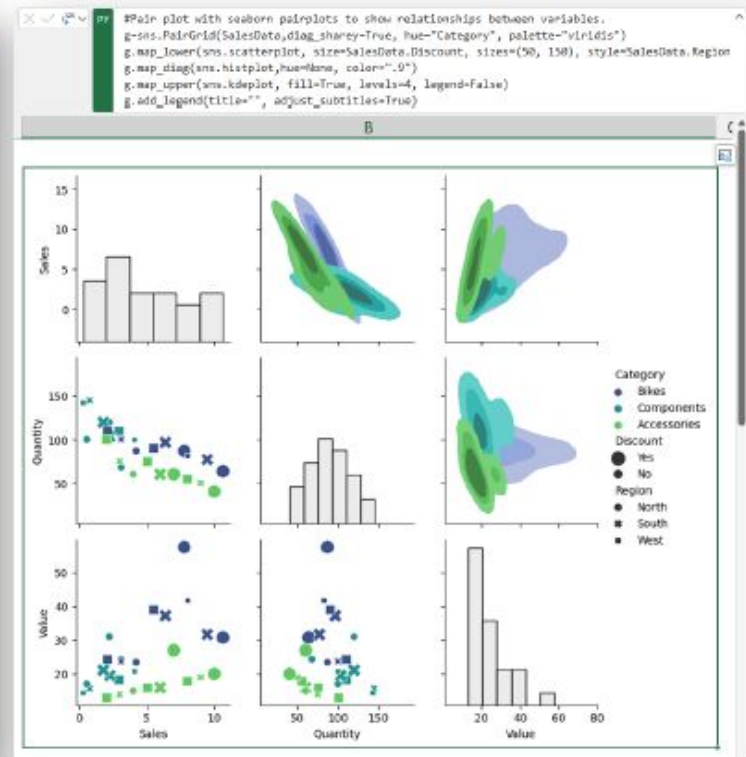
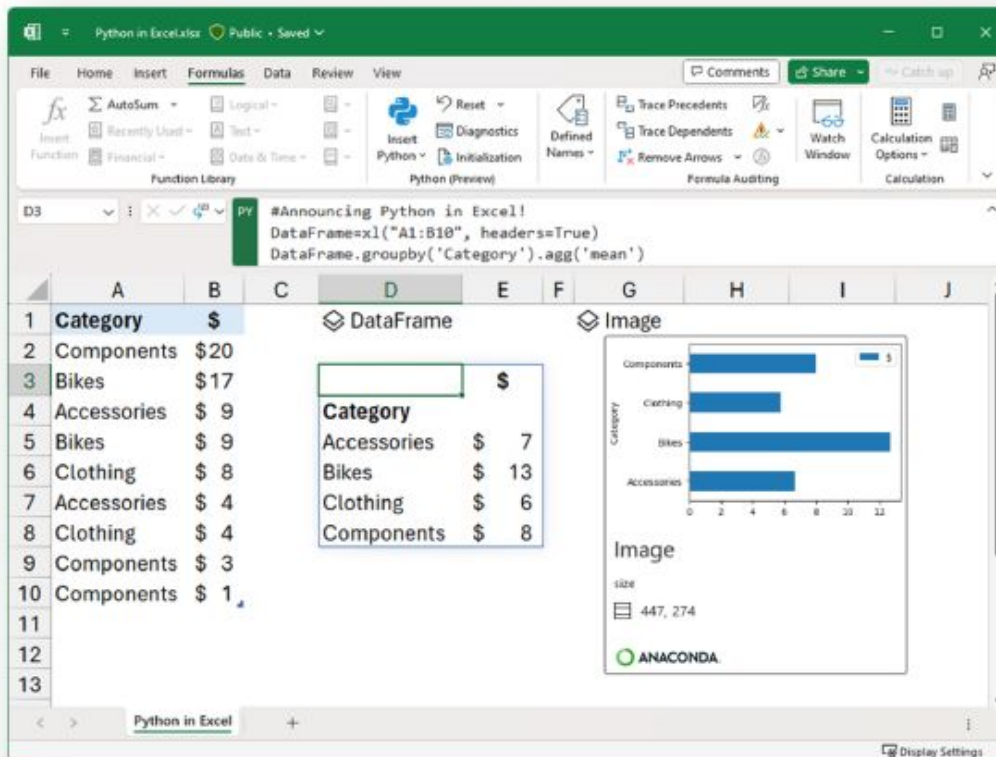
Feature	Maximum limit
Total number of rows and columns on a worksheet	1,048,576 rows by 16,384 columns
Column width	255 characters
Row height	409 points
Page breaks	1,026 horizontal and vertical

1. Microsoft. Excel Specifications and limits. <https://support.microsoft.com/en-us/office/excel-specifications-and-limits-1672b34d-7043-467e-8e27-269d656771c3>

Business Process Automation with Python

Motivation – Extensibility

- Python in Excel ¹



1. *Announcing Python in Excel: Combining the power of Python and the flexibility of Excel.* TECHCOMMUNITY.MICROSOFT.COM.
<https://techcommunity.microsoft.com/t5/excel-blog/announcing-python-in-excel-combining-the-power-of-python-and-the/ba-p/3893439>

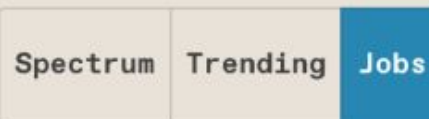
Business Process Automation with Python

Motivation – Jobs

- Python → 1 of the top languages for work

Top Programming Languages 2024

Click a button to see a differently weighted ranking



1. *The Top Programming Languages 2024*. (2024 August 22). IEEE Spectrum.
<https://spectrum.ieee.org/top-programming-languages-2024>

Business Process Automation with Python

Python runtime

- The language and basic runtime environment
- Just like Apple iOS, it keeps updating (e.g., Python 3.12.2)
 - Python 2 was no longer supported from 2020¹

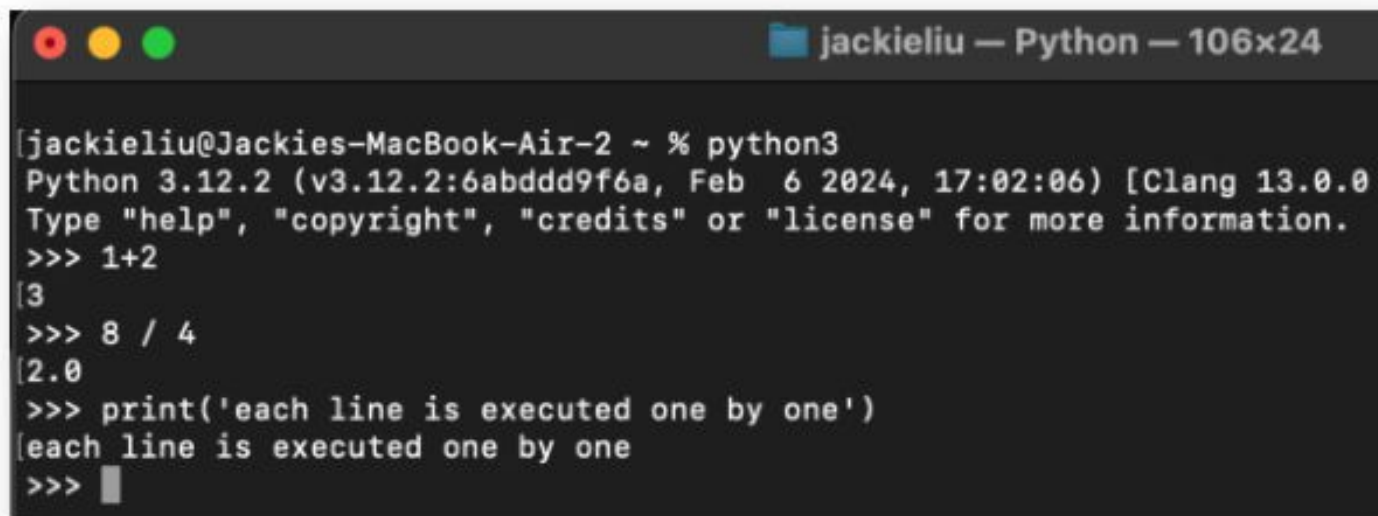


1. Sunsetting Python 2. Python.org. <https://www.python.org/doc/sunset-python-2/>

Business Process Automation with Python

Open Local Python Interpreter

1. In Desktop, click 'Startup' button
2. Type "Anaconda Prompt (Anaconda3)"
3. Enter "python"
4. Input a line of codes -> Press Enter -> Run



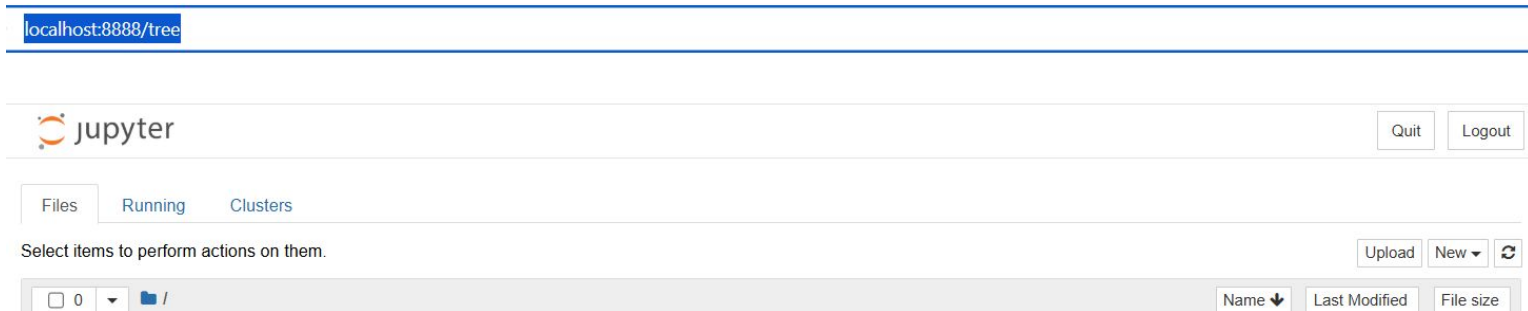
```
jackieliu — Python — 106x24

[jackieliu@Jackies-MacBook-Air-2 ~ % python3
Python 3.12.2 (v3.12.2:6abddd9f6a, Feb  6 2024, 17:02:06) [Clang 13.0.0
Type "help", "copyright", "credits" or "license" for more information.
>>> 1+2
3
>>> 8 / 4
2.0
>>> print('each line is executed one by one')
each line is executed one by one
>>> 
```

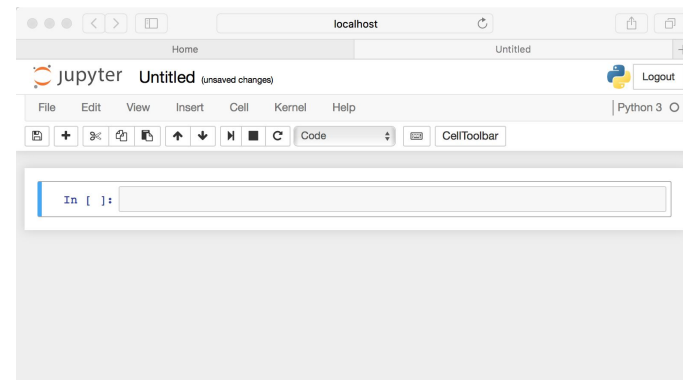
Python Coding Environment Practise - Jupyter notebook

Open Jupyter notebook

1. In Desktop, click 'Startup' button
2. Type "Anaconda Prompt (Anaconda3)"
3. Type "jupyter notebook" to trigger your first notebook
4. <http://localhost:8888/tree> is opened



5. In right top side, click New -> Python 3
6. Renamed file name to "first_pythonlab"



Business Process Automation with Python

Variables

- From the previous lecture
 - Variables → Storage of **values**
 - **Values** → carry data types

	A	B
1		
2		ABC
3		DEF



	A	B
1		
2		DEF
3		ABC



Business Process Automation with Python

Variables

- Declaration
 - In VBA, declarations (“dim”) are recommended
 - Dim year As **Integer**
 - year = 2024
 - In Python, variables are created during **assignment**
 - year = 2024
 - To assign without an actual value
 - year = None



Business Process Automation with Python

Value Check

- VBA
 - Debug.Print()
- Python
 - print()
 - Last line in a block

```
Sub HighlightCell()  
    If Range("B1").Value > 50 Then  
        Range("B1").Interior.Color = vbYellow  
    Else  
        Range("B1").Interior.Color = vbGreen  
    End If  
    Debug.Print (Range("B1"))  
End Sub
```

```
▶ print(1 + 2)           # This gives 3  
  print(1 + 2 + 3)       # This gives 6
```

3
6

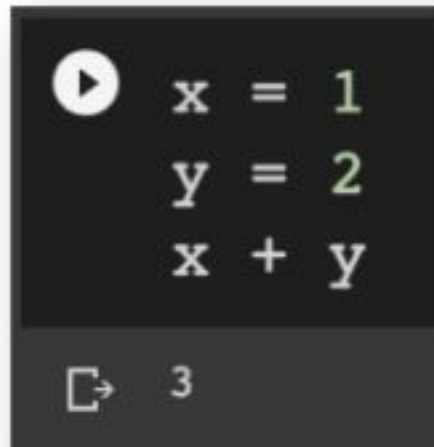
```
▶ 1 + 2                 # This is not shown  
  1 + 2 + 3             # This is shown
```

6

Business Process Automation with Python

Primitive Data Types - Numeric

- Int

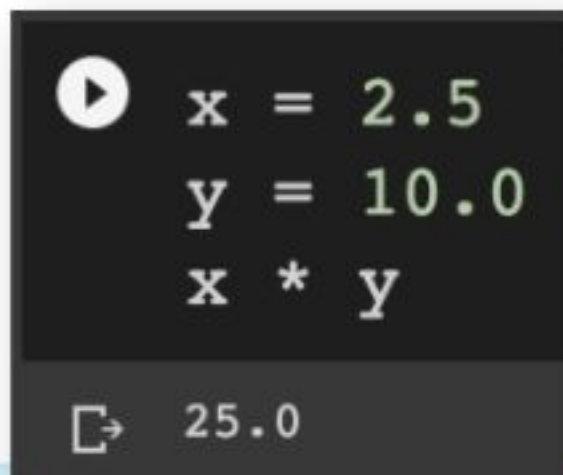


```
x = 1
y = 2
x + y
```

3

A dark-themed code editor window showing three lines of Python code: `x = 1`, `y = 2`, and `x + y`. The code is in a light green monospace font. Below the code, a light gray bar displays the result of the expression, `3`, preceded by a small icon of a box with an arrow pointing right.

- Float



```
x = 2.5
y = 10.0
x * y
```

25.0

A dark-themed code editor window showing three lines of Python code: `x = 2.5`, `y = 10.0`, and `x * y`. The code is in a light green monospace font. Below the code, a light gray bar displays the result of the expression, `25.0`, preceded by a small icon of a box with an arrow pointing right.

Primitive Data Types - Numeric

- Complex (For scientific calculations)

THE QUADRATIC FORMULA

© CHILMATH.COM

If $ax^2 + bx + c = 0$ but $a \neq 0$

then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT

- $b^2 - 4ac > 0$ two real solutions
- $b^2 - 4ac = 0$ one real solutions
- $b^2 - 4ac < 0$ zero real solutions



$$x = 3 + 4j$$

$$y = 2 + 2j$$

$$x - y$$

$$\Rightarrow (1+2j)$$

$$i = \sqrt{-1}$$

Business Process Automation with Python

Primitive Data Types - Numeric

- Common Functions
 - `round()` / `pow()` / `abs()`

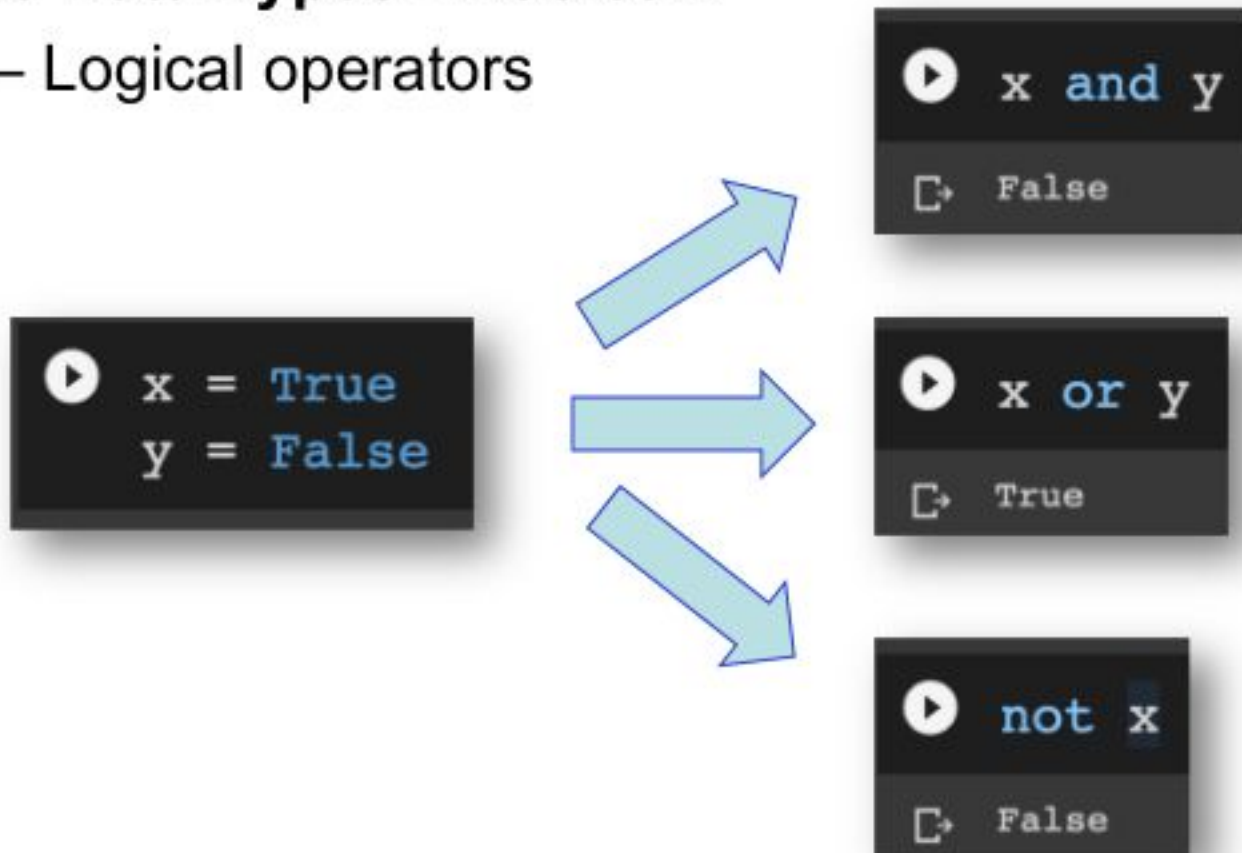
```
▶ print(round(9876.54321, 2))    # Round to 2 decimal places
  print(round(9876.54321, 0))    # Round to nearest integer
  print(pow(2, 3))               # 2 * 2 * 2 = 8
  print(abs(-5))                 # Negative numbers become positive
```

```
□ 9876.54
  9877.0
  8
  5
```

Business Process Automation with Python

Primitive Data Types - Boolean

- Bool – Logical operators



Primitive Data Types - Boolean

- Bool – comparators
 - Generates True/ False

Operator	Name	Example
==	Equal	5 == 5
!=	Not equal	26 != 3
>	Greater than	100 > 67
<	Less than	89 < 216
>=	Greater than or equal to	90 >= 54
<=	Less than or equal to	23 <= 77

Primitive Data Types - Text

- String
 - Representation

```
▶ x = 'This is a string'           # Single quote
  y = "This is also a string"      # Double quote
  z = '''This is a very long string
spanning across more than 1 line''' # Multi-line

print(x)
print(y)
print(z)
```

```
☞ This is a string
   This is also a string
   This is a very long string
   spanning across more than 1 line
```

Business Process Automation with Python

Primitive Data Types - Text

- String
 - Concatenation → “+” → glue 2 strings together

```
▶ x = 'This is a string'  
  y = "This is also a string"
```

```
▶ print(x + ' and ' + y) # Adding 2 strings together
```

```
☐ This is a string and This is also a string
```

Business Process Automation with Python

Primitive Data Types - Text

- String
 - Concatenation → Number vs Text



```
lobster_price = 936
lobster_text = str(lobster_price)
print(2 * lobster_price) # 1872
print(2 * lobster_text) # 936936
```

1872
936936

```
print(10 * '=')
print('WELCOME')
print(10 * '=')
```

```
=====
WELCOME
=====
```

1. 星島日報. (2023, May 6). 臺灣中菜館驚現「天價龍蝦」兩隻竟索價90萬元 酒樓親解釋. Singtaousa.com; 星島日報.
<https://www.singtaousa.com/2023-05-06/%e8%8d%83%e7%81%a3%e4%b8%ad%e8%8f%9c%e9%a4%a8%e9%a9%9a%e7%8f%be%e3%80%8c%e5%a4%a9%e5%83%b9%e9%be%8d%e8%9d%a6%e3%80%8d-%e5%85%a9%e9%9a%bb%e7%ab%9f%e7%b4%a2%e5%83%b990%e8%90%ac%e5%85%83-%e9%85%92/4488305>

Business Process Automation with Python

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
 - A smart way to format strings

```
▶ x = 4  
  y = 8.8  
  f'x is {x} and y is {y}; y divided by x is {y / x}'  
□→ 'x is 4 and y is 8.8; y divided by x is 2.2'
```

1. *Input and Output*. Python Documentation. <https://docs.python.org/3/tutorial/inputoutput.html>

Business Process Automation with Python

Primitive Data Types - String formatting

- F-String (Python 3.6 or newer)
 - A smart way to format strings (advanced)

```
▶ z = 1234.56789  
print(z)  
print(f'{z:.2f}')print(f'{z:,.2f}')
```

```
☐→ 1234.56789  
    1234.57  
    1,234.57
```

1. *Input and Output*. Python Documentation. <https://docs.python.org/3/tutorial/inputoutput.html>

Primitive Data Types - String

- Common Functions
 - upper() / lower() / replace()

```
▶ # Common functions  
test_string = 'Hello Hong Kong!'  
print(test_string.upper())  
print(test_string.lower())  
print(test_string.replace('Hello', 'Bello'))
```

```
☞ HELLO HONG KONG!  
hello hong kong!  
Bello Hong Kong!
```

Business Process Automation with Python

Primitive Data Types - Conversion

- **VBA**

- CInt() / CDbl() / Format()

- **Python**

- int() / float() / str() / bool()



```
int(4.5)
```

 4

```
float(5)
```

 5.0

```
str(5.55)
```

 '5.55'

```
bool('Some values')
```

 True

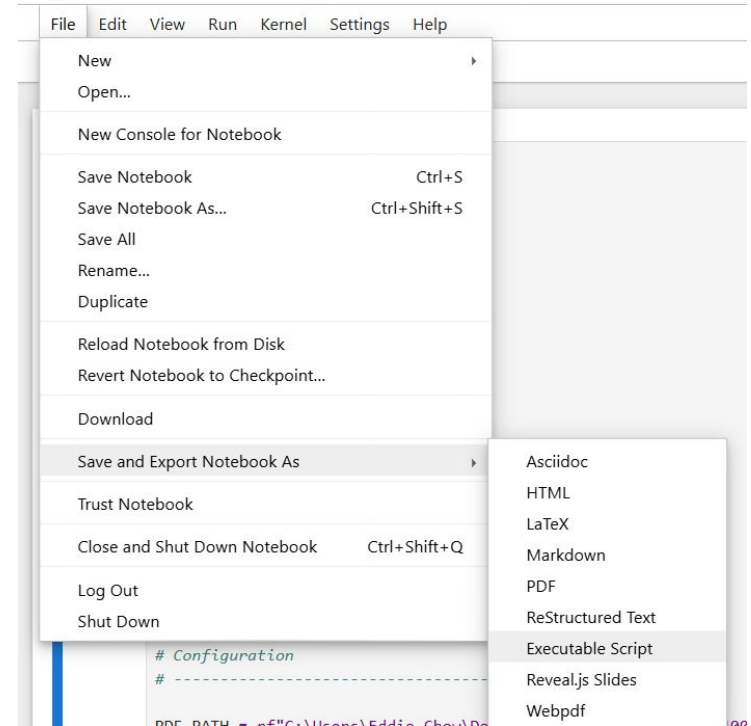
```
bool('')
```

 False

Running python as a file

Download the .py file

1. Click “File” -> “Save and Export Notebook As”
2. Click “Executable Script” and “first_pythonlab.py” is download



Run the .py file

1. Open Anaconda Prompt and enter “Downloads”
2. type command “first_pythonlab.py”

```
Anaconda Prompt (Anaconda) x + v
(base) C:\Users\Eddie Chow>cd Downloads
(base) C:\Users\Eddie Chow\Downloads>python firstlab.py
```

Business Process Automation with Python

Practice 1 – Market Capitalization

- Compute market capitalization of 2 companies
- Show the difference between the two
- Format the result up to 2 decimal places


$$\text{Market Capitalization Formula} = \text{Current Market Price Per Share} \times \text{Total Number of Outstanding Shares}$$


- **Starting notebook:**

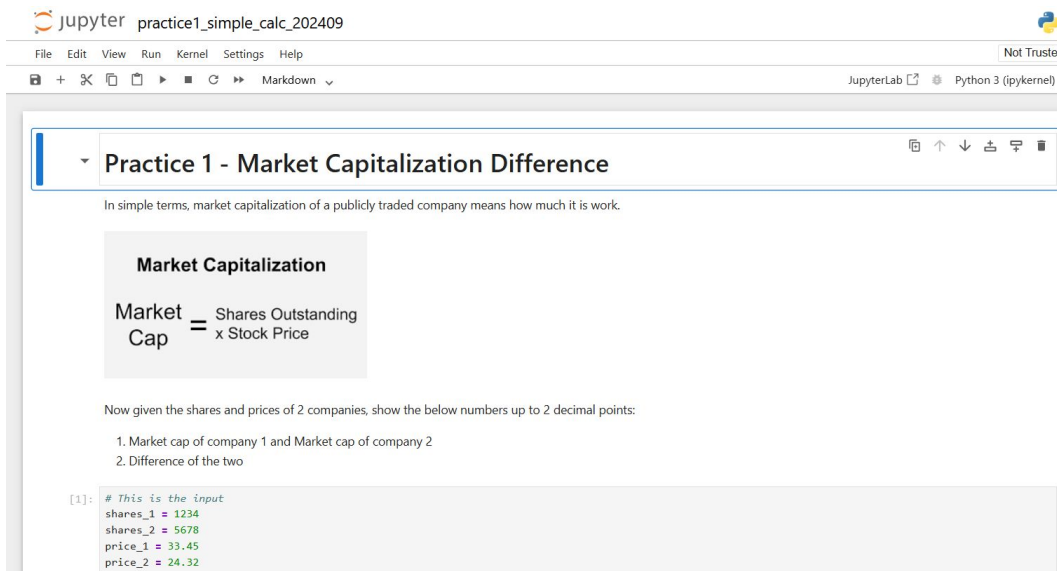
https://github.com/innoviai/IPA_Courses-October2026/blob/main/practice1_simple_calc_202409.ipynb

How to download and edit jupyter notebook

1. Open
https://github.com/innoviai/IPA_Courses-October2026/blob/main/practice1_simple_calc_202409.ipynb
2. Click button “Download raw file”
3. Don’t direct click the practice1_simple_calc_202409.ipynb
4. In jupyter notebook, click “Upload” and select practice1_simple_calc_202409.ipynb



5. Then click practice1_simple_calc_202409.ipynb to edit the content



Business Process Automation with Python

Download and Install Local Python Interpreter

1. Download Anaconda Distribution at <https://www.anaconda.com/download>

BUILD WITH PYTHON

Anaconda Distribution

The gold standard for Python data science and development — trusted packages and the tools you need to build in secure, reliable environments.

 **50M+ users**
worldwide

 **30 seconds***
to get access

[Get Started with Anaconda Distribution](#)

[Returning user? Sign in](#) [Skip to download](#)

2. Click “Skip to download”

Choose Your Download

WindowsMacLinux

Anaconda Distribution

Package and environment management with conda, Anaconda Navigator desktop app, and 600+ packages. Everything you need for data science.

GRAPHICAL INSTALLER

[Windows 64-Bit Graphical Installer](#)

🔧 Graphical install instructions

COMMAND LINE

📄 CLI install instructions

Miniconda

Lightweight installer that only includes conda, Python, and their dependencies. Install only what you need.

GRAPHICAL INSTALLER

[Windows 64-Bit Graphical Installer](#)

🔧 Graphical install instructions

COMMAND LINE

📄 CLI install instructions

Need help choosing an installer? [Compare Anaconda Distribution vs. Miniconda](#) →

3. Install in your own PC

How to run Python – Jupyter Notebook

- Google Colaboratory (Colab)
 - Jupyter Notebook powered by Google Cloud ¹
 - <https://colab.research.google.com/>
 - Benefits ²
 - “Zero configuration required”
 - Easy sharing
 - Free access: CPU & GPU



1. Google Colab FAQ. <https://research.google.com/colaboratory/faq.html>

2. Google Colaboratory. <https://colab.research.google.com>

Cloud Software Introduction

Colaboratory, or Colab for short, is a Google Research product, which allows developers to write and execute Python code through their browser. Google Colab is an excellent tool for deep learning tasks. It is a hosted Jupyter notebook that requires no setup and has an excellent free version, which gives free access to Google computing resources such as GPUs and TPUs.

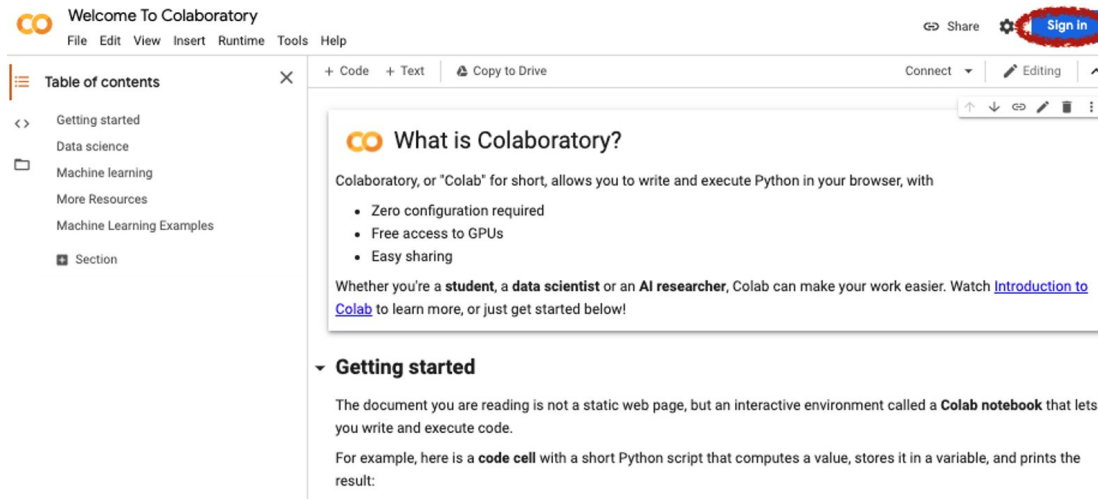
Why Should I Use Google Colab?

There are several reasons to opt to use Google Colab instead of a plain Jupyter Notebook instance:

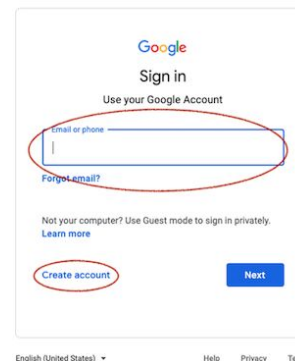
- Pre-Installed Libraries
- Saved on the Cloud
- Collaboration
- Free GPU and TPU Use

Google Colab Setup

1. Visit the [Google Colab](http://colab.research.google.com/) (<http://colab.research.google.com/>) page, which will direct you to the [Google Colaboratory Welcome Page](#).
2. Click the **Sign in** button on the right top.

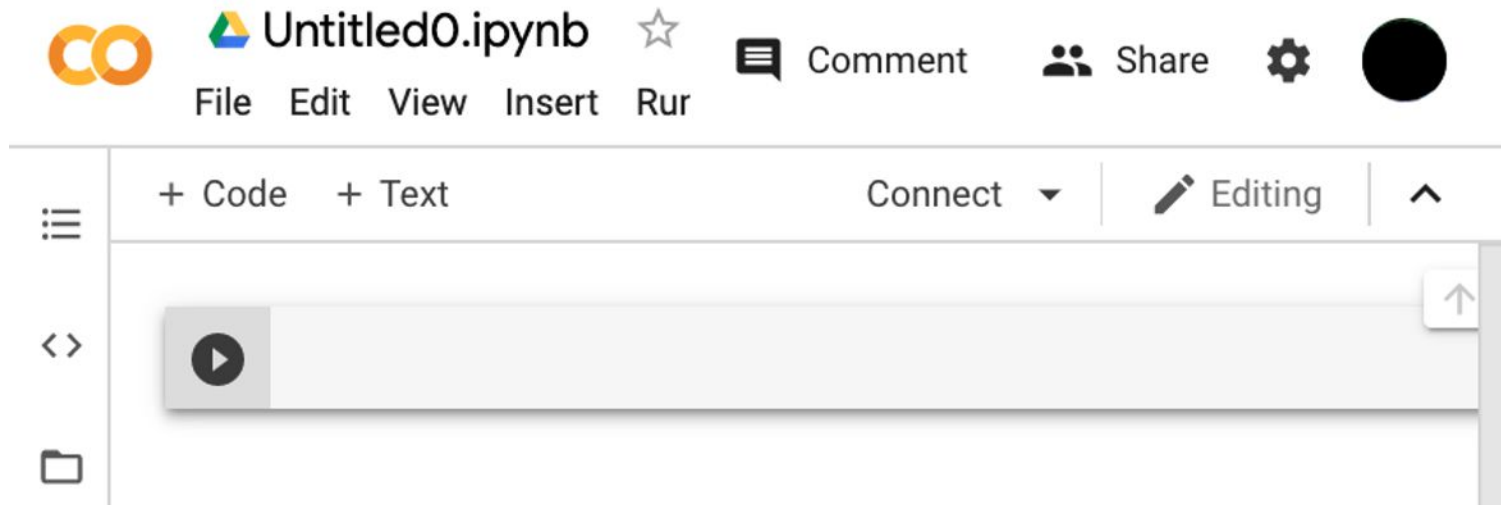


1. Sign in with your Gmail account.



Google Colab Setup

4. As soon as you complete the sign-in process, you are ready to use Google Colab.
4. You may easily create a new Colab notebook on this page by clicking *File> New notebook*.



Summary on running Python

Method 1 — Python Interpreter

- Install Anaconda Distribution in <https://www.anaconda.com/download>
- Open a terminal or command prompt.
- Type `python` and press Enter.
- Enter Python commands one by one and see the output immediately.

Method 2 — Colab

- Go to colab.research.google.com.
- Sign in with your Google account.
- Create a new notebook.
- Write and run Python code in cells using Shift+Enter.

Method 3 — Jupyter notebook

- Run `jupyter notebook` in terminal.
- Open the browser interface.
- Create a new notebook and run cells with Shift+Enter.

Method 4 — Script Mode

- Write code in a text editor and save it as a `.py` file.
- Run it from the terminal using `python filename.py`.
- Example: `python main.py`.

Python Interpreter

- Fast to start, with low resource use
- Small Experiments Quick feedback
- Better to test python library
- Use PC CPU/GPU Resources.

Colab

- Runs in the cloud -> no local setup
- Easy to share and collaborate
- Free GPU/TPU Resources
- Data store in cloud with lower security
- Works well with Google Drive

Jupyter notebook

- Better interactive exploration.
- Open the browser interface.
- Better Documentation along code
- Data store locally, great security.

Method 4 — Script Mode

- Best for larger programs because code is saved in a structured file
- Easier to reuse, test, and deploy as a real application or module

Additional basic python learning

<https://www.w3schools.com/python/>

PYTHON TUTORIAL	
Python HOME	
Python Intro	
Python Get Started	
Python Syntax	▼
Python Output	▼
Python Comments	▼
Python Variables	▼
Python Data Types	▼
Python Numbers	▼
Python Casting	▼
Python Strings	▼
Python Booleans	▼
Python Operators	▼
Python Lists	▼
Python Tuples	▼
Python Sets	▼
Python Dictionaries	▼
Python If...Else	▼
Python Match	▼
Python While Loops	▼
Python For Loops	▼

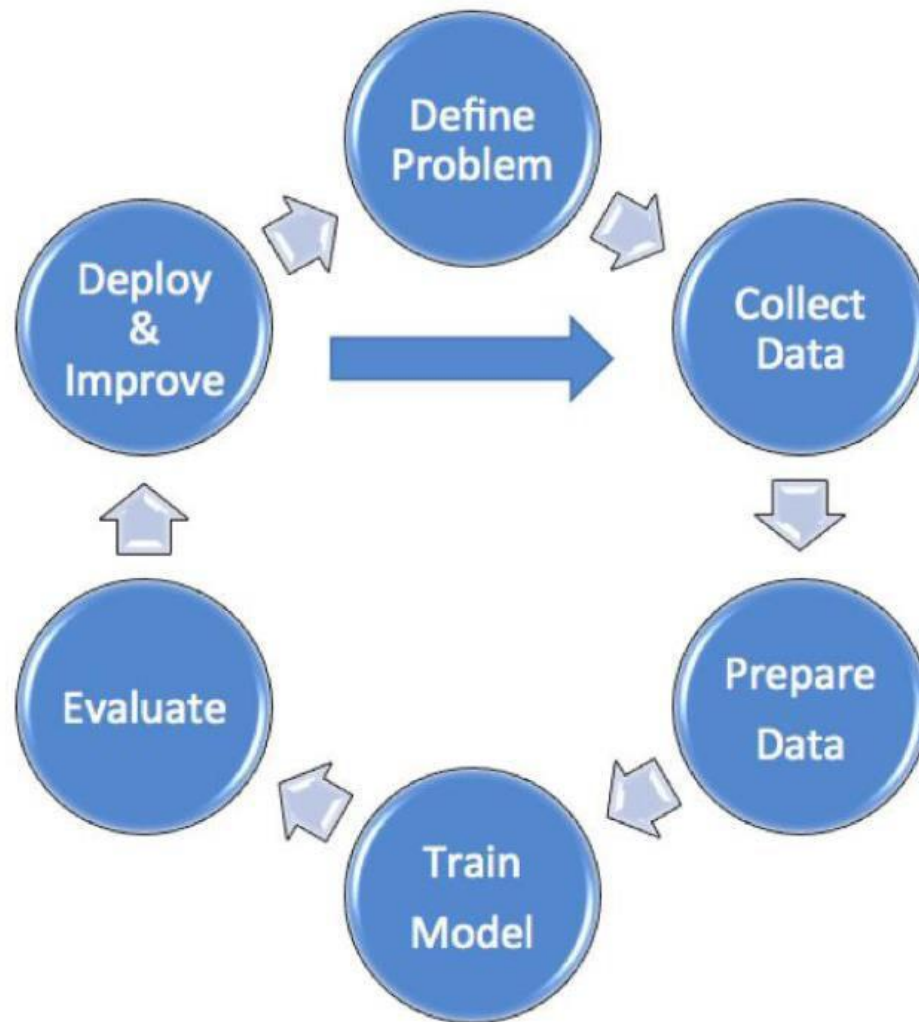
<https://www.tutorialspoint.com/python/index.htm>

- Python - Private Variables
- Python - Data Types
- Python - Type Casting
- Python - Unicode System
- Python - Literals
- Python - Operators
- Python - Arithmetic Operators
- Python - Comparison Operators
- Python - Assignment Operators
- Python - Logical Operators
- Python - Bitwise Operators
- Python - Membership Operators
- Python - Identity Operators
- Python - Walrus Operator
- Python - Operator Precedence

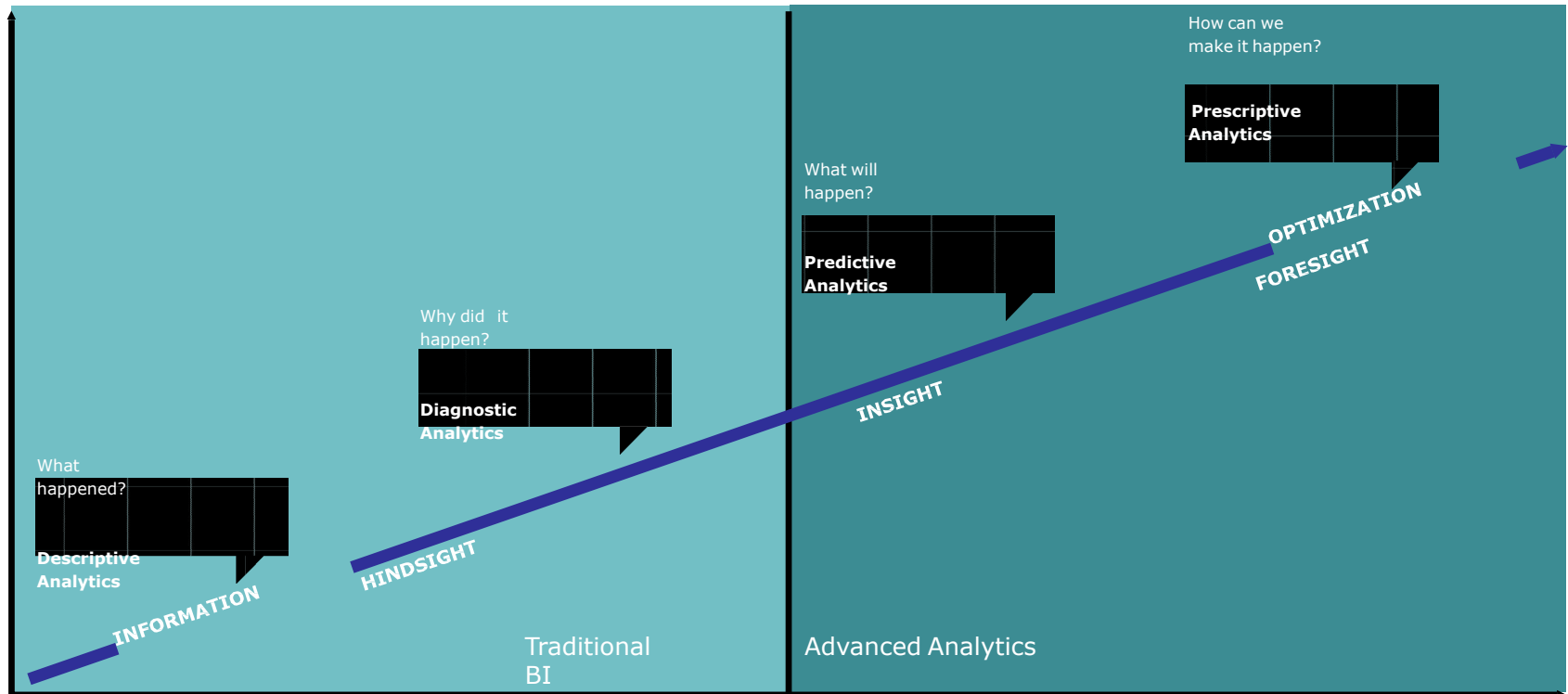


What is Process Automation, RPA, IPA?
Overview of the technological building blocks related to process automation
Contemporary tools for process automation
Challenges and opportunities of business process automation
Business implications of process automation

What is Business Analytics & Machine Learning?



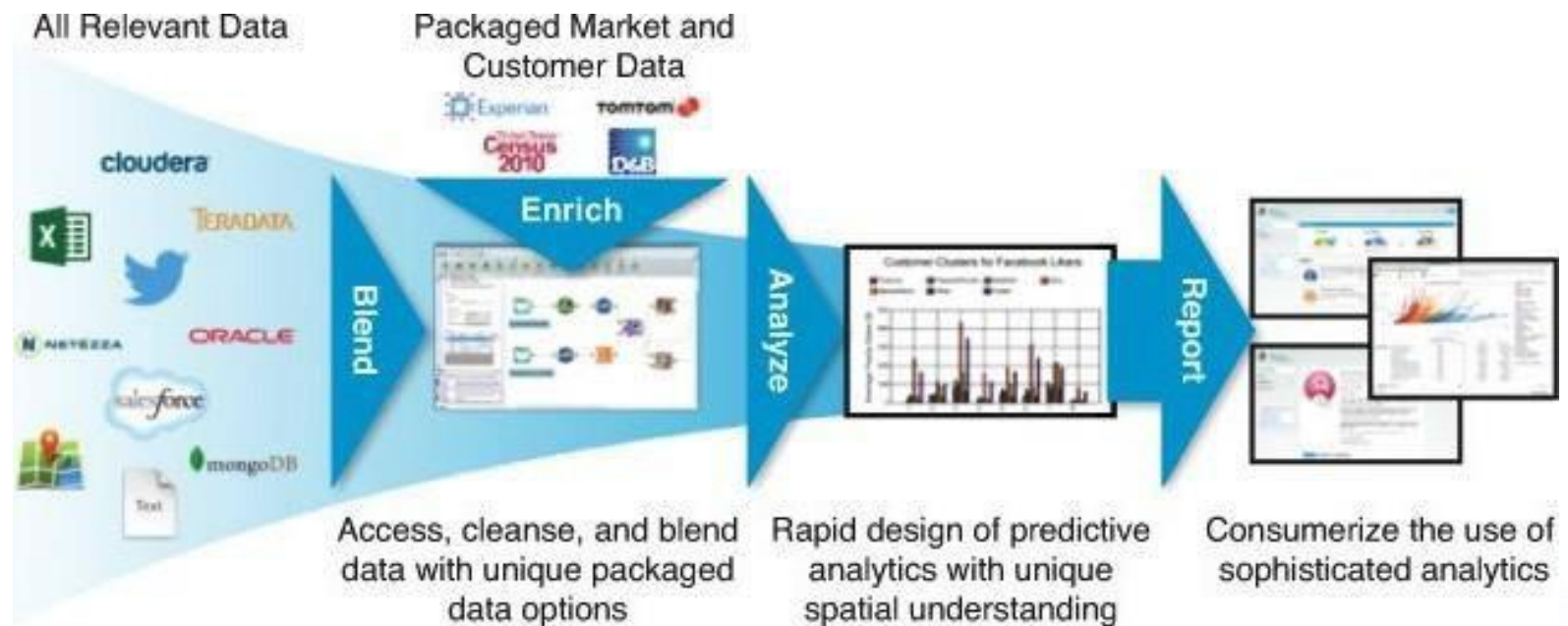
From Digitalization To Data-Driven Decision Making



Modern BI Process - Blend, Enrich, Analyze, and Report

Predictive Analytics: By applying machine learning algorithms to historical project data, organizations can forecast outcomes more accurately and mitigate risks in project delivery

Business Intelligence: Automating the preparation of data for analytics tools allows organizations to gain actionable insights more rapidly, enhancing their decision-making capabilities.



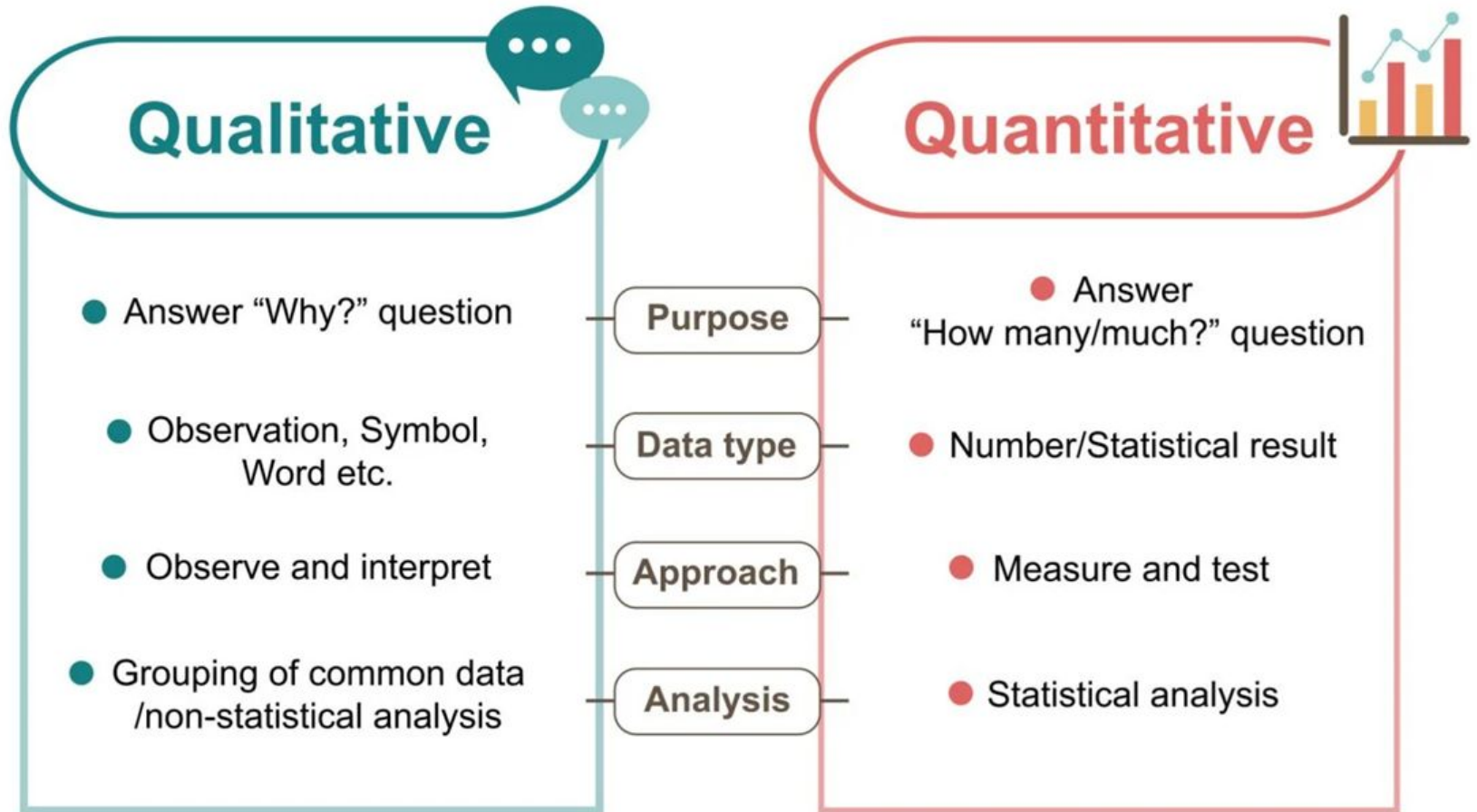
Data-Driven Business Strategy

A **data-driven business strategy** is a systematic plan that guides business decisions, operational improvements, and growth using verified data analysis rather than habit or intuition.

Data-Driven Decision-Making Process



Type of Research Design



Introduction to Business Process Automation

Tools for Automation – Getting a taste

- Visualping (<https://visualping.io/>)
 - Websites monitoring



<https://hkuspace.hku.hk> » prog » cert-for-

Certificate for Module (Busin

(2) Mr Jackie Liu. Mr. Jackie Liu is veter

top global investment banks. With over 1

-module-busi... ⋮

Business Process Automation with VBA ...

an quantitative strategist currently working in one of the
10 years of ...



hku.hk

<https://hkuspace.hku.hk> » prog » cert-for-module-busi... ⋮

Certificate for Module (Business Process Automation with VBA ...

(2) Mr Jackie Liu. Mr. Jackie Liu is veteran quantitative strategist currently working in one of the
top global investment banks. With over 10 years of ...

Tools for Automation – Considerations

- **Cost**

- **One-time cost**

- Initial effort of adaptation
 - Implementation and rollout

- **Running cost**

- License, e.g., Power Automate: \$Millions/year¹
 - Infrastructure, e.g., Cloud storage
 - Training, e.g., Automation Anywhere, 10+ hours²
 - Support, e.g., bot re-configuration

Cost		
Production	Gold	Maint.
60 🍌	240 🍌	1 🍌

1. RPA Pricing: Comparison of Leading RPA Vendors' Fees in 2023. <https://research.aimultiple.com/rpa-pricing/>

2. RPA Course Learning Trails | Automation Anywhere University. <https://university.automationanywhere.com/training/rpa-learning-trails/>

Introduction to Business Process Automation

Tools for Automation – Considerations

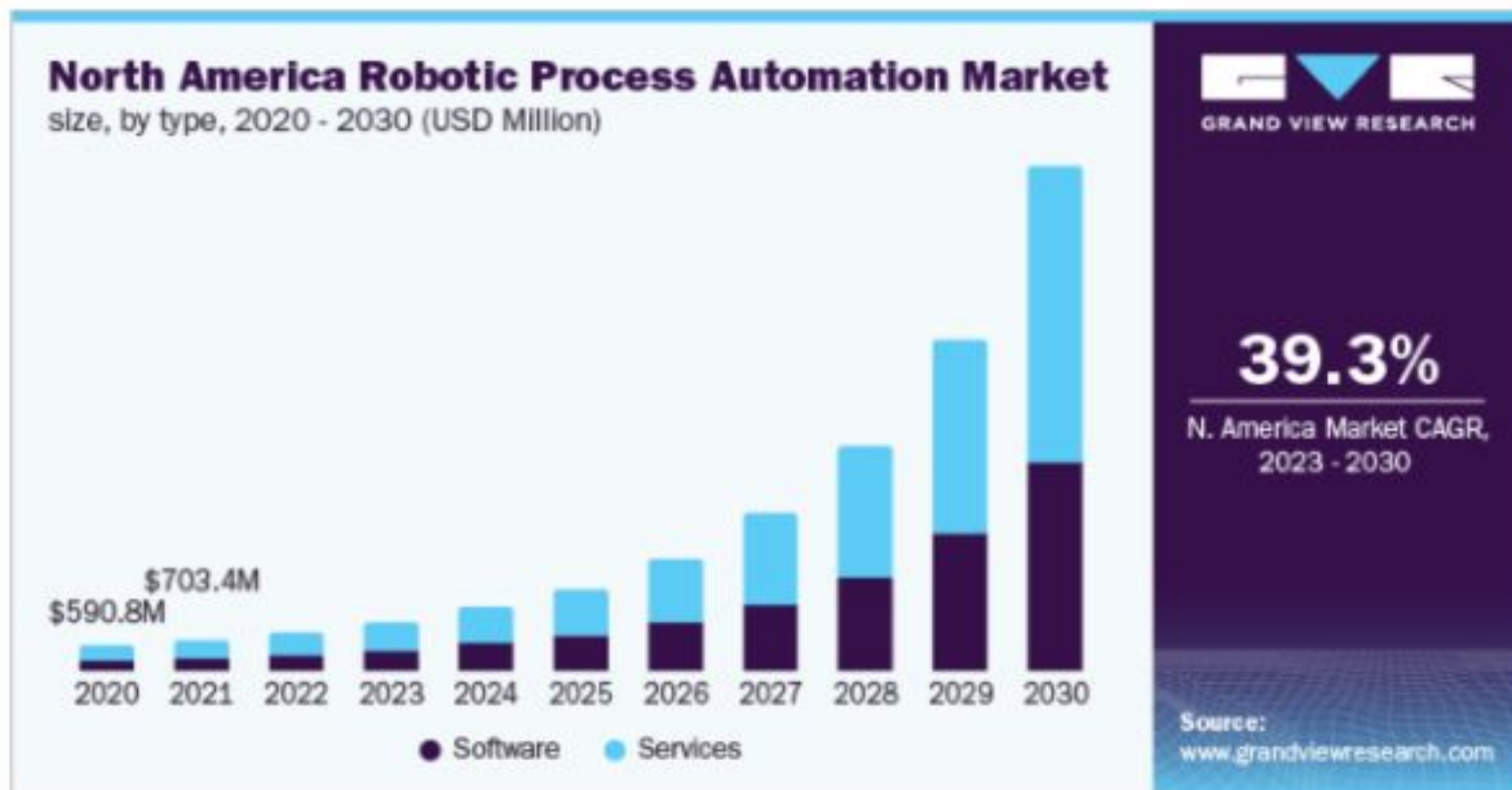
- **Flexibility**
 - Specialized vs Generalized, e.g., XCEPTOR → Tax
- **Scalability**
 - Desktop vs Enterprise, e.g., Excel VBA
 - Version Control, e.g., Use of repository
- **Time to delivery**
 - “Perfect” solution does not exist



1. Smeets, M., Erhard, R., & Kaußler, T. (2021, July 30). *Robotic Process Automation (RPA) in the Financial Sector: Technology - Implementation - Success for Decision Makers and Users*. Springer. <https://doi.org/10.1007/978-3-658-32974-7>

Introduction to Business Process Automation

Opportunities of BPA – The Money



1. *Robotic Process Automation Market Size & Share Report 2030*. Robotic Process Automation Market Size & Share Report 2030.
<https://www.grandviewresearch.com/industry-analysis/robotic-process-automation-rpa-market>

Opportunities of BPA – Sectors

Industries	% of RPA Solutions
Banking, Financial Services & Insurance	51 %
Business Process Outsourcing (BPO)	14 %
Manufacturing – Consumer Packaged Goods (CPG)	7 %
Professional, Legal & Accountancy Services	7 %
Retail Trade	7 %
Technology (IT, Internet, SAAS)	7 %
Utilities	7 %

1. Top 15 RPA Use Cases & Examples in Banking in 2023. <https://research.aimultiple.com/banking-rpa/>

Opportunities of BPA – Sectors

- **Healthcare**
 - Monitoring, drug dispensing
- **Retail**
 - Auto-checkout, smart shelf, inventory management
- **Marketing**
 - Lead generation
 - social media management
- **Finance**
 - Fraud detection, Credit management, Regtech



Introduction to Business Process Automation

Opportunities of BPA – Regtech

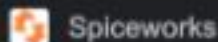
- Follow the money



Wells Fargo ordered to pay \$3.7 billion for 'illegal activity' including unjust foreclosures and vehicle repossessions

The CFPB said the more than \$2 billion in customer refunds Wells Fargo has been ordered to pay includes more than \$1.3 billion to consumers hurt...

20 Dec 2022



SEC Penalizes Major Wall Street Firms \$1.97B For Using Unauthorized Messaging Apps

If there are allegations of wrongdoing or misconduct, we must be ... Morgan Stanley recently agreed to a \$35 million fine by the SEC for...

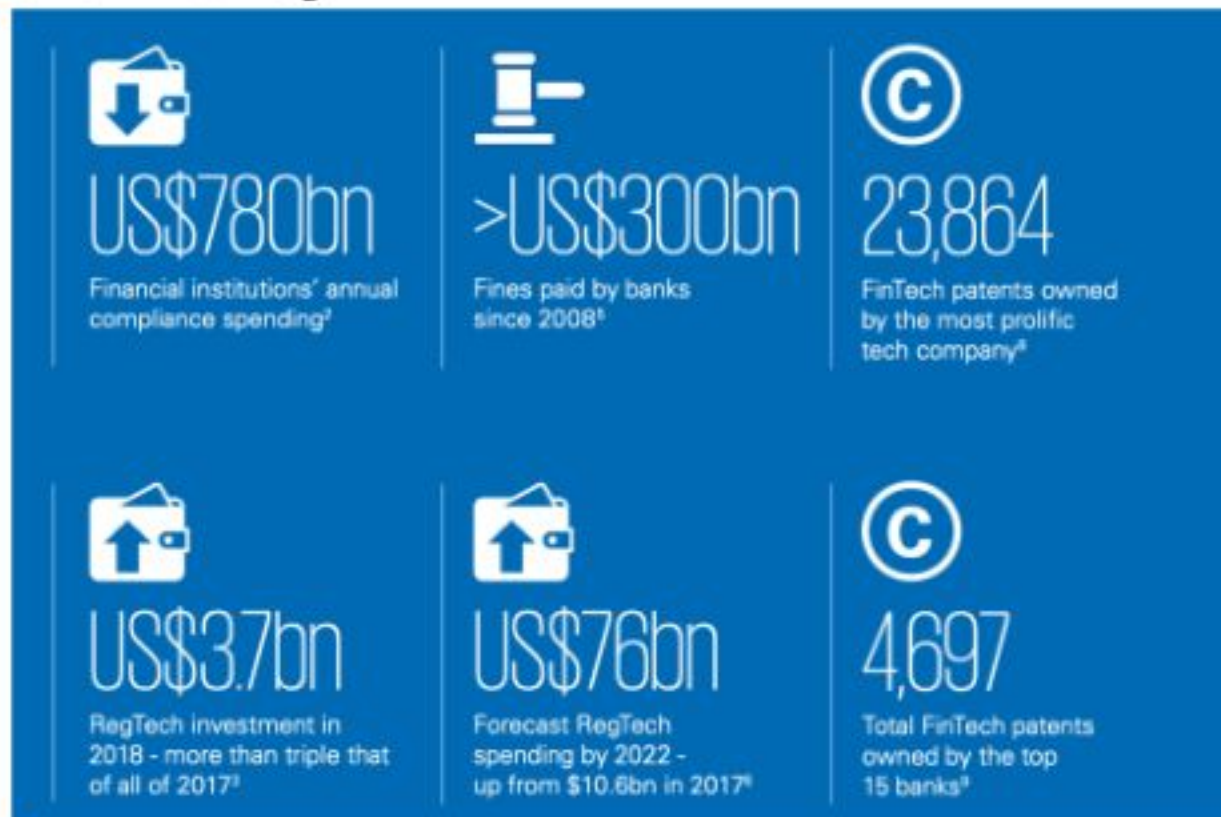
28 Sept 2022



Introduction to Business Process Automation

Opportunities of BPA – Regtech

- Follow the money



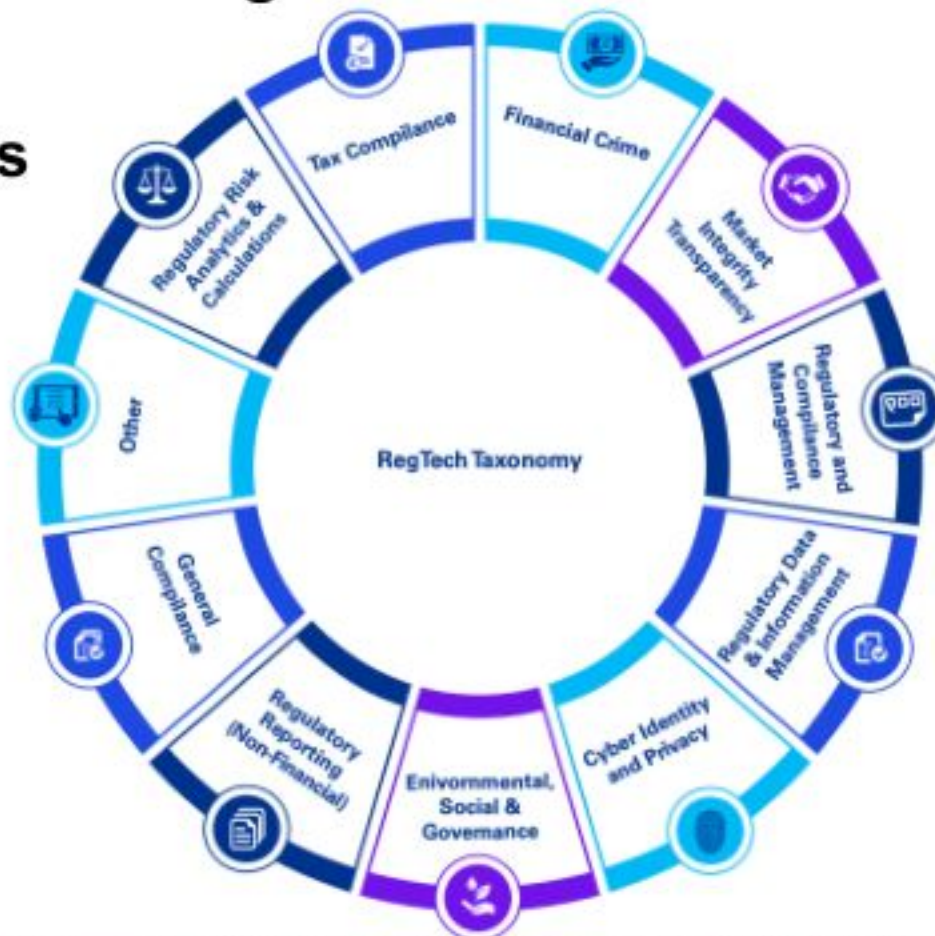
1. KPMG. (2019, June). *There's a revolution coming.*

<https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

- Common themes



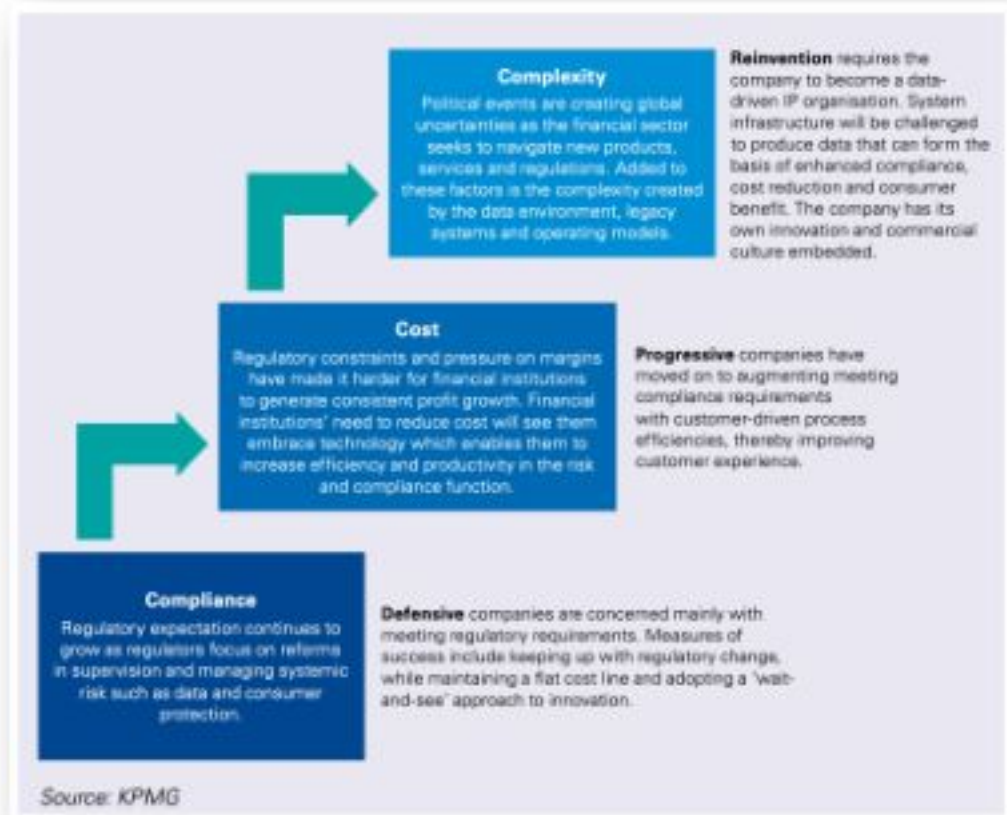
1. KPMG. (2022, November). <https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf>.
<https://assets.kpmg.com/content/dam/kpmg/uk/pdf/2022/11/innovate-finance-regtech-industry-and-adoption.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

Adoption phases:

1. Defensive
2. Progressive
3. Reinvention



1. KPMG. (2019, June). *There's a revolution coming*.
<https://assets.kpmg.com/content/dam/kpmg/cn/pdf/en/2019/06/embracing-the-challenge-of-the-new-regtech-era.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
 - **RPA:** automating rule-based processes
 - **Network Analysis:** Company holdings and exposures



1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Opportunities of BPA – Regtech

- **HKMA 3-Year Roadmap**
 - **Speech-to-text (STT):** Audio to time-stamped text
 - **Sentiment Analysis:** Realtime news monitoring



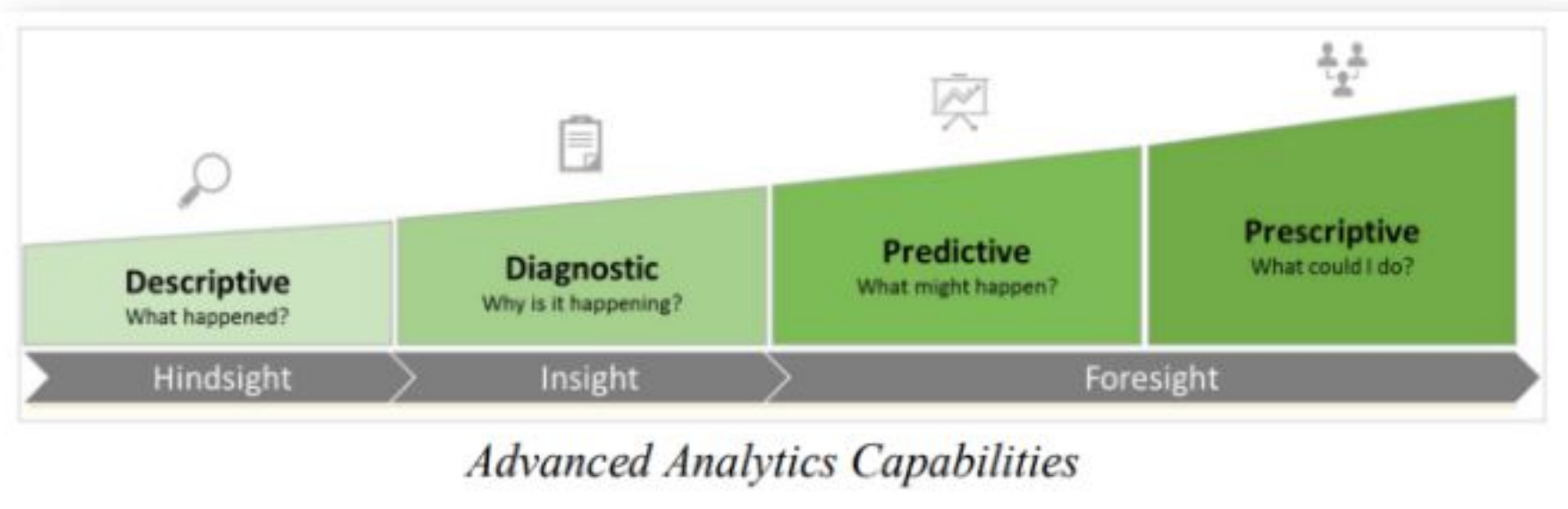
Illustrative Sentiment Analysis Heat Map

1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Introduction to Business Process Automation

Opportunities of BPA – Regtech

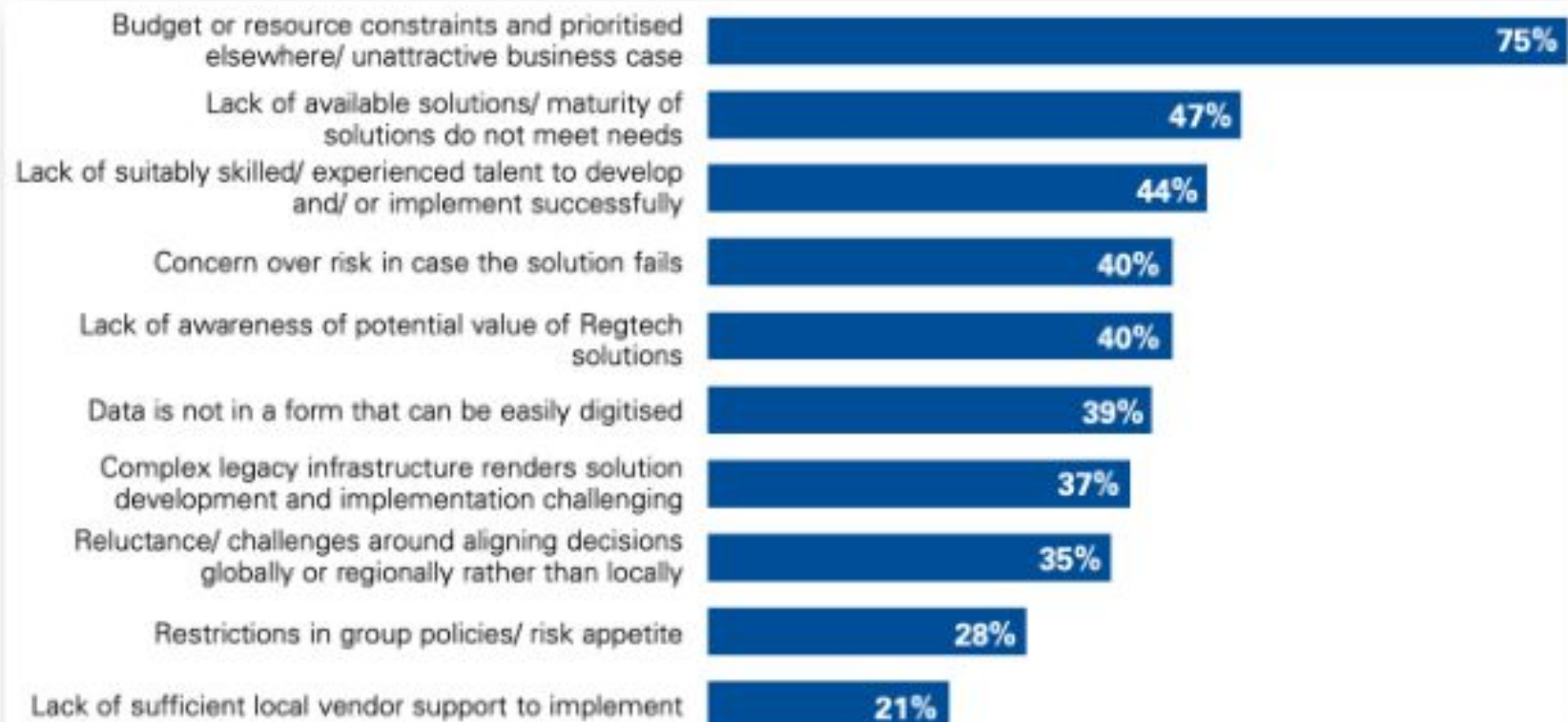
- **Analytics enabled by BPA**



1. HKMA. (2021, June). *Regtech Watch Issue No. 7*.
<https://www.hkma.gov.hk/media/eng/doc/key-information/guidelines-and-circular/2021/20210617e1a1.pdf>

Introduction to Business Process Automation

Challenges of BPA – Insight from the banks

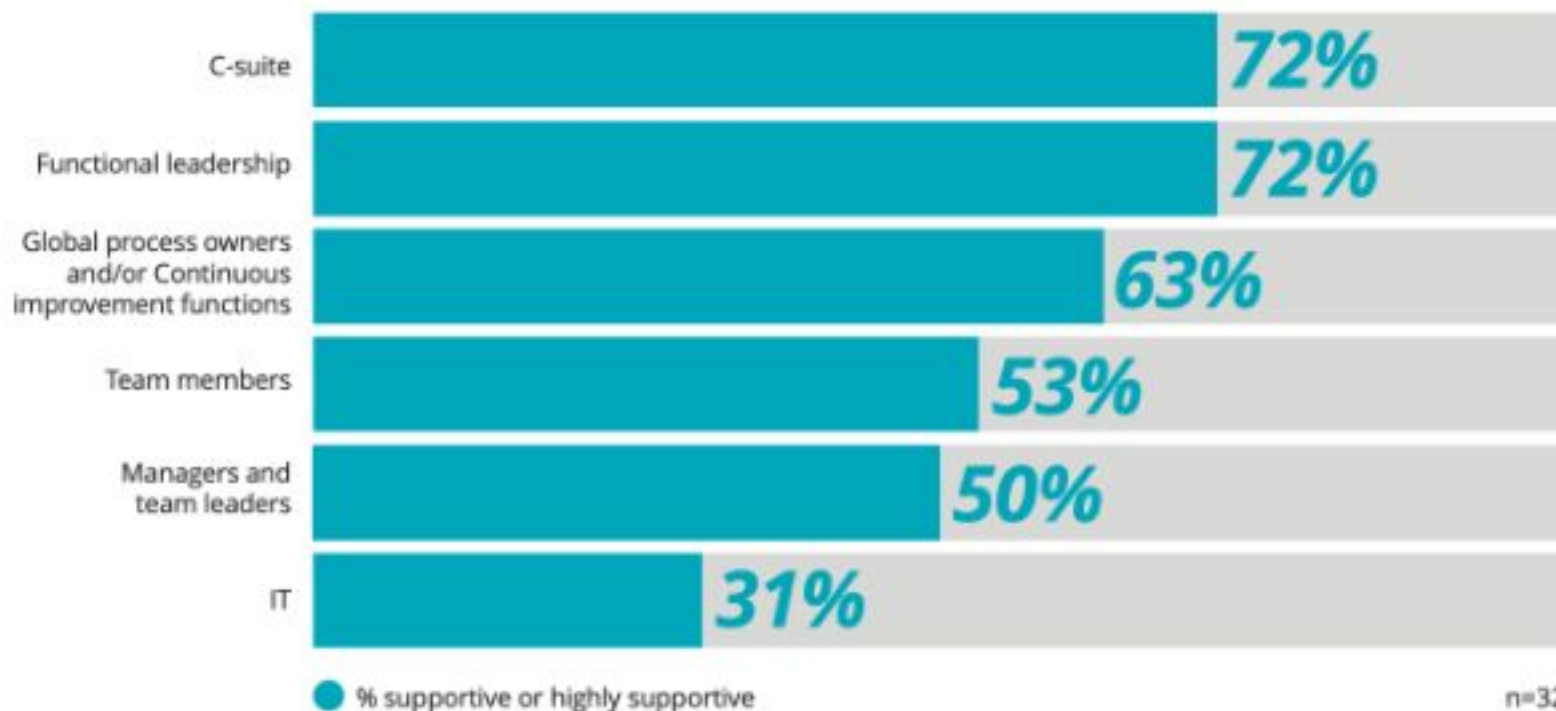


1. KPMG & HKMA. (2020). *Transforming Risk Management and Compliance: Harnessing the Power of Regtech*. <https://www.hkma.gov.hk/media/chi/doc/key-information/press-release/2020/20201102c3a1.pdf>

Introduction to Business Process Automation

Challenges of BPA – Worries from the workers?

Figure 5: How supportive of the RPA implementation were your stakeholder groups?

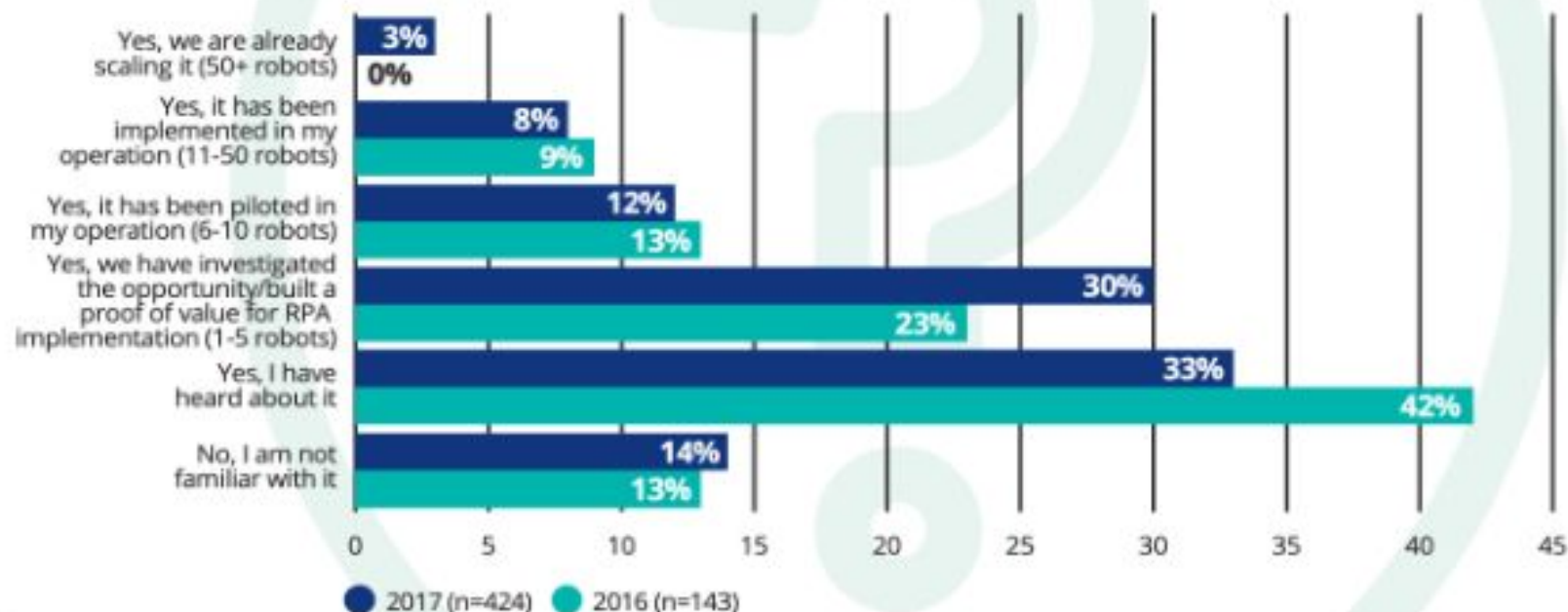


1. Deloitte. (2018). *The robots are ready. Are you?*
<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

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Challenges of BPA – Hesitant to scale up

Figure 2: Are you familiar with RPA?



1. Deloitte. (2018). *The robots are ready. Are you?*
<https://www2.deloitte.com/content/dam/Deloitte/tr/Documents/technology/deloitte-robots-are-ready.pdf>

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Challenges of BPA

- **Shortage**
 - Budget / Resource/ Talent
- **Concerns**
 - Risk of rushing / failing
 - Moral / Structural unemployment

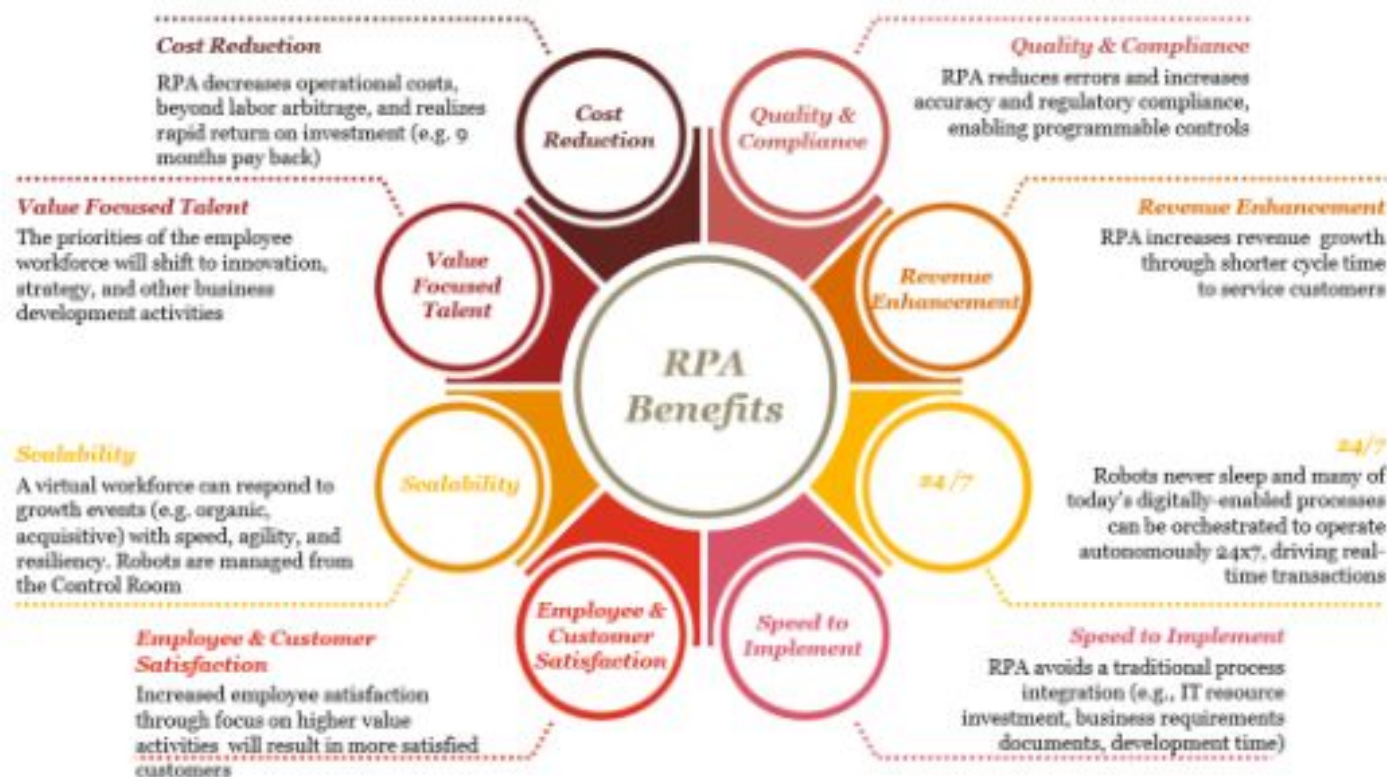


* Success depends on planning
(More on project management later)

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Implications of BPA – The Good

- Cost, Efficiency, Customer Experience



1. PwC Robotics Process Automation Solutions. PwC.
<https://www.pwchk.com/en/services/entrepreneurial-and-private-business/new-technology-digitalisation-and-transformation/process-automation-solutions.html>

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Implications of BPA – The Bad

- Automation complacency / dependency
- Risk of Technical Issues / Downtime

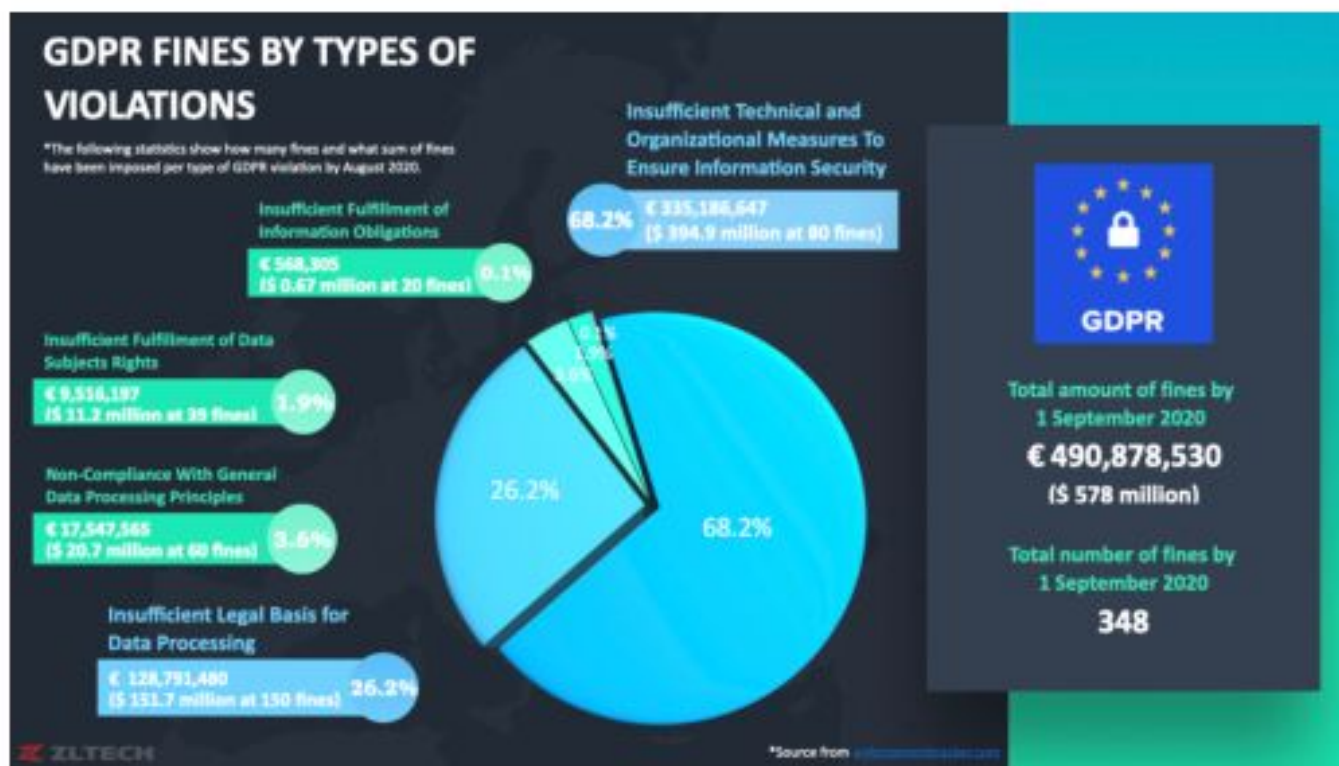


Image: Civil Aviation Safety Authority

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Implications of BPA – The Ugly

- Data Collection and Analysis ¹
- Vulnerability to Cyberattacks



1. Chen, B. (2020, September 9). *GDPR Recap: 28 Nations, 348 Fines, Half a Billion Euros*. ZL Tech.
<https://www.zltech.com/blog/gdpr-recap-28-nations-343-fines-half-a-billion-euros/>



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THANK YOU

